In [1]:
```
import glob
import os
import pandas as pd
import numpy as np
from sklearn.metrics import accuracy_score, precision_score, recall_score, c
pd.set_option('display.max_columns', None)

events = pd.read_parquet("../data/gdelt/events/6_final/events_dataset_v2.par
events = events[events['period'] <= '202406']
```

In [2]:
```
print(len(events.index))
print(events.columns)
events.tail(3)
```

```
71293
Index(['ADMIN0', 'ADMIN1', 'ADMIN2', 'CS_score', 'period', 'SQLDATE',
       'EventCode', 'SOURCEURL', 'NumMentions', 'NumSources', 'NumArticles',
       'valid_SOURCEURL', 'clean_text_list', 'NER_admin0_list',
       'NER_admin1_list', 'NER_admin2_list', 'compound_score_list',
       'neg_score_list', 'neu_score_list', 'pos_score_list',
       'fatalities_freq_list', 'displaced_freq_list', 'detained_freq_list',
       'injured_freq_list', 'sexual_violence_freq_list', 'torture_freq_lis
t',
       'economic_shocks_freq_list', 'agriculture_freq_list',
       'weather_freq_list', 'food_insecurity_freq_list',
       'fatalities_count_list', 'displaced_count_list', 'detained_count_lis
t',
       'injured_count_list', 'sexual_violence_count_list',
       'torture_count_list', 'sentiment.compound_list', 'sentiment.neg_lis
t',
       'sentiment.neu_list', 'sentiment.pos_list', 'pred_impact_type_list',
       'pred_urgency_list', 'pred_resource_food_list',
       'pred_resource_water_list', 'pred_resource_cash_aid_list',
       'pred_resource_healthcare_list', 'pred_resource_shelter_list',
       'pred_resource_livelihoods_list', 'pred_resource_education_list',
       'pred_resource_infrastructure_list', 'pred_resource_none_list'],
      dtype='object')
```

Out[2]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode	
271887	nigeria	zamfara	talata mafara	NaN	201907	[]	[20.0, 20.0, 43.0, 73.0, 46.0, 100.0, 43.0, 20...	[
271898	nigeria	zamfara	tsafe	NaN	201601	[]	[100.0, 42.0, 36.0, 100.0, 36.0, 20.0, 20.0, 1...	[http://
272100	niger	zinder	tanout	NaN	202308	[]	[42.0, 46.0, 42.0, 10.0, 10.0, 42.0, 172.0, 20...	[h

In [3]:
```
# Forecasting Configuration
# ============================
# Set the forecast horizon: predict CS_score N periods ahead
# Options: 1 (nowcasting), 2, 3, or 4 (forecasting)
# Literature shows GDELT features are more valuable for forecasting (2-4 per
# because baseline (previous_CS) becomes less powerful with longer horizons

FORECAST_HORIZON = 3  # Predict 3 periods ahead (can be changed to 2 or 4)

print(f"=== Forecasting Configuration ===")
print(f"Forecast Horizon: {FORECAST_HORIZON} period(s) ahead")
if FORECAST_HORIZON == 1:
    print("Mode: NOWCASTING (predicting next period)")
else:
    print(f"Mode: FORECASTING (predicting {FORECAST_HORIZON} periods ahead)"
print(f"\nWhy forecasting helps GDELT features:")
print(f"  - Baseline 'previous_CS' becomes less predictive ({FORECAST_HORIZO
print(f"  - GDELT temporal patterns can capture early warning signals")
print(f"  - More realistic for early warning systems\n")
```

```
=== Forecasting Configuration ===
Forecast Horizon: 3 period(s) ahead
Mode: FORECASTING (predicting 3 periods ahead)

Why forecasting helps GDELT features:
  - Baseline 'previous_CS' becomes less predictive (3 periods old)
  - GDELT temporal patterns can capture early warning signals
  - More realistic for early warning systems
```

In [4]:
```
df = events.copy()
print(f"Total rows: {len(df.index)}")
print(f"Period range: {min(df['period'])} to {max(df['period'])}")
print(f"Columns: {len(df.columns)}")

# Drop rows where ADMIN0 is missing
mask_missing = df['ADMIN0'].isna() | (df['ADMIN0'] == 'None') | (df['ADMIN0'
```

```
if mask_missing.sum() > 0:
    print(f"Dropping {mask_missing.sum()} rows with ADMIN0=None")
    df = df[~mask_missing].copy()

print(f"Final rows: {len(df)}")
df.tail()
```

```
Total rows: 71293
Period range: 201601 to 202406
Columns: 51
Dropping 429 rows with ADMIN0=None
Final rows: 70864
```

Out[4]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode	
271876	nigeria	zamfara	talata mafara	NaN	201804	[]	[10.0, 42.0, 71.0, 43.0, 43.0, 20.0, 43.0, 43....	[I
271883	nigeria	zamfara	talata mafara	NaN	201901	[]	[42.0, 10.0, 10.0, 20.0, 10.0, 10.0, 10.0, 100...	
271887	nigeria	zamfara	talata mafara	NaN	201907	[]	[20.0, 20.0, 43.0, 73.0, 46.0, 100.0, 43.0, 20...	[
271898	nigeria	zamfara	tsafe	NaN	201601	[]	[100.0, 42.0, 36.0, 100.0, 36.0, 20.0, 20.0, 1...	[http://
272100	niger	zinder	tanout	NaN	202308	[]	[42.0, 46.0, 42.0, 10.0, 10.0, 42.0, 172.0, 20...	[h

In [5]:

```
key_cols = ["ADMIN0", "ADMIN1", "ADMIN2", "period"]

def check_duplicates(df, name):
    dupes = (
        df
        .groupby(key_cols)
        .size()
        .reset_index(name="n")
        .query("n > 1")
    )
    print(f"{name}: {len(dupes)} duplicated keys")
    return dupes

dupes = check_duplicates(df, "events")
```

```
events: 19 duplicated keys
```

In [6]:

```python
# BETWEEN-ASSESSMENT FORECASTING SETUP
# ====================================
# CRITICAL: This implements the correct "between-assessment forecasting" set
#
# Key principle: Observed IPC defines targets; unobserved months are predict
#
# Target: y_next = next observed IPC assessment (forward-lagged label)
# Feature: IPC_last = last observed IPC before this row (forward-filled, cor
#
# Rules:
# - Only use IPC_last as IPC feature (no rolling, no multiple lags, no trend
# - CS_score is NOT the direct target (y_next is)
# - Keep ALL rows (including last periods without future assessments)
# - Filter to rows with valid y_next only when training models

print("="*70)
print("BETWEEN-ASSESSMENT FORECASTING SETUP")
print("="*70)
print("\nKey principle: Observed IPC defines targets; unobserved months are

# Create region identifiers
df['region'] = df['ADMIN0'] + '-' + df['ADMIN1']
df['district'] = df['ADMIN0'] + '-' + df['ADMIN1'] + '-' + df['ADMIN2']

# Sort by district (ADMIN2) and period to ensure deterministic operations
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period']).reset_index(dr

# CS_score is already in the events dataset - just ensure numeric
df['CS_score'] = pd.to_numeric(df['CS_score'], errors='coerce')

# Create IPC_last: last observed IPC before this row (forward-filled)
# This is the ONLY IPC feature allowed - it's stale and constant between ass
print("1. Creating IPC_last (last observed IPC, forward-filled)...")
df['IPC_last'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)['CS_
    lambda x: x.ffill()
)

# Create y_next: next observed IPC assessment (forward-lagged label)
# This is the target variable - we predict the next assessment outcome
print("2. Creating y_next (next observed IPC assessment - THE TARGET)...")

def get_next_assessment(group):
    """Get the next observed IPC assessment for each row"""
    cs_values = group['CS_score'].values
    result = []

    for i in range(len(cs_values)):
        # Look forward to find next non-null value
        found = False
        for j in range(i + 1, len(cs_values)):
            if pd.notna(cs_values[j]) and 1 <= cs_values[j] <= 5:
                result.append(cs_values[j])
                found = True
                break
```

```
        if not found:
            result.append(np.nan)

    return pd.Series(result, index=group.index)

df['y_next'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False, group_
    get_next_assessment
)

# Keep ALL rows - don't drop periods without future assessments
# Rows without y_next (like 202402) can still be used for feature engineerir
# Filter to rows with valid y_next only when training models
rows_with_y_next = df['y_next'].notna().sum()
rows_without_y_next = df['y_next'].isna().sum()
df = df[df['IPC_last'].isin([1, 2, 3, 4, 5])]
print(f"   Rows with y_next (can be used for training): {rows_with_y_next}")
print(f"   Rows without y_next (last periods, feature engineering only): {ro
print(f"   Total rows kept: {len(df)} (no periods dropped)")

# Clean y_next where it exists: ensure valid range (1-5) and convert to int
# Only clean where y_next is not NaN
mask_valid_y_next = df['y_next'].notna()
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].ro
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].cl
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].as

# Summary statistics
print(f"\n=== Summary ===")
print(f"IPC_last: last observed IPC (forward-filled, constant between assess
print(f"y_next: next observed IPC assessment (THE TARGET)")
print(f"\nIPC_last distribution:")
print(df['IPC_last'].value_counts().sort_index())
print(f"\ny_next distribution (TARGET):")
print(df['y_next'].value_counts().sort_index())
print(f"\nRows with valid y_next: {df['y_next'].notna().sum()}")
print(f"Rows without y_next (last periods): {df['y_next'].isna().sum()}")
print(f"Current DataFrame shape: {df.shape}")
print(f"Period range: {df['period'].min()} to {df['period'].max()}")

# Show example for verification
print(f"\n=== Example (first ADMIN2) ===")
if len(df) > 0:
    example_admin2 = df['ADMIN2'].iloc[0]
    example = df[df['ADMIN2'] == example_admin2].head(12)[['ADMIN2', 'perioc
    print(example.to_string(index=False))
    print(f"\nNote: IPC_last is constant between assessments, y_next is the

print("\n" + "="*70)
```

```
======================================================================
BETWEEN-ASSESSMENT FORECASTING SETUP
======================================================================

Key principle: Observed IPC defines targets; unobserved months are predictio
n times.

1. Creating IPC_last (last observed IPC, forward-filled)...
2. Creating y_next (next observed IPC assessment - THE TARGET)...
   Rows with y_next (can be used for training): 66446
   Rows without y_next (last periods, feature engineering only): 4418
   Total rows kept: 69645 (no periods dropped)

=== Summary ===
IPC_last: last observed IPC (forward-filled, constant between assessments)
y_next: next observed IPC assessment (THE TARGET)

IPC_last distribution:
IPC_last
1.0    34059
2.0    18841
3.0    14936
4.0     1805
5.0        4
Name: count, dtype: int64

y_next distribution (TARGET):
y_next
1.0    30949
2.0    18339
3.0    14603
4.0     1754
5.0        4
Name: count, dtype: int64

Rows with valid y_next: 65649
Rows without y_next (last periods): 3996
Current DataFrame shape: (69645, 55)
Period range: 201602 to 202406

=== Example (first ADMIN2) ===
ADMIN2 period  CS_score  IPC_last  y_next
  bale 201602       1.0       1.0     1.0
  bale 201606       1.0       1.0     1.0
  bale 201610       1.0       1.0     1.0
  bale 201702       1.0       1.0     1.0
  bale 201706       1.0       1.0     1.0
  bale 201710       1.0       1.0     1.0
  bale 202012       NaN       1.0     1.0
  bale 202106       1.0       1.0     1.0
  bale 202110       1.0       1.0     1.0
  bale 202202       1.0       1.0     1.0
  bale 202306       1.0       1.0     1.0
  bale 202310       1.0       1.0     1.0

Note: IPC_last is constant between assessments, y_next is the next assessmen
```

```
t value

================================================================================
```

```
/var/folders/3y/cdj5wjs107z_64y5f9ws4cj40000gn/T/ipykernel_47748/3370721709.
py:60: FutureWarning: DataFrameGroupBy.apply operated on the grouping column
s. This behavior is deprecated, and in a future version of pandas the groupi
ng columns will be excluded from the operation. Either pass `include_groups=
False` to exclude the groupings or explicitly select the grouping columns af
ter groupby to silence this warning.
  df['y_next'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False, grou
p_keys=False).apply(
```

In [7]:
```
# Article coverage by IPC score and ADMIN0
# ========================================

df_ipc = df[df['IPC_last'].notna()].copy()
df_ipc['has_articles'] = df_ipc['valid_SOURCEURL'].apply(
    lambda x: x.size > 0 if isinstance(x, np.ndarray) else bool(x) if x is r
)

total = len(df_ipc)
print(f"Total rows (with IPC_last): {total:,}")
print(f"Rows with articles: {df_ipc['has_articles'].sum():,} ({df_ipc['has_a

# Breakdown by IPC_last
print("\nBreakdown by IPC_last:")
breakdown = df_ipc.groupby('IPC_last').agg(
    total=('has_articles', 'size'),
    with_articles=('has_articles', 'sum'),
)
breakdown['pct_articles'] = (breakdown['with_articles'] / breakdown['total']
print(breakdown)

# Breakdown by ADMIN0
print("\nBreakdown by ADMIN0:")
breakdown_admin0 = df_ipc.groupby('ADMIN0').agg(
    total=('has_articles', 'size'),
    with_articles=('has_articles', 'sum'),
    admin2_total=('ADMIN2', 'nunique'),
)
admin2_cov = df_ipc[df_ipc['has_articles']].groupby('ADMIN0')['ADMIN2'].nuni
breakdown_admin0 = breakdown_admin0.join(admin2_cov, how='left').fillna({'ad
breakdown_admin0['admin2_with_coverage'] = breakdown_admin0['admin2_with_cov
breakdown_admin0['pct_admin2_covered'] = (breakdown_admin0['admin2_with_cove
breakdown_admin0['pct_articles'] = (breakdown_admin0['with_articles'] / brea
breakdown_admin0 = breakdown_admin0.sort_values('total', ascending=False)
print(breakdown_admin0)

# Admin1 propagation gain
admin1_coverage = df_ipc.groupby(['ADMIN0', 'ADMIN1', 'period'])['has_articl
o, p = df_ipc['has_articles'].sum(), admin1_coverage.sum()
print(f"\nAdmin1 propagation: {o:,} ({o/total*100:.1f}%) -> {p:,} ({p/total*

# Breakdown by ADMIN0 after admin1 propagation
print("\nBreakdown by ADMIN0 (after admin1 propagation):")
df_ipc['has_articles_admin1'] = admin1_coverage.values
```

```
breakdown_prop = df_ipc.groupby('ADMIN0').agg(
    total=('has_articles', 'size'),
    admin2_total=('ADMIN2', 'nunique'),
    pct_admin2=('has_articles', 'mean'),
    pct_admin1=('has_articles_admin1', 'mean'),
)
breakdown_prop['pct_admin2'] = (breakdown_prop['pct_admin2'] * 100).round(1)
breakdown_prop['pct_admin1'] = (breakdown_prop['pct_admin1'] * 100).round(1)
breakdown_prop['gain_pp'] = (breakdown_prop['pct_admin1'] - breakdown_prop['
breakdown_prop = breakdown_prop.sort_values('total', ascending=False)
print(breakdown_prop)
```

Total rows (with IPC_last): 69,645
Rows with articles: 14,939 (21.5%)

Breakdown by IPC_last:

IPC_last	total	with_articles	pct_articles
1.0	34059	6964	20.4
2.0	18841	4243	22.5
3.0	14936	3274	21.9
4.0	1805	454	25.2
5.0	4	4	100.0

Breakdown by ADMIN0:

ADMIN0	total	with_articles	admin2_total	admin2_with_coverage	pct_admin2_covered
nigeria	19989	3133	729	364	49.9
kenya	7183	1467	269	171	63.6
yemen	6586	980	300	130	43.3
sudan	4522	942	167	51	30.5
mozambique	3861	502	154	58	37.7
uganda	3142	870	112	76	67.9
democratic republic of the congo	2772	242	181	41	22.7
ethiopia	2428	862	76	53	69.7
zimbabwe	2414	1211	74	64	86.5
somalia	2339	799	74	51	68.9
niger	1917	303	72	24	33.3
south sudan	1879	477	79	28	35.4
chad	1831	290	68	21	30.9
burundi	1498	92	111		
mali	1464	459	50		
malawi	854	430	27		
cameroon	838	176	58		
zambia	837	355	89		
burkina faso	794	260	45		
liberia	723	216	131		
madagascar	671	271	22		
mauritania	505	165	45		
guinea	333	212	34		
sierra leone	265	225	13		

```
burundi                                          22               19.8
mali                                             40               80.0
malawi                                           25               92.6
cameroon                                         36               62.1
zambia                                           53               59.6
burkina faso                                     28               62.2
liberia                                          18               13.7
madagascar                                       20               90.9
mauritania                                       12               26.7
guinea                                           12               35.3
sierra leone                                     10               76.9

                                 pct_articles
ADMIN0
nigeria                                  15.7
kenya                                    20.4
yemen                                    14.9
sudan                                    20.8
mozambique                               13.0
uganda                                   27.7
democratic republic of the congo          8.7
ethiopia                                 35.5
zimbabwe                                 50.2
somalia                                  34.2
niger                                    15.8
south sudan                              25.4
chad                                     15.8
burundi                                   6.1
mali                                     31.4
malawi                                   50.4
cameroon                                 21.0
zambia                                   42.4
burkina faso                             32.7
liberia                                  29.9
madagascar                               40.4
mauritania                               32.7
guinea                                   63.7
sierra leone                             84.9

Admin1 propagation: 14,939 (21.5%) -> 46,010 (66.1%), gain +31,071 (44.6%)

Breakdown by ADMIN0 (after admin1 propagation):
                                 total  admin2_total  pct_admin2  pct_admin
1  \
ADMIN0
nigeria                          19989           729        15.7        76.
8
kenya                             7183           269        20.4        54.
3
yemen                             6586           300        14.9        67.
7
sudan                             4522           167        20.8        58.
4
mozambique                        3861           154        13.0        54.
6
uganda                            3142           112        27.7        99.
```

democratic republic of the congo	2772	181	8.7	31.6
ethiopia	2428	76	35.5	82.5
zimbabwe	2414	74	50.2	94.2
somalia	2339	74	34.2	64.0
niger	1917	72	15.8	49.0
south sudan	1879	79	25.4	75.6
chad	1831	68	15.8	21.9
burundi	1498	111	6.1	23.6
mali	1464	50	31.4	68.2
malawi	854	27	50.4	98.2
cameroon	838	58	21.0	59.3
zambia	837	89	42.4	71.3
burkina faso	794	45	32.7	44.1
liberia	723	131	29.9	40.8
madagascar	671	22	40.4	64.4
mauritania	505	45	32.7	38.8
guinea	333	34	63.7	69.4
sierra leone	265	13	84.9	89.8

ADMIN0	gain_pp
nigeria	61.1
kenya	33.9
yemen	52.8
sudan	37.6
mozambique	41.6
uganda	71.3
democratic republic of the congo	22.9
ethiopia	47.0
zimbabwe	44.0
somalia	29.8
niger	33.2
south sudan	50.2
chad	6.1
burundi	17.5
mali	36.8
malawi	47.8

```
cameroon                    38.3
zambia                      28.9
burkina faso                11.4
liberia                     10.9
madagascar                  24.0
mauritania                   6.1
guinea                       5.7
sierra leone                 4.9
```

In [8]:
```
print(min(df['period']))
print(max(df['period']))
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period'])
df.tail()
```

```
201602
202406
```

Out[8]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode
70859	zimbabwe	midlands	zvishavane urban	2.0	202206	[]	[70.0, 70.0, 42.0]
70860	zimbabwe	midlands	zvishavane urban	2.0	202210	[]	[10.0, 173.0, 173.0, 10.0, 42.0]
70861	zimbabwe	midlands	zvishavane urban	2.0	202302	[]	[173.0, 112.0]
70862	zimbabwe	midlands	zvishavane urban	2.0	202306	[20230604.0]	[173.0, 71.0, 10.0, 30.0, 43.0, 71.0]
70863	zimbabwe	midlands	zvishavane urban	2.0	202402	[]	[173.0, 42.0, 10.0]

Feature engineering

In [9]:
```
# COMPREHENSIVE FEATURE ENGINEERING
# ==================================
# This cell performs ALL feature engineering for IPC and GDELT Events featur
# At this point: lists are NOT aggregated yet, y_next and IPC_last exist

print("="*70)
print("COMPREHENSIVE FEATURE ENGINEERING")
print("="*70)
print(f"\nStarting with {len(df)} rows")
print(f"Columns before feature engineering: {len(df.columns)}\n")
```

```python
# ============================================================================
# 1. AGGREGATE GDELT LIST FEATURES
# ============================================================================
print("1. Aggregating GDELT list features...")

# Helper functions (already defined in previous cell, but ensure they exist)
def safe_list_agg(lst, func):
    """Safely aggregate a list, handling None, empty lists, and non-numeric
    if lst is None:
        return np.nan
    if isinstance(lst, np.ndarray):
        if lst.size == 0:
            return np.nan
        lst = lst.tolist()
    try:
        if pd.isna(lst):
            return np.nan
    except (ValueError, TypeError):
        pass
    if isinstance(lst, (int, float)):
        return float(lst)
    if isinstance(lst, str):
        try:
            if lst.startswith('[') or lst.startswith('('):
                lst = eval(lst)
            else:
                return float(lst)
        except:
            return np.nan
    if not isinstance(lst, (list, tuple)):
        return np.nan
    if len(lst) == 0:
        return np.nan
    try:
        numeric_lst = []
        for x in lst:
            try:
                if pd.isna(x) or x is None:
                    continue
            except (ValueError, TypeError):
                pass
            try:
                val = float(x)
                if not np.isinf(val) and not np.isnan(val):
                    numeric_lst.append(val)
            except (ValueError, TypeError):
                continue
        if len(numeric_lst) == 0:
            return np.nan
        result = func(numeric_lst)
        return float(result) if not np.isnan(result) and not np.isinf(result
    except Exception as e:
        return np.nan

def safe_list_count(x):
```

```
    """Safely count elements in a list/array"""
    if x is None:
        return 0
    if isinstance(x, np.ndarray):
        return x.size if x.size > 0 else 0
    if isinstance(x, (list, tuple)):
        return len(x)
    try:
        if pd.isna(x):
            return 0
        return 1
    except (ValueError, TypeError):
        return 1

def aggregate_list_features(df, list_cols, prefix=""):
    """Aggregate list columns into meaningful statistical features for NLP-d

    For noisy NLP features, we keep only:
    - mean: central tendency
    - count: volume indicator
    - sum: total (especially useful for frequencies/counts)
    - max: peak intensity (only for crisis indicators)

    We skip: min (usually 0), std (variance not meaningful for noisy signals
    """
    new_cols = {}
    cols_to_drop = []

    for col in list_cols:
        if col not in df.columns:
            continue

        base_name = col.replace('_list', '')
        if prefix:
            base_name = f"{prefix}_{base_name}"

        # For text columns (NER, clean_text), only create count
        if 'NER' in col or 'clean_text' in col:
            new_cols[f"{base_name}_count"] = df[col].apply(safe_list_count)
        else:
            # For numeric columns, create meaningful aggregations only
            # Mean: central tendency (most important for noisy signals)
            new_cols[f"{base_name}_mean"] = df[col].apply(lambda x: safe_lis

            # Count: volume indicator (how many articles/events)
            new_cols[f"{base_name}_count"] = df[col].apply(safe_list_count)

            # Sum: total (especially useful for frequencies and counts)
            new_cols[f"{base_name}_sum"] = df[col].apply(lambda x: safe_list

            # Max: peak intensity (only for crisis indicators where peaks ma
            # Check if this is a crisis-related feature
            is_crisis_feature = any(x in col.lower() for x in ['fatalities',
                                                               'violence',
            if is_crisis_feature:
                new_cols[f"{base_name}_max"] = df[col].apply(lambda x: safe_
```

```
            # Skip: min (usually 0), std (variance not meaningful for noisy

        cols_to_drop.append(col)

    if new_cols:
        new_df = pd.DataFrame(new_cols, index=df.index)
        df = pd.concat([df, new_df], axis=1)

    if cols_to_drop:
        df = df.drop(columns=cols_to_drop)

    return df

# Find and aggregate list columns (events only, no merge suffix)
list_cols = [c for c in df.columns if '_list' in c]
print(f"   Found {len(list_cols)} list columns")

if list_cols:
    print("   Aggregating events features...")
    df = aggregate_list_features(df, list_cols, prefix="evt")

# ================================================================================
# 2. CLEAN AND PREPARE IPC FEATURES
# ================================================================================
print("\n2. Cleaning IPC features...")

# Clean IPC_last: round to nearest integer and ensure valid range
df['IPC_last'] = df['IPC_last'].round()
# Keep IPC_last as float for now (can filter invalid values later if needed)
# Don't drop rows here - keep all periods

print(f"   IPC_last range: {df['IPC_last'].min():.1f} to {df['IPC_last'].max
print(f"   Valid IPC_last (1-5): {(df['IPC_last'].between(1, 5, inclusive='b

# ================================================================================
# 3. CREATE TEMPORAL FEATURES FOR GDELT
# ================================================================================
print("\n3. Creating temporal features for GDELT...")

# Focus on most reliable features for temporal patterns
# Prioritize: sentiment, crisis indicators, food security
key_gdelt_features = [
    # Sentiment (most reliable NLP signals)
    'evt_compound_score_mean', 'evt_sentiment.compound_mean',

    # Crisis indicators (high signal-to-noise)
    'evt_fatalities_freq_mean', 'evt_displaced_freq_mean',

    # Food security (directly relevant to IPC)
    'evt_food_insecurity_freq_mean',

    # Economic/agricultural (relevant but less reliable)
    'evt_economic_shocks_freq_mean', 'evt_agriculture_freq_mean',
]
```

```
# Filter to features that actually exist
key_gdelt_features = [f for f in key_gdelt_features if f in df.columns]

print(f"   Creating temporal features for {len(key_gdelt_features)} key GDEL

# Ensure sorted for lag operations
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period']).reset_index(dr

# Create rolling windows (smoothing noise - most useful for NLP features)
# Use 3 and 6 periods to capture short and medium-term trends
for window in [3, 6]:
    for feat in key_gdelt_features:
        df[f"{feat}_rolling_{window}"] = df.groupby(['ADMIN0', 'ADMIN1', 'AD
            lambda x: x.rolling(window=window, min_periods=1).mean()
        )

# Create lags (1 and 2 periods only - reduce noise)
# Lags can capture trends but too many lags add noise
for lag in [1, 2]:
    for feat in key_gdelt_features:
        df[f"{feat}_lag{lag}"] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'],

# Create escalation indicators (binary - less noisy than continuous change)
# Use rolling mean comparison instead of raw change to reduce noise
for feat in key_gdelt_features:
    rolling_3 = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)[feat]
        lambda x: x.rolling(window=3, min_periods=1).mean()
    )
    rolling_6 = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)[feat]
        lambda x: x.rolling(window=6, min_periods=2).mean()
    )
    # Escalation: current 3-period average > 6-period average
    df[f"{feat}_escalation"] = (rolling_3 > rolling_6).astype(int)

# Skip: change, pct_change, anomaly (too noisy for NLP-derived features)

print(f"   Created temporal features: rolling windows (3, 6 periods), lags (

# ============================================================================
# 4. CREATE FEATURE INDICATORS AND INTERACTIONS
# ============================================================================
print("\n4. Creating feature indicators and interactions...")

# GDELT feature availability indicator
gdelt_cols = [c for c in df.columns if c.startswith('evt_')]
if gdelt_cols:
    has_gdelt = df[gdelt_cols].notna().any(axis=1)
    df['has_gdelt'] = has_gdelt.astype(int)
else:
    df['has_gdelt'] = 0

# Sentiment features from events

# Combine crisis indicators
crisis_features = [c for c in df.columns if any(x in c for x in ['fatalities
                   and c.endswith('_mean') and c.startswith('evt_')]
```

```python
if crisis_features:
    df['crisis_intensity'] = df[crisis_features].fillna(0).sum(axis=1)

# Food security combined
food_features = [c for c in df.columns if 'food_insecurity' in c and c.endsw
                 and c.startswith('evt_')]
if food_features:
    df['food_security_combined'] = df[food_features].fillna(0).mean(axis=1)

print(f"   Created interaction and combined features")

# ============================================================================
# 5. CREATE GEOGRAPHIC DUMMY VARIABLES
# ============================================================================
print("\n5. Creating geographic dummy variables...")

# Only create dummies for base geographic identifiers (ADMIN0, ADMIN1, ADMIN
# Note: 'region' and 'district' are redundant (derived from ADMIN0+ADMIN1 ar
geographic_cols = []
for col in ['ADMIN0', 'ADMIN1', 'ADMIN2']:
    if col in df.columns:
        # Create dummy variables (one-hot encoding)
        dummies = pd.get_dummies(df[col], prefix=col, drop_first=False)
        df = pd.concat([df, dummies], axis=1)
        geographic_cols.extend(dummies.columns.tolist())
        print(f"   Created {len(dummies.columns)} dummy variables for {col}"

geographic_features = geographic_cols
print(f"   Total geographic features: {len(geographic_features)}")
print(f"   Note: 'region' and 'district' are kept as categorical identifiers

# ============================================================================
# 6. SUMMARY
# ============================================================================
print("\n" + "="*70)
print("FEATURE ENGINEERING COMPLETE")
print("="*70)

print(f"\nFinal dataset: {len(df)} rows")
print(f"Total columns: {len(df.columns)}")

# Feature counts
ipc_features = ['IPC_last']

# GDELT base features: evt_ prefixed features
gdelt_base = [c for c in df.columns if c.startswith('evt_')
              and not any(x in c for x in ['_lag', '_rolling', '_escalation'

# GDELT temporal features: evt_ prefixed features with temporal patterns
gdelt_temporal = [c for c in df.columns if c.startswith('evt_')
                  and any(x in c for x in ['_lag', '_rolling', '_escalatior

# Geographic features were created as dummies in section 5 above
# geographic_features is already set to geographic_cols

# Geographic features: only ADMIN0, ADMIN1, ADMIN2 are converted to dummies
```

```
# 'region' and 'district' are kept as categorical identifiers for grouping b
# Exclude original categorical columns from feature lists (but keep them in
geographic_categorical_cols = ['ADMIN0', 'ADMIN1', 'ADMIN2', 'region', 'dist

# Exclude all non-feature columns (geographic categoricals, metadata, scores
excluded_cols = (ipc_features + gdelt_base + gdelt_temporal + geographic_fea
                 ['y_next', 'has_gdelt', 'has_articles',
                  'CS_score',
                  'period', 'SQLDATE', 'EventCode', 'SOURCEURL', 'NumMentior
                  'NumArticles', 'valid_SOURCEURL'] +
                 geographic_categorical_cols +
                 # Also exclude any remaining list columns that weren't aggr
                 [c for c in df.columns if '_list' in c])
other_features = [c for c in df.columns if c not in excluded_cols]

print(f"\nFeature breakdown:")
print(f"  - IPC features: {len(ipc_features)}")
print(f"  - GDELT base features: {len(gdelt_base)} (mean, count, sum, max fo
print(f"  - GDELT temporal features: {len(gdelt_temporal)} (rolling windows,
print(f"  - Geographic features: {len(geographic_features)} (dummy variables
print(f"  - Other features: {len(other_features)}")
print(f"  - Total features: {len(ipc_features) + len(gdelt_base) + len(gdelt
print(f"\nNote: Simplified feature set for noisy NLP-derived signals:")
print(f"  - Base: mean, count, sum (all features) + max (crisis features onl
print(f"  - Temporal: rolling windows (3, 6 periods), lags (1, 2 periods), e
print(f"  - Removed: min, std, change, pct_change, anomaly (too noisy for NL

print(f"\nRows with y_next: {df['y_next'].notna().sum()} ({df['y_next'].notr
print(f"Rows with IPC_last: {df['IPC_last'].notna().sum()} ({df['IPC_last'].
print(f"Rows with GDELT features: {df['has_gdelt'].sum()} ({df['has_gdelt'].

# ============================================================================
# 6. DROP PROCESSED/METADATA COLUMNS
# ============================================================================
print("\n6. Dropping processed metadata columns...")

cols_to_drop = ['SQLDATE', 'EventCode', 'SOURCEURL', 'NumMentions',
                'NumSources', 'NumArticles', 'valid_SOURCEURL']
# Only drop columns that exist
cols_to_drop = [c for c in cols_to_drop if c in df.columns]
if cols_to_drop:
    df.drop(cols_to_drop, axis=1, inplace=True)
    print(f"   Dropped {len(cols_to_drop)} metadata columns")

print("\n" + "="*70)
print("READY FOR MODELING")
print("="*70)
print("\nFeature sets available:")
print(f"  - IPC only: {len(ipc_features)} IPC features + {len(geographic_fea
print(f"  - IPC + GDELT base: {len(ipc_features)} IPC + {len(gdelt_base)} GD
print(f"  - IPC + GDELT + temporal: {len(ipc_features)} IPC + {len(gdelt_bas
print("="*70)
```

```
================================================================================
COMPREHENSIVE FEATURE ENGINEERING
================================================================================

Starting with 69645 rows
Columns before feature engineering: 55

1. Aggregating GDELT list features...
   Found 39 list columns
   Aggregating events features...

2. Cleaning IPC features...
   IPC_last range: 1.0 to 5.0
   Valid IPC_last (1-5): 69645 rows

3. Creating temporal features for GDELT...
   Creating temporal features for 7 key GDELT features
   Created temporal features: rolling windows (3, 6 periods), lags (1, 2 per
iods), escalation indicators

4. Creating feature indicators and interactions...
   Created interaction and combined features

5. Creating geographic dummy variables...
   Created 24 dummy variables for ADMIN0
   Created 338 dummy variables for ADMIN1
   Created 2969 dummy variables for ADMIN2
   Total geographic features: 3331
   Note: 'region' and 'district' are kept as categorical identifiers but not
converted to dummies (redundant with ADMIN0+ADMIN1+ADMIN2)

================================================================================
FEATURE ENGINEERING COMPLETE
================================================================================

Final dataset: 69645 rows
Total columns: 3504

Feature breakdown:
  - IPC features: 1
  - GDELT base features: 119 (mean, count, sum, max for crisis)
  - GDELT temporal features: 35 (rolling windows, lags, escalation)
  - Geographic features: 3331 (dummy variables)
  - Other features: 2
  - Total features: 3488

Note: Simplified feature set for noisy NLP-derived signals:
  - Base: mean, count, sum (all features) + max (crisis features only)
  - Temporal: rolling windows (3, 6 periods), lags (1, 2 periods), escalatio
n indicators
  - Removed: min, std, change, pct_change, anomaly (too noisy for NLP featur
es)

Rows with y_next: 65649 (94.3%)
Rows with IPC_last: 69645 (100.0%)
Rows with GDELT features: 69645 (100.0%)
```

```
6. Dropping processed metadata columns...
   Dropped 7 metadata columns

======================================================================
READY FOR MODELING
======================================================================

Feature sets available:
  - IPC only: 1 IPC features + 3331 geographic features
  - IPC + GDELT base: 1 IPC + 119 GDELT base + 3331 geographic
  - IPC + GDELT + temporal: 1 IPC + 119 GDELT base + 35 GDELT temporal + 333
1 geographic
======================================================================
```

In [10]:
```
print(min(df['period']))
print(max(df['period']))
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period'])
df.tail()
```

```
201602
202406
```

Out[10]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	region	district	IPC
69640	zimbabwe	midlands	zvishavane urban	2.0	202206	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69641	zimbabwe	midlands	zvishavane urban	2.0	202210	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69642	zimbabwe	midlands	zvishavane urban	2.0	202302	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69643	zimbabwe	midlands	zvishavane urban	2.0	202306	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69644	zimbabwe	midlands	zvishavane urban	2.0	202402	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	

Configure train and test, and loops

In [11]:
```
# ======================================================================
# NAIVE BASELINE MODELS — TRANSITION PREDICTION
# ======================================================================
```

```
# Target: binary — did CS_score WORSEN vs the previous assessment?
#   worsened = 1  if CS_score(T) > CS_score(T-1)
#   worsened = 0  otherwise (stable or improved)
#
# Baselines:
#   No-Change      — always predicts 0 (no worsening)
#   Always-Worsen  — always predicts 1 (worst-case alert)
#   SPLY-Transition — predicts 1 if area worsened at same period last year
# ============================================================================

from sklearn.metrics import accuracy_score, precision_score, recall_score, f

admin_cols = ['ADMIN0', 'ADMIN1', 'ADMIN2']

assessment_periods_bl = sorted(
    df.loc[df['CS_score'].between(1, 5), 'period'].unique()
)

# Precompute CS_score lookup per (admin2, period) for transition calculation
cs_lookup = (df.loc[df['CS_score'].between(1, 5), admin_cols + ['period', 'C
             .drop_duplicates(admin_cols + ['period']))

print(f"Evaluating transition baselines on {len(assessment_periods_bl)} asse

baseline_names = ['No-Change', 'Always-Worsen', 'SPLY-Transition']
period_results = {name: [] for name in baseline_names}

for test_period in assessment_periods_bl:
    test = df[(df['period'] == test_period) & df['CS_score'].between(1, 5)].
    train_periods = [p for p in assessment_periods_bl if p < test_period]

    if len(test) == 0 or len(train_periods) < 2:
        continue

    prev_period = train_periods[-1]

    # Get previous assessment CS_score for each area → compute transition ta
    prev_scores = (cs_lookup[cs_lookup['period'] == prev_period]
                   [admin_cols + ['CS_score']]
                   .rename(columns={'CS_score': 'prev_CS'}))
    test = test.merge(prev_scores, on=admin_cols, how='left')

    # Only evaluate areas that have a previous score (otherwise transition i
    test = test[test['prev_CS'].notna()].copy()
    if len(test) == 0:
        continue

    y_test = (test['CS_score'] > test['prev_CS']).astype(int)
    n_worsened = y_test.sum()
    print(f"  {test_period}: {len(test):,} areas, {n_worsened} worsened ({10

    # No-Change: always predict 0
    no_change_pred = pd.Series(0, index=y_test.index)

    # Always-Worsen: always predict 1
    always_worsen_pred = pd.Series(1, index=y_test.index)
```

```
    # SPLY-Transition: did area worsen at same period last year?
    target_sply = str(int(test_period) - 100)
    sply_period = min(train_periods, key=lambda p: abs(int(p) - int(target_s
    # Need the period before sply_period to compute that year's transition
    sply_prev_idx = assessment_periods_bl.index(sply_period) - 1
    if sply_prev_idx >= 0:
        sply_prev_period = assessment_periods_bl[sply_prev_idx]
        sply_scores = (cs_lookup[cs_lookup['period'] == sply_period]
                       [admin_cols + ['CS_score']].rename(columns={'CS_score
        sply_prev_scores = (cs_lookup[cs_lookup['period'] == sply_prev_perio
                            [admin_cols + ['CS_score']].rename(columns={'CS_
        sply_merged = (test[admin_cols].merge(sply_scores, on=admin_cols, ho
                       .merge(sply_prev_scores, on=admin_cols, how='left'))
        sply_transition = (sply_merged['sply_CS'] > sply_merged['sply_prev_C
    else:
        sply_transition = pd.Series(0, index=y_test.index)

    preds = {'No-Change': no_change_pred, 'Always-Worsen': always_worsen_pre
             'SPLY-Transition': sply_transition}

    for name in baseline_names:
        period_results[name].append({
            'period': test_period,
            'n_test': len(test),
            'n_worsened': n_worsened,
            'accuracy': accuracy_score(y_test, preds[name]),
            'precision': precision_score(y_test, preds[name], average='binar
            'recall': recall_score(y_test, preds[name], average='binary', ze
            'f1': f1_score(y_test, preds[name], average='binary', zero_divis
        })

# ── Summary ──────────────────────────────────────────────────────────
total_samples = sum(period_results['No-Change'][i]['n_test'] for i in range(
total_worsened = sum(period_results['No-Change'][i]['n_worsened'] for i in r
print(f"\nOverall: {total_samples:,} test samples, {total_worsened:,} worser

rows = []
for name in baseline_names:
    res = period_results[name]
    rows.append({
        'Model': name,
        'Accuracy': np.mean([r['accuracy'] for r in res]),
        'Precision': np.mean([r['precision'] for r in res]),
        'Recall': np.mean([r['recall'] for r in res]),
        'F1 (worsen)': np.mean([r['f1'] for r in res]),
        'N Periods': len(res),
        'Total Test': sum(r['n_test'] for r in res),
    })

baseline_summary = pd.DataFrame(rows)
print(baseline_summary.to_string(index=False))
```

```
Evaluating transition baselines on 26 assessment periods

  201610: 2,633 areas, 106 worsened (4.0%)
  201702: 2,447 areas, 294 worsened (12.0%)
  201706: 2,271 areas, 427 worsened (18.8%)
  201710: 2,573 areas, 151 worsened (5.9%)
  201802: 2,441 areas, 138 worsened (5.7%)
  201806: 2,438 areas, 317 worsened (13.0%)
  201810: 1,078 areas, 48 worsened (4.5%)
  201902: 1,081 areas, 62 worsened (5.7%)
  201906: 2,400 areas, 432 worsened (18.0%)
  201910: 2,400 areas, 206 worsened (8.6%)
  202002: 1,476 areas, 81 worsened (5.5%)
  202006: 1,588 areas, 213 worsened (13.4%)
  202010: 2,491 areas, 299 worsened (12.0%)
  202102: 2,489 areas, 251 worsened (10.1%)
  202106: 2,489 areas, 560 worsened (22.5%)
  202110: 2,535 areas, 144 worsened (5.7%)
  202202: 2,535 areas, 467 worsened (18.4%)
  202206: 2,490 areas, 381 worsened (15.3%)
  202210: 2,399 areas, 300 worsened (12.5%)
  202302: 1,376 areas, 131 worsened (9.5%)
  202306: 1,376 areas, 419 worsened (30.5%)
  202310: 1,919 areas, 61 worsened (3.2%)
  202402: 1,919 areas, 333 worsened (17.4%)
  202406: 1,919 areas, 567 worsened (29.5%)

Overall: 50,763 test samples, 6,388 worsened (12.6%)

          Model  Accuracy  Precision   Recall  F1 (worsen)  N Periods  Total
Test
      No-Change  0.874340   0.000000 0.000000     0.000000         24
50763
  Always-Worsen  0.125660   0.125660 1.000000     0.215809         24
50763
SPLY-Transition  0.829052   0.219227 0.184893     0.184651         24
50763
```

ML from here

In [12]:
```python
n_training_periods = 20     # months of training data before each test asses
n_eval_periods = 26          # number of IPC assessment periods to evaluate

# Optional: set to None to auto-detect latest assessment, or specify e.g. 20
start_period = None
```

In [13]:
```python
# ============================================================================
# ROLLING WINDOW EVALUATION — TRANSITION PREDICTION
# ============================================================================
# Target: binary — did CS_score WORSEN vs the previous assessment?
#   worsened = 1  if CS_score(T) > CS_score(T-1)
#   worsened = 0  otherwise (stable or improved)
#
# Training: y_train = (y_next > IPC_last) — did next assessment worsen?
```

```python
# Testing:  y_test  = (CS_score > prev_CS_score) — did this assessment worse
# IPC_last leakage fix still applied (replace with previous assessment's sco
# ================================================================================

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.base import clone
from sklearn.metrics import accuracy_score, precision_score, recall_score, f
from catboost import CatBoostClassifier

all_periods = sorted(df['period'].unique())

# ── 1. Identify IPC assessment periods ──────────────────────────────────────
ipc_obs_per_period = df[df['CS_score'].between(1, 5)].groupby('period').size
assessment_periods = sorted(ipc_obs_per_period.index.tolist())

print("="*70)
print("IPC ASSESSMENT PERIODS")
print("="*70)
print(f"\nTotal periods in data: {len(all_periods)}  ({all_periods[0]} → {al
print(f"Assessment periods (with CS_score data): {len(assessment_periods)}\r
for p in assessment_periods:
    print(f"  {p}:  {ipc_obs_per_period[p]:>5,} scored")

# ── 2. Select evaluation assessment periods (backwards from latest) ─────────
if start_period is not None:
    candidates = [p for p in assessment_periods if p <= start_period]
else:
    candidates = list(assessment_periods)

eval_assessments = candidates[-n_eval_periods:]  # latest N

print(f"\n{'='*70}")
print("ROLLING WINDOW EVALUATION SETUP")
print(f"{'='*70}")
print(f"\nConfiguration:")
print(f"  - Training window:       {n_training_periods} periods before test
print(f"  - Assessments to test:   {len(eval_assessments)} (of {len(assessme
print(f"  - Assessment periods:    {len(assessment_periods)}")

# ── 3. Create evaluation splits ─────────────────────────────────────────────
evaluation_splits = []

for i, test_period in enumerate(reversed(eval_assessments)):
    test_idx = all_periods.index(test_period)
    train_end_idx = test_idx - 1  # everything BEFORE test period
    train_start_idx = max(0, train_end_idx - n_training_periods + 1)

    if train_end_idx < 0:
        print(f"\n  ⚠️  Assessment {test_period}: no training data available
        continue

    train_periods = all_periods[train_start_idx:train_end_idx + 1]

    evaluation_splits.append({
```

```
        'eval_id': i + 1,
        'test_period': test_period,
        'n_scored': int(ipc_obs_per_period.get(test_period, 0)),
        'train_periods': train_periods,
        'train_start': train_periods[0],
        'train_end': train_periods[-1],
        'n_train_periods': len(train_periods),
    })

print(f"\nEvaluation schedule (latest → earliest):\n")
for s in evaluation_splits:
    print(f"  Eval {s['eval_id']}: test {s['test_period']}  "
          f"({s['n_scored']:,} scored)  |  "
          f"train {s['train_start']} → {s['train_end']}  "
          f"({s['n_train_periods']} periods)")

print(f"\n{'='*70}")
print(f"Created {len(evaluation_splits)} evaluation splits")
print("="*70)

# ============================================================================
# DEFINE FEATURE SETS
# ============================================================================
if 'ipc_features' in globals():
    ipc_feat = [c for c in ipc_features if c in df.columns]
else:
    ipc_feat = ['IPC_last']

_temporal_suffixes = ('_rolling_', '_lag', '_escalation')

if 'gdelt_features' in globals():
    gdelt_base_feat = [c for c in gdelt_features if c in df.columns]
else:
    gdelt_base_feat = [c for c in df.columns
                       if c.startswith('evt_') and not any(s in c for s in _

if 'temporal_features' in globals():
    gdelt_temporal_feat = [c for c in temporal_features if c in df.columns]
else:
    gdelt_temporal_feat = [c for c in df.columns
                           if c.startswith('evt_') and any(s in c for s in _

if 'geographic_features' in globals():
    geographic_feat = [c for c in geographic_features if c in df.columns]
else:
    geographic_feat = [c for c in df.columns if c.startswith('geo_')]

feature_sets = {
    # With geographic dummies (location fixed effects)
    'IPC only':               ipc_feat + geographic_feat,
    'IPC + GDELT base':       ipc_feat + gdelt_base_feat + geographic_feat,
    'IPC + GDELT + temporal': ipc_feat + gdelt_base_feat + gdelt_temporal_fe
    'GDELT only':             gdelt_base_feat + gdelt_temporal_feat + geogra
    # Without geographic dummies (pure feature signal)
    'IPC only (no geo)':               ipc_feat,
    'IPC + GDELT base (no geo)':       ipc_feat + gdelt_base_feat,
```

```
    'IPC + GDELT + temporal (no geo)': ipc_feat + gdelt_base_feat + gdelt_te
    'GDELT only (no geo)':             gdelt_base_feat + gdelt_temporal_feat
}

print("\nFeature sets available:")
for name, features in feature_sets.items():
    n_ipc = len([c for c in features if c in ipc_feat])
    n_gdelt_base = len([c for c in features if c in gdelt_base_feat])
    n_gdelt_temp = len([c for c in features if c in gdelt_temporal_feat])
    n_geo = len([c for c in features if c in geographic_feat])
    print(f"  - {name}: {n_ipc} IPC + {n_gdelt_base} GDELT base + {n_gdelt_t

# ============================================================================
# DEFINE MODELS
# ============================================================================
models = {
    'Logistic Regression': {
        'model': LogisticRegression(max_iter=1000, random_state=42, solver='
        'needs_scaling': True,
    },
    'Random Forest': {
        'model': RandomForestClassifier(
            n_estimators=150, max_depth=12, min_samples_split=15,
            min_samples_leaf=8, max_features='sqrt',
            class_weight='balanced', random_state=42, n_jobs=-1
        ),
        'needs_scaling': False,
    },
    'CatBoost': {
        'model': CatBoostClassifier(
            iterations=300, depth=4, learning_rate=0.05,
            loss_function='MultiClass', auto_class_weights='Balanced',
            l2_leaf_reg=5, colsample_bylevel=0.7,
            random_seed=42, verbose=0
        ),
        'needs_scaling': False,
    },
}

print(f"\nModels to evaluate: {', '.join(models.keys())}")

# ============================================================================
# TRAIN AND EVALUATE
# ============================================================================
print("\n" + "="*70)
print("TRAINING AND EVALUATION")
print("="*70)

all_results = {}  # keyed by (model_name, feature_set_name)
rf_importances = {}  # keyed by feature_set_name → list of (feature_cols, im

for model_name, model_cfg in models.items():
    for feature_set_name, feature_cols in feature_sets.items():
        result_key = (model_name, feature_set_name)
        print(f"\n{'='*70}")
        print(f"MODEL: {model_name}  |  FEATURE SET: {feature_set_name}")
```

```
        print(f"{'='*70}")
        print(f"Using {len(feature_cols)} features")

        feature_results = []

        for split in evaluation_splits:
            print(f"\n  ── Eval {split['eval_id']}: Assessment period {split

            # ── Training set: all prior periods with valid y_next ────────
            #    (includes inter-assessment rows with GDELT features)
            train_df = df[
                (df['period'].isin(split['train_periods'])) &
                (df['y_next'].notna())
            ].copy()

            # ── Test set: rows from assessment period with valid CS_score -
            test_df = df[
                (df['period'] == split['test_period']) &
                (df['CS_score'].between(1, 5))
            ].copy()

            print(f"  Training samples:   {len(train_df):,}")
            print(f"  Test samples:       {len(test_df):,}  (admin2 areas ir

            if len(train_df) == 0:
                print(f"  ⚠️  Skipping — no training data")
                continue
            if len(test_df) == 0:
                print(f"  ⚠️  Skipping — no test data")
                continue

            # ── Compute transition targets ──────────────────────────────
            # Training: worsened = (y_next > IPC_last)
            y_train = (train_df['y_next'] > train_df['IPC_last']).astype(int

            # Testing: worsened = (CS_score > prev_assessment's CS_score)
            train_assessments = sorted(
                p for p in assessment_periods if p in split['train_periods']
            )
            if not train_assessments:
                print(f"  ⚠️  Skipping — no previous assessment to compute t
                continue

            prev_period = train_assessments[-1]
            prev_scores = (
                df.loc[(df['period'] == prev_period) & df['CS_score'].betwee
                       ['ADMIN0', 'ADMIN1', 'ADMIN2', 'CS_score']]
                .drop_duplicates(['ADMIN0', 'ADMIN1', 'ADMIN2'])
                .rename(columns={'CS_score': '_prev_CS'})
            )
            test_df = test_df.merge(prev_scores, on=['ADMIN0', 'ADMIN1', 'AD
            test_df = test_df[test_df['_prev_CS'].notna()].copy()

            if len(test_df) == 0:
                print(f"  ⚠️  Skipping — no areas with previous assessment s
                continue
```

```
            y_test = (test_df['CS_score'] > test_df['_prev_CS']).astype(int)
            n_worsened = y_test.sum()
            print(f"  Transition target: {n_worsened} worsened / {len(test_d

            # ── Prepare features ──────────────────────────────────────
            X_train = train_df[feature_cols].fillna(0)
            X_test = test_df[feature_cols].fillna(0)

            # ── Fix IPC_last leakage: replace with previous assessment's sc
            if 'IPC_last' in feature_cols:
                X_test['IPC_last'] = test_df['_prev_CS'].fillna(0).values

            # ── Scale (only for models that need it) ──────────────────
            if model_cfg['needs_scaling']:
                scaler = StandardScaler()
                X_train_fit = scaler.fit_transform(X_train)
                X_test_fit = scaler.transform(X_test)
            else:
                X_train_fit = X_train
                X_test_fit = X_test

            # ── Train ─────────────────────────────────────────────────
            print(f"  Training {model_name}...")
            model = clone(model_cfg['model'])
            model.fit(X_train_fit, y_train)

            y_pred = model.predict(X_test_fit)

            # ── Capture Random Forest feature importances ─────────────
            if model_name == 'Random Forest':
                if feature_set_name not in rf_importances:
                    rf_importances[feature_set_name] = []
                rf_importances[feature_set_name].append(
                    pd.Series(model.feature_importances_, index=feature_cols
                )

            # ── Metrics (binary: positive class = worsened) ───────────
            accuracy  = accuracy_score(y_test, y_pred)
            precision = precision_score(y_test, y_pred, average='binary', ze
            recall    = recall_score(y_test, y_pred, average='binary', zero_
            f1        = f1_score(y_test, y_pred, average='binary', zero_divi
            cm        = confusion_matrix(y_test, y_pred)

            feature_results.append({
                'eval_id': split['eval_id'],
                'test_period': split['test_period'],
                'n_train': len(train_df),
                'n_test': len(test_df),
                'accuracy': accuracy,
                'precision': precision,
                'recall': recall,
                'f1': f1,
                'confusion_matrix': cm
            })
```

```
            print(f"\n  Results  (n_test={len(test_df):,}):")
            print(f"    Accuracy:  {accuracy:.4f}")
            print(f"    Precision: {precision:.4f}")
            print(f"    Recall:    {recall:.4f}")
            print(f"    F1 Score:  {f1:.4f}")
            print(f"\n  Confusion Matrix:\n    {cm}")

        all_results[result_key] = feature_results

# ============================================================================
# SUMMARY
# ============================================================================
if any(len(v) > 0 for v in all_results.values()):
    print("\n" + "="*70)
    print("SUMMARY: TRANSITION PREDICTION — AVERAGE RESULTS BY MODEL & FEATU
    print("="*70)
    print("(Precision/Recall/F1 are for the WORSENED class)")

    summary_data = []
    for (model_name, feature_set_name), results in all_results.items():
        if len(results) == 0:
            continue

        avg_acc  = np.mean([r['accuracy']  for r in results])
        avg_prec = np.mean([r['precision'] for r in results])
        avg_rec  = np.mean([r['recall']    for r in results])
        avg_f1   = np.mean([r['f1']        for r in results])
        total_test = sum(r['n_test'] for r in results)

        print(f"\n{model_name} — {feature_set_name}:")
        print("-" * 70)
        print(f"  Average Metrics (across {len(results)} assessments, {total
        print(f"    Accuracy:       {avg_acc:.4f}")
        print(f"    Precision (W):  {avg_prec:.4f}")
        print(f"    Recall (W):     {avg_rec:.4f}")
        print(f"    F1 (worsen):    {avg_f1:.4f}")

        print(f"\n  Per assessment:")
        for r in results:
            print(f"    {r['test_period']}:  n={r['n_test']:>5,}  acc={r['ac

        n_features = len(feature_sets[feature_set_name])
        n_ipc = len([c for c in feature_sets[feature_set_name] if c in ipc_f
        n_gdelt_base = len([c for c in feature_sets[feature_set_name] if c i
        n_gdelt_temp = len([c for c in feature_sets[feature_set_name] if c i
        n_geo = len([c for c in feature_sets[feature_set_name] if c in geogr

        summary_data.append({
            'Model': model_name,
            'Feature Set': feature_set_name,
            'N Features': n_features,
            'IPC': n_ipc,
            'GDELT Base': n_gdelt_base,
            'GDELT Temporal': n_gdelt_temp,
            'Geographic': n_geo,
            'Accuracy': avg_acc,
```

```
            'Prec (W)': avg_prec,
            'Recall (W)': avg_rec,
            'F1 (worsen)': avg_f1,
            'Total Test': total_test,
            'N Evals': len(results),
        })

    summary_df = pd.DataFrame(summary_data)
    summary_df = summary_df.sort_values('F1 (worsen)', ascending=False)

    print(f"\n{'='*70}")
    print("SUMMARY TABLE")
    print("="*70)
    print(summary_df.to_string(index=False))

else:
    print("\n⚠️  No evaluations completed — check data availability")

# ============================================================================
# RANDOM FOREST — AVERAGE FEATURE IMPORTANCES
# ============================================================================
if rf_importances:
    print("\n" + "="*70)
    print("RANDOM FOREST: AVERAGE FEATURE IMPORTANCES")
    print("="*70)

    for fs_name, imp_list in rf_importances.items():
        avg_imp = pd.concat(imp_list, axis=1).mean(axis=1).sort_values(ascer
        print(f"\n── {fs_name} (averaged over {len(imp_list)} splits) ──")
        print(f"{'Feature':<55} {'Importance':>10}")
        print("-" * 67)
        for feat, imp in avg_imp.head(25).items():
            print(f"  {feat:<53} {imp:>10.4f}")
        if len(avg_imp) > 25:
            print(f"  ... ({len(avg_imp) - 25} more features omitted)")
```

```
================================================================================
IPC ASSESSMENT PERIODS
================================================================================

Total periods in data: 101  (201602 → 202406)
Assessment periods (with CS_score data): 26

  201602:  2,639 scored
  201606:  2,635 scored
  201610:  2,748 scored
  201702:  2,449 scored
  201706:  2,573 scored
  201710:  2,577 scored
  201802:  2,441 scored
  201806:  2,440 scored
  201810:  1,081 scored
  201902:  2,400 scored
  201906:  2,400 scored
  201910:  2,400 scored
  202002:  1,588 scored
  202006:  2,570 scored
  202010:  2,491 scored
  202102:  2,489 scored
  202106:  2,536 scored
  202110:  2,535 scored
  202202:  2,535 scored
  202206:  2,568 scored
  202210:  2,399 scored
  202302:  1,376 scored
  202306:  2,613 scored
  202310:  1,919 scored
  202402:  2,376 scored
  202406:  1,919 scored

================================================================================
ROLLING WINDOW EVALUATION SETUP
================================================================================

Configuration:
  - Training window:       20 periods before test assessment
  - Assessments to test:   26 (of 26 total)
  - Assessment periods:    26

  ⚠️  Assessment 201602: no training data available — skipping

Evaluation schedule (latest → earliest):

  Eval 1: test 202406  (1,919 scored)  |  train 202210 → 202405  (20 period
s)
  Eval 2: test 202402  (2,376 scored)  |  train 202206 → 202401  (20 period
s)
  Eval 3: test 202310  (1,919 scored)  |  train 202202 → 202309  (20 period
s)
  Eval 4: test 202306  (2,613 scored)  |  train 202110 → 202305  (20 period
s)
  Eval 5: test 202302  (1,376 scored)  |  train 202106 → 202301  (20 period
```

```
s)
  Eval 6: test 202210  (2,399 scored)  |  train 202102 → 202209  (20 period
s)
  Eval 7: test 202206  (2,568 scored)  |  train 202010 → 202205  (20 period
s)
  Eval 8: test 202202  (2,535 scored)  |  train 202006 → 202201  (20 period
s)
  Eval 9: test 202110  (2,535 scored)  |  train 202002 → 202109  (20 period
s)
  Eval 10: test 202106  (2,536 scored)  |  train 201910 → 202105  (20 period
s)
  Eval 11: test 202102  (2,489 scored)  |  train 201906 → 202101  (20 period
s)
  Eval 12: test 202010  (2,491 scored)  |  train 201902 → 202009  (20 period
s)
  Eval 13: test 202006  (2,570 scored)  |  train 201810 → 202005  (20 period
s)
  Eval 14: test 202002  (1,588 scored)  |  train 201806 → 202001  (20 period
s)
  Eval 15: test 201910  (2,400 scored)  |  train 201802 → 201909  (20 period
s)
  Eval 16: test 201906  (2,400 scored)  |  train 201710 → 201905  (20 period
s)
  Eval 17: test 201902  (2,400 scored)  |  train 201706 → 201901  (20 period
s)
  Eval 18: test 201810  (1,081 scored)  |  train 201702 → 201809  (20 period
s)
  Eval 19: test 201806  (2,440 scored)  |  train 201610 → 201805  (20 period
s)
  Eval 20: test 201802  (2,441 scored)  |  train 201606 → 201801  (20 period
s)
  Eval 21: test 201710  (2,577 scored)  |  train 201602 → 201709  (20 period
s)
  Eval 22: test 201706  (2,573 scored)  |  train 201602 → 201705  (16 period
s)
  Eval 23: test 201702  (2,449 scored)  |  train 201602 → 201701  (12 period
s)
  Eval 24: test 201610  (2,748 scored)  |  train 201602 → 201609  (8 period
s)
  Eval 25: test 201606  (2,635 scored)  |  train 201602 → 201605  (4 period
s)

======================================================================
Created 25 evaluation splits
======================================================================

Feature sets available:
  - IPC only: 1 IPC + 0 GDELT base + 0 GDELT temporal + 3331 geographic = 33
32 total
  - IPC + GDELT base: 1 IPC + 119 GDELT base + 0 GDELT temporal + 3331 geogr
aphic = 3451 total
  - IPC + GDELT + temporal: 1 IPC + 119 GDELT base + 35 GDELT temporal + 333
1 geographic = 3486 total
  - GDELT only: 0 IPC + 119 GDELT base + 35 GDELT temporal + 3331 geographic
= 3485 total
  - IPC only (no geo): 1 IPC + 0 GDELT base + 0 GDELT temporal + 0 geographi
```

```
c = 1 total
  - IPC + GDELT base (no geo): 1 IPC + 119 GDELT base + 0 GDELT temporal + 0
geographic = 120 total
  - IPC + GDELT + temporal (no geo): 1 IPC + 119 GDELT base + 35 GDELT tempo
ral + 0 geographic = 155 total
  - GDELT only (no geo): 0 IPC + 119 GDELT base + 35 GDELT temporal + 0 geog
raphic = 154 total

Models to evaluate: Logistic Regression, Random Forest, CatBoost

======================================================================
TRAINING AND EVALUATION
======================================================================

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC only
======================================================================
Using 3332 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.9651
    Precision: 0.9125
    Recall:    0.9753
    F1 Score:  0.9429

  Confusion Matrix:
    [[1299   53]
 [  14  553]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7962
    Precision: 0.4598
    Recall:    0.9970
    F1 Score:  0.6294

  Confusion Matrix:
    [[1196  390]
 [   1  332]]

  —— Eval 3: Assessment period 202310 ——
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training Logistic Regression...
```

```
 Results  (n_test=1,919):
   Accuracy:  0.9291
   Precision: 0.3096
   Recall:    1.0000
   F1 Score:  0.4729

 Confusion Matrix:
   [[1722  136]
[   0   61]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.9033
   Precision: 0.7609
   Recall:    0.9952
   F1 Score:  0.8625

 Confusion Matrix:
   [[826 131]
[  2 417]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8227
   Precision: 0.3493
   Recall:    1.0000
   F1 Score:  0.5178

 Confusion Matrix:
   [[1001  244]
[   0  131]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.9400
   Precision: 0.6814
   Recall:    0.9767
   F1 Score:  0.8027

 Confusion Matrix:
```

```
   [[1962  137]
[   7  293]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Logistic Regression...

 Results  (n_test=2,490):
   Accuracy:  0.9382
   Precision: 0.7137
   Recall:    0.9948
   F1 Score:  0.8311

 Confusion Matrix:
   [[1957  152]
[   2  379]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8864
   Precision: 0.6198
   Recall:    0.9914
   F1 Score:  0.7628

 Confusion Matrix:
   [[1784  284]
[   4  463]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.9105
   Precision: 0.3881
   Recall:    1.0000
   F1 Score:  0.5592

 Confusion Matrix:
   [[2164  227]
[   0  144]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...
```

```
 Results  (n_test=2,489):
   Accuracy:  0.9289
   Precision: 0.7613
   Recall:    0.9964
   F1 Score:  0.8631

 Confusion Matrix:
   [[1754  175]
[   2  558]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8935
   Precision: 0.4863
   Recall:    0.9920
   F1 Score:  0.6527

 Confusion Matrix:
   [[1975  263]
[   2  249]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.9426
   Precision: 0.6773
   Recall:    0.9967
   F1 Score:  0.8065

 Confusion Matrix:
   [[2050  142]
[   1  298]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8715
   Precision: 0.5108
   Recall:    0.9953
   F1 Score:  0.6752

 Confusion Matrix:
```

```
   [[1172  203]
[   1  212]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.9112
   Precision: 0.3810
   Recall:    0.9877
   F1 Score:  0.5498

 Confusion Matrix:
   [[1265  130]
[   1   80]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.9513
   Precision: 0.6386
   Recall:    0.9951
   F1 Score:  0.7780

 Confusion Matrix:
   [[2078  116]
[   1  205]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.9454
   Precision: 0.7702
   Recall:    0.9931
   F1 Score:  0.8675

 Confusion Matrix:
   [[1840  128]
[   3  429]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...
```

```
 Results  (n_test=1,081):
   Accuracy:  0.9288
   Precision: 0.4460
   Recall:    1.0000
   F1 Score:  0.6169

 Confusion Matrix:
   [[942  77]
[  0  62]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.8432
   Precision: 0.2212
   Recall:    1.0000
   F1 Score:  0.3623

 Confusion Matrix:
   [[861 169]
[  0  48]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.8733
   Precision: 0.5064
   Recall:    1.0000
   F1 Score:  0.6723

 Confusion Matrix:
   [[1812  309]
[   0  317]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8206
   Precision: 0.2396
   Recall:    1.0000
   F1 Score:  0.3866

 Confusion Matrix:
```

```
   [[1865  438]
[   0  138]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.9250
   Precision: 0.4386
   Recall:    0.9934
   F1 Score:  0.6085

 Confusion Matrix:
   [[2230  192]
[   1  150]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.9665
   Precision: 0.8489
   Recall:    1.0000
   F1 Score:  0.9183

 Confusion Matrix:
   [[1768   76]
[   0  427]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.9452
   Precision: 0.6869
   Recall:    1.0000
   F1 Score:  0.8144

 Confusion Matrix:
   [[2019  134]
[   0  294]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Logistic Regression...
```

```
  Results  (n_test=2,633):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[2527    0]
 [   0  106]]

  —— Eval 25: Assessment period 201606 ——
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[2326    0]
 [   0  309]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT base
======================================================================
Using 3451 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.9635
    Precision: 0.9135
    Recall:    0.9683
    F1 Score:  0.9401

  Confusion Matrix:
    [[1300   52]
 [  18  549]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7983
```

```
  Precision: 0.4623
  Recall:    0.9940
  F1 Score:  0.6311

 Confusion Matrix:
   [[1201  385]
[   2  331]]

 —— Eval 3: Assessment period 202310 ——
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.9291
   Precision: 0.3077
   Recall:    0.9836
   F1 Score:  0.4688

 Confusion Matrix:
   [[1723  135]
[   1   60]]

 —— Eval 4: Assessment period 202306 ——
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.9019
   Precision: 0.7620
   Recall:    0.9857
   F1 Score:  0.8595

 Confusion Matrix:
   [[828 129]
[  6 413]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8241
   Precision: 0.3512
   Recall:    1.0000
   F1 Score:  0.5198

 Confusion Matrix:
   [[1003  242]
[   0  131]]
```

```
── Eval 6: Assessment period 202210 ──
Training samples:   13,866
Test samples:       2,399  (admin2 areas in 202210)
Transition target: 300 worsened / 2399 (12.5%)
Training Logistic Regression...

Results  (n_test=2,399):
  Accuracy:  0.9387
  Precision: 0.6792
  Recall:    0.9667
  F1 Score:  0.7978

Confusion Matrix:
  [[1962  137]
[  10  290]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training Logistic Regression...

Results  (n_test=2,490):
  Accuracy:  0.9357
  Precision: 0.7105
  Recall:    0.9790
  F1 Score:  0.8234

Confusion Matrix:
  [[1957  152]
[   8  373]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training Logistic Regression...

Results  (n_test=2,535):
  Accuracy:  0.8856
  Precision: 0.6194
  Recall:    0.9829
  F1 Score:  0.7599

Confusion Matrix:
  [[1786  282]
[   8  459]]

── Eval 9: Assessment period 202110 ──
Training samples:   12,950
Test samples:       2,535  (admin2 areas in 202110)
Transition target: 144 worsened / 2535 (5.7%)
Training Logistic Regression...

Results  (n_test=2,535):
  Accuracy:  0.9097
```

```
  Precision: 0.3836
  Recall:    0.9722
  F1 Score:  0.5501

 Confusion Matrix:
   [[2166  225]
[   4  140]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.9269
   Precision: 0.7632
   Recall:    0.9786
   F1 Score:  0.8576

 Confusion Matrix:
   [[1759  170]
[  12  548]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8955
   Precision: 0.4911
   Recall:    0.9880
   F1 Score:  0.6561

 Confusion Matrix:
   [[1981  257]
[   3  248]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.9410
   Precision: 0.6759
   Recall:    0.9766
   F1 Score:  0.7989

 Confusion Matrix:
   [[2052  140]
[   7  292]]
```

```
── Eval 13: Assessment period 202006 ──
Training samples:   11,506
Test samples:       2,570  (admin2 areas in 202006)
Transition target: 213 worsened / 1588 (13.4%)
Training Logistic Regression...

Results  (n_test=1,588):
  Accuracy:  0.8703
  Precision: 0.5086
  Recall:    0.9718
  F1 Score:  0.6677

Confusion Matrix:
  [[1175  200]
[   6  207]]

── Eval 14: Assessment period 202002 ──
Training samples:   12,388
Test samples:       1,588  (admin2 areas in 202002)
Transition target: 81 worsened / 1476 (5.5%)
Training Logistic Regression...

Results  (n_test=1,476):
  Accuracy:  0.9133
  Precision: 0.3842
  Recall:    0.9630
  F1 Score:  0.5493

Confusion Matrix:
  [[1270  125]
[   3   78]]

── Eval 15: Assessment period 201910 ──
Training samples:   12,533
Test samples:       2,400  (admin2 areas in 201910)
Transition target: 206 worsened / 2400 (8.6%)
Training Logistic Regression...

Results  (n_test=2,400):
  Accuracy:  0.9513
  Precision: 0.6422
  Recall:    0.9757
  F1 Score:  0.7746

Confusion Matrix:
  [[2082  112]
[   5  201]]

── Eval 16: Assessment period 201906 ──
Training samples:   12,605
Test samples:       2,400  (admin2 areas in 201906)
Transition target: 432 worsened / 2400 (18.0%)
Training Logistic Regression...

Results  (n_test=2,400):
  Accuracy:  0.9425
```

```
  Precision: 0.7712
  Recall:    0.9676
  F1 Score:  0.8583

 Confusion Matrix:
   [[1844  124]
[  14  418]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.9251
   Precision: 0.4307
   Recall:    0.9516
   F1 Score:  0.5930

 Confusion Matrix:
   [[941  78]
[  3  59]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.8414
   Precision: 0.2057
   Recall:    0.8958
   F1 Score:  0.3346

 Confusion Matrix:
   [[864 166]
[  5  43]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.8749
   Precision: 0.5097
   Recall:    0.9905
   F1 Score:  0.6731

 Confusion Matrix:
   [[1819  302]
[   3  314]]
```

```
 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8230
   Precision: 0.2403
   Recall:    0.9855
   F1 Score:  0.3864

 Confusion Matrix:
   [[1873  430]
[   2  136]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.9250
   Precision: 0.4375
   Recall:    0.9735
   F1 Score:  0.6037

 Confusion Matrix:
   [[2233  189]
[   4  147]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.9608
   Precision: 0.8435
   Recall:    0.9719
   F1 Score:  0.9032

 Confusion Matrix:
   [[1767   77]
[  12  415]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.9403
```

```
    Precision: 0.6779
    Recall:    0.9592
    F1 Score:  0.7944

  Confusion Matrix:
    [[2019  134]
 [  12  282]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Logistic Regression...

  Results  (n_test=2,633):
    Accuracy:  0.9909
    Precision: 0.8796
    Recall:    0.8962
    F1 Score:  0.8879

  Confusion Matrix:
    [[2514   13]
 [  11   95]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[2326    0]
 [   0  309]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.9640
    Precision: 0.9150
    Recall:    0.9683
    F1 Score:  0.9409
```

```
 Confusion Matrix:
   [[1301   51]
[  18  549]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.8072
   Precision: 0.4732
   Recall:    0.9820
   F1 Score:  0.6387

 Confusion Matrix:
   [[1222  364]
[   6  327]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.9302
   Precision: 0.3048
   Recall:    0.9344
   F1 Score:  0.4597

 Confusion Matrix:
   [[1728  130]
[   4   57]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.9033
   Precision: 0.7719
   Recall:    0.9690
   F1 Score:  0.8593

 Confusion Matrix:
   [[837 120]
[ 13 406]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
```

```
Transition target: 131 worsened / 1376 (9.5%)
Training Logistic Regression...

Results  (n_test=1,376):
  Accuracy:  0.8299
  Precision: 0.3581
  Recall:    0.9924
  F1 Score:  0.5263

Confusion Matrix:
  [[1012  233]
[   1  130]]

── Eval 6: Assessment period 202210 ──
Training samples:   13,866
Test samples:       2,399  (admin2 areas in 202210)
Transition target: 300 worsened / 2399 (12.5%)
Training Logistic Regression...

Results  (n_test=2,399):
  Accuracy:  0.9350
  Precision: 0.6714
  Recall:    0.9400
  F1 Score:  0.7833

Confusion Matrix:
  [[1961  138]
[  18  282]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training Logistic Regression...

Results  (n_test=2,490):
  Accuracy:  0.9337
  Precision: 0.7085
  Recall:    0.9633
  F1 Score:  0.8165

Confusion Matrix:
  [[1958  151]
[  14  367]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training Logistic Regression...

Results  (n_test=2,535):
  Accuracy:  0.8868
  Precision: 0.6247
  Recall:    0.9657
  F1 Score:  0.7586
```

```
 Confusion Matrix:
   [[1797  271]
[  16  451]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.9128
   Precision: 0.3928
   Recall:    0.9792
   F1 Score:  0.5606

 Confusion Matrix:
   [[2173  218]
[   3  141]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.9257
   Precision: 0.7645
   Recall:    0.9679
   F1 Score:  0.8542

 Confusion Matrix:
   [[1762  167]
[  18  542]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8980
   Precision: 0.4969
   Recall:    0.9721
   F1 Score:  0.6577

 Confusion Matrix:
   [[1991  247]
[   7  244]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
```

```
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.9418
   Precision: 0.6825
   Recall:    0.9632
   F1 Score:  0.7989

 Confusion Matrix:
   [[2058  134]
[  11  288]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8722
   Precision: 0.5124
   Recall:    0.9671
   F1 Score:  0.6699

 Confusion Matrix:
   [[1179  196]
[   7  206]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.9119
   Precision: 0.3805
   Recall:    0.9630
   F1 Score:  0.5455

 Confusion Matrix:
   [[1268  127]
[   3   78]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.9500
   Precision: 0.6414
   Recall:    0.9466
   F1 Score:  0.7647
```

```
 Confusion Matrix:
   [[2085  109]
[  11  195]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.9425
   Precision: 0.7722
   Recall:    0.9653
   F1 Score:  0.8580

 Confusion Matrix:
   [[1845  123]
[  15  417]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.9260
   Precision: 0.4338
   Recall:    0.9516
   F1 Score:  0.5960

 Confusion Matrix:
   [[942  77]
[  3  59]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.8451
   Precision: 0.2069
   Recall:    0.8750
   F1 Score:  0.3347

 Confusion Matrix:
   [[869 161]
[  6  42]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
```

```
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.8774
   Precision: 0.5148
   Recall:    0.9874
   F1 Score:  0.6768

 Confusion Matrix:
   [[1826  295]
[   4  313]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8296
   Precision: 0.2454
   Recall:    0.9710
   F1 Score:  0.3918

 Confusion Matrix:
   [[1891  412]
[   4  134]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.9265
   Precision: 0.4431
   Recall:    0.9801
   F1 Score:  0.6103

 Confusion Matrix:
   [[2236  186]
[   3  148]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.9595
   Precision: 0.8317
   Recall:    0.9836
   F1 Score:  0.9013
```

```
  Confusion Matrix:
    [[1759   85]
 [   7  420]]

  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Transition target: 294 worsened / 2447 (12.0%)
  Training Logistic Regression...

  Results  (n_test=2,447):
    Accuracy:  0.9322
    Precision: 0.6531
    Recall:    0.9286
    F1 Score:  0.7669

  Confusion Matrix:
    [[2008  145]
 [  21  273]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Logistic Regression...

  Results  (n_test=2,633):
    Accuracy:  0.9848
    Precision: 0.7845
    Recall:    0.8585
    F1 Score:  0.8198

  Confusion Matrix:
    [[2502   25]
 [  15   91]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[2326    0]
 [   0  309]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: GDELT only
======================================================================
```

```
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7181
    Precision: 0.5126
    Recall:    0.9330
    F1 Score:  0.6617

  Confusion Matrix:
    [[849 503]
 [ 38 529]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6790
    Precision: 0.3477
    Recall:    0.9700
    F1 Score:  0.5119

  Confusion Matrix:
    [[980 606]
 [ 10 323]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.5248
    Precision: 0.0609
    Recall:    0.9672
    F1 Score:  0.1146

  Confusion Matrix:
    [[948 910]
 [  2  59]]

  ── Eval 4: Assessment period 202306 ──
  Training samples:   12,765
  Test samples:       2,613  (admin2 areas in 202306)
  Transition target: 419 worsened / 1376 (30.5%)
  Training Logistic Regression...
```

```
 Results  (n_test=1,376):
   Accuracy:  0.6846
   Precision: 0.4909
   Recall:    0.9690
   F1 Score:  0.6517

 Confusion Matrix:
   [[536 421]
[ 13 406]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.5342
   Precision: 0.1680
   Recall:    0.9847
   F1 Score:  0.2870

 Confusion Matrix:
   [[606 639]
[  2 129]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.6398
   Precision: 0.2500
   Recall:    0.9400
   F1 Score:  0.3950

 Confusion Matrix:
   [[1253  846]
[  18  282]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Logistic Regression...

 Results  (n_test=2,490):
   Accuracy:  0.6534
   Precision: 0.3008
   Recall:    0.9554
   F1 Score:  0.4576

 Confusion Matrix:
   [[1263  846]
```

```
[  17  364]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.6903
   Precision: 0.3707
   Recall:    0.9764
   F1 Score:  0.5374

 Confusion Matrix:
   [[1294  774]
[  11  456]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.5917
   Precision: 0.1169
   Recall:    0.9444
   F1 Score:  0.2081

 Confusion Matrix:
   [[1364 1027]
[   8  136]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.7260
   Precision: 0.4493
   Recall:    0.9643
   F1 Score:  0.6129

 Confusion Matrix:
   [[1267  662]
[  20  540]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...
```

```
 Results  (n_test=2,489):
   Accuracy:  0.6754
   Precision: 0.2320
   Recall:    0.9602
   F1 Score:  0.3736

 Confusion Matrix:
   [[1440  798]
[  10  241]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.6837
   Precision: 0.2691
   Recall:    0.9532
   F1 Score:  0.4197

 Confusion Matrix:
   [[1418  774]
[  14  285]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.7582
   Precision: 0.3508
   Recall:    0.9437
   F1 Score:  0.5115

 Confusion Matrix:
   [[1003  372]
[  12  201]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.6436
   Precision: 0.1298
   Recall:    0.9630
   F1 Score:  0.2287

 Confusion Matrix:
   [[872 523]
```

```
[  3  78]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.7013
   Precision: 0.2208
   Recall:    0.9806
   F1 Score:  0.3604

 Confusion Matrix:
   [[1481  713]
[   4  202]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8087
   Precision: 0.4842
   Recall:    0.9560
   F1 Score:  0.6428

 Confusion Matrix:
   [[1528  440]
[  19  413]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.7706
   Precision: 0.1837
   Recall:    0.8710
   F1 Score:  0.3034

 Confusion Matrix:
   [[779 240]
[  8  54]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...
```

```
 Results  (n_test=1,078):
   Accuracy:  0.6178
   Precision: 0.0991
   Recall:    0.9375
   F1 Score:  0.1793

 Confusion Matrix:
   [[621 409]
[  3  45]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.7404
   Precision: 0.3323
   Recall:    0.9874
   F1 Score:  0.4972

 Confusion Matrix:
   [[1492  629]
[   4  313]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.6936
   Precision: 0.1510
   Recall:    0.9565
   F1 Score:  0.2609

 Confusion Matrix:
   [[1561  742]
[   6  132]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.6794
   Precision: 0.1497
   Recall:    0.9536
   F1 Score:  0.2588

 Confusion Matrix:
   [[1604  818]
```

```
[   7  144]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.8027
   Precision: 0.4877
   Recall:    0.9742
   F1 Score:  0.6500

 Confusion Matrix:
   [[1407  437]
[  11  416]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.8672
   Precision: 0.4729
   Recall:    0.9184
   F1 Score:  0.6243

 Confusion Matrix:
   [[1852  301]
[  24  270]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Logistic Regression...

 Results  (n_test=2,633):
   Accuracy:  0.8861
   Precision: 0.2392
   Recall:    0.8396
   F1 Score:  0.3724

 Confusion Matrix:
   [[2244  283]
[  17   89]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Transition target: 309 worsened / 2635 (11.7%)
 Training Logistic Regression...
```

```
  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[2326    0]
 [   0  309]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6681
    Precision: 0.4556
    Recall:    0.6332
    F1 Score:  0.5299

  Confusion Matrix:
    [[923 429]
 [208 359]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.5638
    Precision: 0.2402
    Recall:    0.6997
    F1 Score:  0.3576

  Confusion Matrix:
    [[849 737]
 [100 233]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7134
    Precision: 0.0080
```

```
  Recall:    0.0656
  F1 Score:  0.0143

 Confusion Matrix:
   [[1365  493]
[  57    4]]

 —— Eval 4: Assessment period 202306 ——
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.7682
   Precision: 0.5862
   Recall:    0.8115
   F1 Score:  0.6807

 Confusion Matrix:
   [[717 240]
[ 79 340]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.6926
   Precision: 0.1958
   Recall:    0.7176
   F1 Score:  0.3077

 Confusion Matrix:
   [[859 386]
[ 37  94]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.6807
   Precision: 0.2109
   Recall:    0.5667
   F1 Score:  0.3074

 Confusion Matrix:
   [[1463  636]
[ 130  170]]

 —— Eval 7: Assessment period 202206 ——
```

```
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Logistic Regression...

 Results  (n_test=2,490):
   Accuracy:  0.6506
   Precision: 0.2237
   Recall:    0.5197
   F1 Score:  0.3128

 Confusion Matrix:
   [[1422  687]
[ 183  198]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.5893
   Precision: 0.2433
   Recall:    0.5824
   F1 Score:  0.3432

 Confusion Matrix:
   [[1222  846]
[ 195  272]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.6517
   Precision: 0.0796
   Recall:    0.4861
   F1 Score:  0.1369

 Confusion Matrix:
   [[1582  809]
[  74   70]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.6472
   Precision: 0.3709
```

```
  Recall:    0.8161
  F1 Score:  0.5100

 Confusion Matrix:
   [[1154  775]
[ 103  457]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.5464
   Precision: 0.1136
   Recall:    0.5139
   F1 Score:  0.1860

 Confusion Matrix:
   [[1231 1007]
[ 122  129]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.5813
   Precision: 0.1702
   Recall:    0.6421
   F1 Score:  0.2691

 Confusion Matrix:
   [[1256  936]
[ 107  192]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.5649
   Precision: 0.1422
   Recall:    0.4460
   F1 Score:  0.2157

 Confusion Matrix:
   [[802 573]
[118  95]]

 —— Eval 14: Assessment period 202002 ——
```

```
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.6640
   Precision: 0.0842
   Recall:    0.5185
   F1 Score:  0.1448

 Confusion Matrix:
   [[938 457]
[ 39  42]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.5004
   Precision: 0.0789
   Recall:    0.4515
   F1 Score:  0.1343

 Confusion Matrix:
   [[1108 1086]
[ 113   93]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.4500
   Precision: 0.1891
   Recall:    0.6250
   F1 Score:  0.2903

 Confusion Matrix:
   [[ 810 1158]
[ 162  270]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.3858
   Precision: 0.0326
```

```
  Recall:    0.3387
  F1 Score:  0.0595

 Confusion Matrix:
   [[396 623]
[ 41  21]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.4425
   Precision: 0.0445
   Recall:    0.5625
   F1 Score:  0.0824

 Confusion Matrix:
   [[450 580]
[ 21  27]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.4401
   Precision: 0.1530
   Recall:    0.7287
   F1 Score:  0.2529

 Confusion Matrix:
   [[ 842 1279]
[  86  231]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.3806
   Precision: 0.0492
   Recall:    0.5435
   F1 Score:  0.0903

 Confusion Matrix:
   [[ 854 1449]
[  63   75]]

 —— Eval 21: Assessment period 201710 ——
```

```
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.4602
   Precision: 0.0778
   Recall:    0.7550
   F1 Score:  0.1410

 Confusion Matrix:
   [[1070 1352]
[  37  114]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.6275
   Precision: 0.2110
   Recall:    0.3583
   F1 Score:  0.2656

 Confusion Matrix:
   [[1272  572]
[ 274  153]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.7331
   Precision: 0.2119
   Recall:    0.4490
   F1 Score:  0.2879

 Confusion Matrix:
   [[1662  491]
[ 162  132]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Logistic Regression...

 Results  (n_test=2,633):
   Accuracy:  0.6312
   Precision: 0.0384
```

```
    Recall:    0.3396
    F1 Score:  0.0690

  Confusion Matrix:
    [[1626  901]
 [  70   36]]

  —— Eval 25: Assessment period 201606 ——
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.6949
    Precision: 0.2050
    Recall:    0.5566
    F1 Score:  0.2997

  Confusion Matrix:
    [[1659  667]
 [ 137  172]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6582
    Precision: 0.4454
    Recall:    0.6402
    F1 Score:  0.5253

  Confusion Matrix:
    [[900 452]
 [204 363]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.5487
    Precision: 0.2359
    Recall:    0.7147
    F1 Score:  0.3547
```

```
 Confusion Matrix:
   [[815 771]
[ 95 238]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.6889
   Precision: 0.0109
   Recall:    0.0984
   F1 Score:  0.0197

 Confusion Matrix:
   [[1316  542]
[  55    6]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.7384
   Precision: 0.5468
   Recall:    0.8234
   F1 Score:  0.6571

 Confusion Matrix:
   [[671 286]
[ 74 345]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.6570
   Precision: 0.1777
   Recall:    0.7176
   F1 Score:  0.2848

 Confusion Matrix:
   [[810 435]
[ 37  94]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
```

```
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.6703
   Precision: 0.2129
   Recall:    0.6067
   F1 Score:  0.3152

 Confusion Matrix:
   [[1426  673]
[ 118  182]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Logistic Regression...

 Results  (n_test=2,490):
   Accuracy:  0.6430
   Precision: 0.2196
   Recall:    0.5223
   F1 Score:  0.3092

 Confusion Matrix:
   [[1402  707]
[ 182  199]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.5775
   Precision: 0.2397
   Recall:    0.5953
   F1 Score:  0.3417

 Confusion Matrix:
   [[1186  882]
[ 189  278]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.6418
   Precision: 0.0765
   Recall:    0.4792
   F1 Score:  0.1319
```

```
 Confusion Matrix:
   [[1558  833]
[  75   69]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.6448
   Precision: 0.3700
   Recall:    0.8232
   F1 Score:  0.5105

 Confusion Matrix:
   [[1144  785]
[  99  461]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.5428
   Precision: 0.1147
   Recall:    0.5259
   F1 Score:  0.1883

 Confusion Matrix:
   [[1219 1019]
[ 119  132]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.5793
   Precision: 0.1724
   Recall:    0.6589
   F1 Score:  0.2732

 Confusion Matrix:
   [[1246  946]
[ 102  197]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
```

```
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.5586
   Precision: 0.1433
   Recall:    0.4601
   F1 Score:  0.2185

 Confusion Matrix:
   [[789 586]
[115  98]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.6504
   Precision: 0.0777
   Recall:    0.4938
   F1 Score:  0.1342

 Confusion Matrix:
   [[920 475]
[ 41  40]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.5071
   Precision: 0.0864
   Recall:    0.4951
   F1 Score:  0.1471

 Confusion Matrix:
   [[1115 1079]
[ 104  102]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.4658
   Precision: 0.1964
   Recall:    0.6366
   F1 Score:  0.3002
```

```
 Confusion Matrix:
   [[ 843 1125]
[ 157  275]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.3895
   Precision: 0.0343
   Recall:    0.3548
   F1 Score:  0.0625

 Confusion Matrix:
   [[399 620]
[ 40  22]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.4481
   Precision: 0.0403
   Recall:    0.5000
   F1 Score:  0.0747

 Confusion Matrix:
   [[459 571]
[ 24  24]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.4655
   Precision: 0.1576
   Recall:    0.7161
   F1 Score:  0.2584

 Confusion Matrix:
   [[ 908 1213]
[  90  227]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
```

```
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.4019
   Precision: 0.0546
   Recall:    0.5870
   F1 Score:  0.0999

 Confusion Matrix:
   [[ 900 1403]
[  57   81]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.8962
   Precision: 0.0606
   Recall:    0.0530
   F1 Score:  0.0565

 Confusion Matrix:
   [[2298  124]
[ 143    8]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.7785
   Precision: 0.1885
   Recall:    0.0539
   F1 Score:  0.0838

 Confusion Matrix:
   [[1745   99]
[ 404   23]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.8288
   Precision: 0.1585
   Recall:    0.0986
   F1 Score:  0.1216
```

```
  Confusion Matrix:
    [[1999  154]
 [ 265   29]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Logistic Regression...

  Results  (n_test=2,633):
    Accuracy:  0.9107
    Precision: 0.0728
    Recall:    0.1038
    F1 Score:  0.0856

  Confusion Matrix:
    [[2387  140]
 [  95   11]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.6956
    Precision: 0.2048
    Recall:    0.5534
    F1 Score:  0.2990

  Confusion Matrix:
    [[1662  664]
 [ 138  171]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6608
    Precision: 0.4493
    Recall:    0.6561
    F1 Score:  0.5333

  Confusion Matrix:
    [[896 456]
 [195 372]]
```

```
── Eval 2: Assessment period 202402 ──
Training samples:   12,107
Test samples:       2,376  (admin2 areas in 202402)
Transition target: 333 worsened / 1919 (17.4%)
Training Logistic Regression...

Results  (n_test=1,919):
  Accuracy:  0.5482
  Precision: 0.2377
  Recall:    0.7267
  F1 Score:  0.3583

Confusion Matrix:
  [[810 776]
[ 91 242]]

── Eval 3: Assessment period 202310 ──
Training samples:   12,645
Test samples:       1,919  (admin2 areas in 202310)
Transition target: 61 worsened / 1919 (3.2%)
Training Logistic Regression...

Results  (n_test=1,919):
  Accuracy:  0.6832
  Precision: 0.0176
  Recall:    0.1639
  F1 Score:  0.0318

Confusion Matrix:
  [[1301  557]
[  51   10]]

── Eval 4: Assessment period 202306 ──
Training samples:   12,765
Test samples:       2,613  (admin2 areas in 202306)
Transition target: 419 worsened / 1376 (30.5%)
Training Logistic Regression...

Results  (n_test=1,376):
  Accuracy:  0.7326
  Precision: 0.5395
  Recall:    0.8305
  F1 Score:  0.6541

Confusion Matrix:
  [[660 297]
[ 71 348]]

── Eval 5: Assessment period 202302 ──
Training samples:   13,772
Test samples:       1,376  (admin2 areas in 202302)
Transition target: 131 worsened / 1376 (9.5%)
Training Logistic Regression...

Results  (n_test=1,376):
```

```
  Accuracy:  0.6432
  Precision: 0.1751
  Recall:    0.7405
  F1 Score:  0.2832

Confusion Matrix:
  [[788 457]
[ 34  97]]

── Eval 6: Assessment period 202210 ──
Training samples:   13,866
Test samples:       2,399  (admin2 areas in 202210)
Transition target: 300 worsened / 2399 (12.5%)
Training Logistic Regression...

Results  (n_test=2,399):
  Accuracy:  0.6661
  Precision: 0.2131
  Recall:    0.6200
  F1 Score:  0.3171

Confusion Matrix:
  [[1412  687]
[ 114  186]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training Logistic Regression...

Results  (n_test=2,490):
  Accuracy:  0.6365
  Precision: 0.2171
  Recall:    0.5276
  F1 Score:  0.3076

Confusion Matrix:
  [[1384  725]
[ 180  201]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training Logistic Regression...

Results  (n_test=2,535):
  Accuracy:  0.5763
  Precision: 0.2421
  Recall:    0.6103
  F1 Score:  0.3467

Confusion Matrix:
  [[1176  892]
[ 182  285]]
```

```
 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.6394
   Precision: 0.0779
   Recall:    0.4931
   F1 Score:  0.1345

 Confusion Matrix:
   [[1550  841]
[  73   71]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.6452
   Precision: 0.3707
   Recall:    0.8268
   F1 Score:  0.5119

 Confusion Matrix:
   [[1143  786]
[  97  463]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.5376
   Precision: 0.1167
   Recall:    0.5458
   F1 Score:  0.1923

 Confusion Matrix:
   [[1201 1037]
[ 114  137]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
```

```
  Accuracy:  0.5773
  Precision: 0.1699
  Recall:    0.6488
  F1 Score:  0.2693

Confusion Matrix:
  [[1244  948]
[ 105  194]]

── Eval 13: Assessment period 202006 ──
Training samples:   11,506
Test samples:       2,570  (admin2 areas in 202006)
Transition target: 213 worsened / 1588 (13.4%)
Training Logistic Regression...

Results  (n_test=1,588):
  Accuracy:  0.5661
  Precision: 0.1561
  Recall:    0.5070
  F1 Score:  0.2387

Confusion Matrix:
  [[791 584]
[105 108]]

── Eval 14: Assessment period 202002 ──
Training samples:   12,388
Test samples:       1,588  (admin2 areas in 202002)
Transition target: 81 worsened / 1476 (5.5%)
Training Logistic Regression...

Results  (n_test=1,476):
  Accuracy:  0.6531
  Precision: 0.0832
  Recall:    0.5309
  F1 Score:  0.1438

Confusion Matrix:
  [[921 474]
[ 38  43]]

── Eval 15: Assessment period 201910 ──
Training samples:   12,533
Test samples:       2,400  (admin2 areas in 201910)
Transition target: 206 worsened / 2400 (8.6%)
Training Logistic Regression...

Results  (n_test=2,400):
  Accuracy:  0.5112
  Precision: 0.0920
  Recall:    0.5291
  F1 Score:  0.1567

Confusion Matrix:
  [[1118 1076]
[  97  109]]
```

```
── Eval 16: Assessment period 201906 ──
Training samples:   12,605
Test samples:       2,400  (admin2 areas in 201906)
Transition target: 432 worsened / 2400 (18.0%)
Training Logistic Regression...

Results  (n_test=2,400):
  Accuracy:  0.4783
  Precision: 0.2007
  Recall:    0.6366
  F1 Score:  0.3052

Confusion Matrix:
  [[ 873 1095]
[ 157  275]]

── Eval 17: Assessment period 201902 ──
Training samples:   12,785
Test samples:       2,400  (admin2 areas in 201902)
Transition target: 62 worsened / 1081 (5.7%)
Training Logistic Regression...

Results  (n_test=1,081):
  Accuracy:  0.3922
  Precision: 0.0344
  Recall:    0.3548
  F1 Score:  0.0628

Confusion Matrix:
  [[402 617]
[ 40  22]]

── Eval 18: Assessment period 201810 ──
Training samples:   13,957
Test samples:       1,081  (admin2 areas in 201810)
Transition target: 48 worsened / 1078 (4.5%)
Training Logistic Regression...

Results  (n_test=1,078):
  Accuracy:  0.4564
  Precision: 0.0378
  Recall:    0.4583
  F1 Score:  0.0698

Confusion Matrix:
  [[470 560]
[ 26  22]]

── Eval 19: Assessment period 201806 ──
Training samples:   14,374
Test samples:       2,440  (admin2 areas in 201806)
Transition target: 317 worsened / 2438 (13.0%)
Training Logistic Regression...

Results  (n_test=2,438):
```

```
  Accuracy:  0.4943
  Precision: 0.1622
  Recall:    0.6940
  F1 Score:  0.2630

Confusion Matrix:
  [[ 985 1136]
[  97  220]]

── Eval 20: Assessment period 201802 ──
Training samples:   14,614
Test samples:       2,441  (admin2 areas in 201802)
Transition target: 138 worsened / 2441 (5.7%)
Training Logistic Regression...

Results  (n_test=2,441):
  Accuracy:  0.4367
  Precision: 0.0629
  Recall:    0.6449
  F1 Score:  0.1146

Confusion Matrix:
  [[ 977 1326]
[  49   89]]

── Eval 21: Assessment period 201710 ──
Training samples:   14,902
Test samples:       2,577  (admin2 areas in 201710)
Transition target: 151 worsened / 2573 (5.9%)
Training Logistic Regression...

Results  (n_test=2,573):
  Accuracy:  0.8368
  Precision: 0.0961
  Recall:    0.2119
  F1 Score:  0.1322

Confusion Matrix:
  [[2121  301]
[ 119   32]]

── Eval 22: Assessment period 201706 ──
Training samples:   11,953
Test samples:       2,573  (admin2 areas in 201706)
Transition target: 427 worsened / 2271 (18.8%)
Training Logistic Regression...

Results  (n_test=2,271):
  Accuracy:  0.7288
  Precision: 0.1942
  Recall:    0.1405
  F1 Score:  0.1630

Confusion Matrix:
  [[1595  249]
[ 367   60]]
```

```
  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Transition target: 294 worsened / 2447 (12.0%)
  Training Logistic Regression...

  Results  (n_test=2,447):
    Accuracy:  0.7965
    Precision: 0.1645
    Recall:    0.1701
    F1 Score:  0.1672

  Confusion Matrix:
    [[1899  254]
 [ 244   50]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Logistic Regression...

  Results  (n_test=2,633):
    Accuracy:  0.8796
    Precision: 0.0622
    Recall:    0.1415
    F1 Score:  0.0865

  Confusion Matrix:
    [[2301  226]
 [  91   15]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.6835
    Precision: 0.1930
    Recall:    0.5340
    F1 Score:  0.2835

  Confusion Matrix:
    [[1636  690]
 [ 144  165]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
```

```
 Test samples:       1,919  (admin2 areas in 202406)
 Transition target: 567 worsened / 1919 (29.5%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.6404
   Precision: 0.2901
   Recall:    0.1499
   F1 Score:  0.1977

 Confusion Matrix:
   [[1144  208]
[ 482   85]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.7379
   Precision: 0.1205
   Recall:    0.0811
   F1 Score:  0.0969

 Confusion Matrix:
   [[1389  197]
[ 306   27]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.8588
   Precision: 0.0513
   Recall:    0.1967
   F1 Score:  0.0814

 Confusion Matrix:
   [[1636  222]
[  49   12]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.6301
   Precision: 0.2704
   Recall:    0.1265
```

```
  F1 Score:  0.1724

 Confusion Matrix:
   [[814 143]
[366  53]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8118
   Precision: 0.1049
   Recall:    0.1298
   F1 Score:  0.1160

 Confusion Matrix:
   [[1100  145]
[ 114   17]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.8220
   Precision: 0.2019
   Recall:    0.1433
   F1 Score:  0.1676

 Confusion Matrix:
   [[1929  170]
[ 257   43]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Logistic Regression...

 Results  (n_test=2,490):
   Accuracy:  0.7743
   Precision: 0.1276
   Recall:    0.0814
   F1 Score:  0.0994

 Confusion Matrix:
   [[1897  212]
[ 350   31]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
```

```
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.7432
   Precision: 0.2514
   Recall:    0.1991
   F1 Score:  0.2222

 Confusion Matrix:
   [[1791  277]
[ 374   93]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8394
   Precision: 0.0744
   Recall:    0.1597
   F1 Score:  0.1015

 Confusion Matrix:
   [[2105  286]
[ 121   23]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.6935
   Precision: 0.2308
   Recall:    0.1554
   F1 Score:  0.1857

 Confusion Matrix:
   [[1639  290]
[ 473   87]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.7843
   Precision: 0.1006
   Recall:    0.1434
```

```
  F1 Score:  0.1182

 Confusion Matrix:
   [[1916  322]
[ 215   36]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.7704
   Precision: 0.1155
   Recall:    0.1371
   F1 Score:  0.1254

 Confusion Matrix:
   [[1878  314]
[ 258   41]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.7878
   Precision: 0.1900
   Recall:    0.1784
   F1 Score:  0.1840

 Confusion Matrix:
   [[1213  162]
[ 175   38]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Logistic Regression...

 Results  (n_test=1,476):
   Accuracy:  0.7961
   Precision: 0.0703
   Recall:    0.2222
   F1 Score:  0.1068

 Confusion Matrix:
   [[1157  238]
[  63   18]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
```

```
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8096
   Precision: 0.1138
   Recall:    0.1796
   F1 Score:  0.1394

 Confusion Matrix:
   [[1906  288]
[ 169   37]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.7329
   Precision: 0.1953
   Recall:    0.1551
   F1 Score:  0.1729

 Confusion Matrix:
   [[1692  276]
[ 365   67]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.8113
   Precision: 0.0723
   Recall:    0.1935
   F1 Score:  0.1053

 Confusion Matrix:
   [[865 154]
[ 50  12]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Logistic Regression...

 Results  (n_test=1,078):
   Accuracy:  0.1818
   Precision: 0.0428
   Recall:    0.8125
```

```
  F1 Score:  0.0813

 Confusion Matrix:
   [[157 873]
[  9  39]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Logistic Regression...

 Results  (n_test=2,438):
   Accuracy:  0.2822
   Precision: 0.1419
   Recall:    0.8959
   F1 Score:  0.2450

 Confusion Matrix:
   [[ 404 1717]
[  33  284]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8193
   Precision: 0.0659
   Recall:    0.1667
   F1 Score:  0.0945

 Confusion Matrix:
   [[1977  326]
[ 115   23]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.8344
   Precision: 0.0991
   Recall:    0.2252
   F1 Score:  0.1377

 Confusion Matrix:
   [[2113  309]
[ 117   34]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
```

```
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Logistic Regression...

 Results  (n_test=2,271):
   Accuracy:  0.7305
   Precision: 0.1987
   Recall:    0.1429
   F1 Score:  0.1662

 Confusion Matrix:
   [[1598  246]
[ 366   61]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Logistic Regression...

 Results  (n_test=2,447):
   Accuracy:  0.8047
   Precision: 0.1783
   Recall:    0.1735
   F1 Score:  0.1759

 Confusion Matrix:
   [[1918  235]
[ 243   51]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Logistic Regression...

 Results  (n_test=2,633):
   Accuracy:  0.8804
   Precision: 0.0664
   Recall:    0.1509
   F1 Score:  0.0922

 Confusion Matrix:
   [[2302  225]
[  90   16]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Transition target: 309 worsened / 2635 (11.7%)
 Training Logistic Regression...

 Results  (n_test=2,635):
   Accuracy:  0.8262
   Precision: 0.1535
   Recall:    0.1068
```

```
    F1 Score:  0.1260

  Confusion Matrix:
    [[2144  182]
 [ 276   33]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC only
======================================================================
Using 3332 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.7608
    Precision: 0.5711
    Recall:    0.7654
    F1 Score:  0.6541

  Confusion Matrix:
    [[1026  326]
 [ 133  434]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6946
    Precision: 0.3496
    Recall:    0.8829
    F1 Score:  0.5009

  Confusion Matrix:
    [[1039  547]
 [  39  294]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.7342
    Precision: 0.0658
    Recall:    0.5574
    F1 Score:  0.1176

  Confusion Matrix:
```

```
   [[1375  483]
[  27   34]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7696
   Precision: 0.5844
   Recall:    0.8425
   F1 Score:  0.6901

 Confusion Matrix:
   [[706 251]
[ 66 353]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.5174
   Precision: 0.1561
   Recall:    0.9237
   F1 Score:  0.2671

 Confusion Matrix:
   [[591 654]
[ 10 121]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6411
   Precision: 0.2117
   Recall:    0.6867
   F1 Score:  0.3236

 Confusion Matrix:
   [[1332  767]
[  94  206]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...
```

```
 Results  (n_test=2,490):
   Accuracy:  0.6635
   Precision: 0.3008
   Recall:    0.9055
   F1 Score:  0.4516

 Confusion Matrix:
   [[1307  802]
[  36  345]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7389
   Precision: 0.4041
   Recall:    0.8801
   F1 Score:  0.5539

 Confusion Matrix:
   [[1462  606]
[  56  411]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.5365
   Precision: 0.0708
   Recall:    0.5903
   F1 Score:  0.1264

 Confusion Matrix:
   [[1275 1116]
[  59   85]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7075
   Precision: 0.4311
   Recall:    0.9393
   F1 Score:  0.5910

 Confusion Matrix:
```

```
   [[1235  694]
[  34  526]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.5745
   Precision: 0.1582
   Recall:    0.7450
   F1 Score:  0.2610

 Confusion Matrix:
   [[1243  995]
[  64  187]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.6303
   Precision: 0.2178
   Recall:    0.8027
   F1 Score:  0.3426

 Confusion Matrix:
   [[1330  862]
[  59  240]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7204
   Precision: 0.3059
   Recall:    0.8545
   F1 Score:  0.4505

 Confusion Matrix:
   [[962 413]
[ 31 182]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...
```

```
 Results  (n_test=1,476):
   Accuracy:  0.6416
   Precision: 0.1014
   Recall:    0.7037
   F1 Score:  0.1773

 Confusion Matrix:
   [[890 505]
[ 24  57]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6550
   Precision: 0.1458
   Recall:    0.6214
   F1 Score:  0.2362

 Confusion Matrix:
   [[1444  750]
[  78  128]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7271
   Precision: 0.3842
   Recall:    0.8565
   F1 Score:  0.5305

 Confusion Matrix:
   [[1375  593]
[  62  370]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8104
   Precision: 0.2218
   Recall:    0.9194
   F1 Score:  0.3574

 Confusion Matrix:
```

```
   [[819 200]
[  5  57]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.5761
   Precision: 0.0835
   Recall:    0.8542
   F1 Score:  0.1521

 Confusion Matrix:
   [[580 450]
[  7  41]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.7445
   Precision: 0.3281
   Recall:    0.9211
   F1 Score:  0.4838

 Confusion Matrix:
   [[1523  598]
[  25  292]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.6821
   Precision: 0.1495
   Recall:    0.9855
   F1 Score:  0.2595

 Confusion Matrix:
   [[1529  774]
[   2  136]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...
```

```
 Results  (n_test=2,573):
   Accuracy:  0.6389
   Precision: 0.1302
   Recall:    0.9073
   F1 Score:  0.2278

 Confusion Matrix:
   [[1507  915]
[  14  137]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...

 Results  (n_test=2,271):
   Accuracy:  0.6812
   Precision: 0.3412
   Recall:    0.7471
   F1 Score:  0.4684

 Confusion Matrix:
   [[1228  616]
[ 108  319]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Random Forest...

 Results  (n_test=2,447):
   Accuracy:  0.7037
   Precision: 0.2680
   Recall:    0.8469
   F1 Score:  0.4072

 Confusion Matrix:
   [[1473  680]
[  45  249]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Random Forest...

 Results  (n_test=2,633):
   Accuracy:  0.7497
   Precision: 0.0940
   Recall:    0.6038
   F1 Score:  0.1626

 Confusion Matrix:
```

```
    [[1910  617]
 [  42   64]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8459
    Precision: 0.4304
    Recall:    0.9709
    F1 Score:  0.5964

  Confusion Matrix:
    [[1929  397]
 [   9  300]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT base
======================================================================
Using 3451 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.8218
    Precision: 0.7517
    Recall:    0.5926
    F1 Score:  0.6627

  Confusion Matrix:
    [[1241  111]
 [ 231  336]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.7535
    Precision: 0.4028
    Recall:    0.8709
    F1 Score:  0.5508

  Confusion Matrix:
    [[1156  430]
 [  43  290]]
```

```
 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.7447
   Precision: 0.0613
   Recall:    0.4918
   F1 Score:  0.1091

 Confusion Matrix:
   [[1399  459]
[  31   30]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.8089
   Precision: 0.6461
   Recall:    0.8234
   F1 Score:  0.7240

 Confusion Matrix:
   [[768 189]
[ 74 345]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7369
   Precision: 0.2404
   Recall:    0.8168
   F1 Score:  0.3715

 Confusion Matrix:
   [[907 338]
[ 24 107]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6874
```

```
  Precision: 0.2365
  Recall:    0.6733
  F1 Score:  0.3501

Confusion Matrix:
  [[1447  652]
[  98  202]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training Random Forest...

Results  (n_test=2,490):
  Accuracy:  0.7072
  Precision: 0.3161
  Recall:    0.7848
  F1 Score:  0.4506

Confusion Matrix:
  [[1462  647]
[  82  299]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training Random Forest...

Results  (n_test=2,535):
  Accuracy:  0.7779
  Precision: 0.4442
  Recall:    0.8180
  F1 Score:  0.5757

Confusion Matrix:
  [[1590  478]
[  85  382]]

── Eval 9: Assessment period 202110 ──
Training samples:   12,950
Test samples:       2,535  (admin2 areas in 202110)
Transition target: 144 worsened / 2535 (5.7%)
Training Random Forest...

Results  (n_test=2,535):
  Accuracy:  0.7558
  Precision: 0.0954
  Recall:    0.3889
  F1 Score:  0.1532

Confusion Matrix:
  [[1860  531]
[  88   56]]
```

```
 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7891
   Precision: 0.5193
   Recall:    0.8411
   F1 Score:  0.6421

 Confusion Matrix:
   [[1493  436]
[  89  471]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7537
   Precision: 0.2033
   Recall:    0.4940
   F1 Score:  0.2880

 Confusion Matrix:
   [[1752  486]
[ 127  124]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7363
   Precision: 0.2177
   Recall:    0.4615
   F1 Score:  0.2958

 Confusion Matrix:
   [[1696  496]
[ 161  138]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.8583
```

```
  Precision: 0.4796
  Recall:    0.6620
  F1 Score:  0.5562

 Confusion Matrix:
   [[1222  153]
[  72  141]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.8001
   Precision: 0.1457
   Recall:    0.5432
   F1 Score:  0.2298

 Confusion Matrix:
   [[1137  258]
[  37   44]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7275
   Precision: 0.1782
   Recall:    0.6019
   F1 Score:  0.2749

 Confusion Matrix:
   [[1622  572]
[  82  124]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7837
   Precision: 0.4421
   Recall:    0.7685
   F1 Score:  0.5613

 Confusion Matrix:
   [[1549  419]
[ 100  332]]
```

```
 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8862
   Precision: 0.3129
   Recall:    0.8226
   F1 Score:  0.4533

 Confusion Matrix:
   [[907 112]
[ 11  51]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.5937
   Precision: 0.0724
   Recall:    0.6875
   F1 Score:  0.1310

 Confusion Matrix:
   [[607 423]
[ 15  33]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.7363
   Precision: 0.3228
   Recall:    0.9369
   F1 Score:  0.4802

 Confusion Matrix:
   [[1498  623]
[  20  297]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.7722
```

```
  Precision: 0.1935
  Recall:    0.9565
  F1 Score:  0.3220

Confusion Matrix:
  [[1753  550]
[   6  132]]

── Eval 21: Assessment period 201710 ──
Training samples:   14,902
Test samples:       2,577  (admin2 areas in 201710)
Transition target: 151 worsened / 2573 (5.9%)
Training Random Forest...

Results  (n_test=2,573):
  Accuracy:  0.7792
  Precision: 0.1855
  Recall:    0.8146
  F1 Score:  0.3022

Confusion Matrix:
  [[1882  540]
[  28  123]]

── Eval 22: Assessment period 201706 ──
Training samples:   11,953
Test samples:       2,573  (admin2 areas in 201706)
Transition target: 427 worsened / 2271 (18.8%)
Training Random Forest...

Results  (n_test=2,271):
  Accuracy:  0.7486
  Precision: 0.3997
  Recall:    0.6721
  F1 Score:  0.5013

Confusion Matrix:
  [[1413  431]
[ 140  287]]

── Eval 23: Assessment period 201702 ──
Training samples:   9,303
Test samples:       2,449  (admin2 areas in 201702)
Transition target: 294 worsened / 2447 (12.0%)
Training Random Forest...

Results  (n_test=2,447):
  Accuracy:  0.7761
  Precision: 0.2883
  Recall:    0.5884
  F1 Score:  0.3870

Confusion Matrix:
  [[1726  427]
[ 121  173]]
```

```
  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Random Forest...

  Results  (n_test=2,633):
    Accuracy:  0.8519
    Precision: 0.0774
    Recall:    0.2453
    F1 Score:  0.1176

  Confusion Matrix:
    [[2217  310]
 [  80   26]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8801
    Precision: 0.4929
    Recall:    0.7896
    F1 Score:  0.6070

  Confusion Matrix:
    [[2075  251]
 [  65  244]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.8385
    Precision: 0.8009
    Recall:    0.6032
    F1 Score:  0.6881

  Confusion Matrix:
    [[1267   85]
 [ 225  342]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
```

```
 Transition target: 333 worsened / 1919 (17.4%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.7603
   Precision: 0.4084
   Recall:    0.8498
   F1 Score:  0.5517

 Confusion Matrix:
   [[1176  410]
[  50  283]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.8030
   Precision: 0.0634
   Recall:    0.3770
   F1 Score:  0.1085

 Confusion Matrix:
   [[1518  340]
[  38   23]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.8227
   Precision: 0.6699
   Recall:    0.8234
   F1 Score:  0.7388

 Confusion Matrix:
   [[787 170]
[ 74 345]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7209
   Precision: 0.2326
   Recall:    0.8397
   F1 Score:  0.3642
```

```
 Confusion Matrix:
   [[882 363]
[ 21 110]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.7086
   Precision: 0.2471
   Recall:    0.6500
   F1 Score:  0.3581

 Confusion Matrix:
   [[1505  594]
[ 105  195]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.7787
   Precision: 0.3655
   Recall:    0.6063
   F1 Score:  0.4561

 Confusion Matrix:
   [[1708  401]
[ 150  231]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7819
   Precision: 0.4486
   Recall:    0.8030
   F1 Score:  0.5756

 Confusion Matrix:
   [[1607  461]
[  92  375]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
```

```
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7373
   Precision: 0.0790
   Recall:    0.3403
   F1 Score:  0.1283

 Confusion Matrix:
   [[1820  571]
[  95   49]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7895
   Precision: 0.5199
   Recall:    0.8411
   F1 Score:  0.6426

 Confusion Matrix:
   [[1494  435]
[  89  471]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7421
   Precision: 0.1635
   Recall:    0.3785
   F1 Score:  0.2284

 Confusion Matrix:
   [[1752  486]
[ 156   95]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7302
   Precision: 0.1745
   Recall:    0.3344
   F1 Score:  0.2294
```

```
 Confusion Matrix:
   [[1719  473]
[ 199  100]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.8432
   Precision: 0.4362
   Recall:    0.5775
   F1 Score:  0.4970

 Confusion Matrix:
   [[1216  159]
[  90  123]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.8089
   Precision: 0.1593
   Recall:    0.5802
   F1 Score:  0.2500

 Confusion Matrix:
   [[1147  248]
[  34   47]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7137
   Precision: 0.1719
   Recall:    0.6117
   F1 Score:  0.2684

 Confusion Matrix:
   [[1587  607]
[  80  126]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
```

```
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7979
   Precision: 0.4611
   Recall:    0.7269
   F1 Score:  0.5642

 Confusion Matrix:
   [[1601  367]
[ 118  314]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8723
   Precision: 0.2625
   Recall:    0.6774
   F1 Score:  0.3784

 Confusion Matrix:
   [[901 118]
[ 20  42]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.5900
   Precision: 0.0736
   Recall:    0.7083
   F1 Score:  0.1333

 Confusion Matrix:
   [[602 428]
[ 14  34]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.7084
   Precision: 0.3006
   Recall:    0.9369
   F1 Score:  0.4552
```

```
 Confusion Matrix:
   [[1430  691]
[  20  297]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8206
   Precision: 0.2201
   Recall:    0.8551
   F1 Score:  0.3501

 Confusion Matrix:
   [[1885  418]
[  20  118]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.8228
   Precision: 0.2202
   Recall:    0.7947
   F1 Score:  0.3448

 Confusion Matrix:
   [[1997  425]
[  31  120]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...

 Results  (n_test=2,271):
   Accuracy:  0.7420
   Precision: 0.3568
   Recall:    0.4637
   F1 Score:  0.4033

 Confusion Matrix:
   [[1487  357]
[ 229  198]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
```

```
  Transition target: 294 worsened / 2447 (12.0%)
  Training Random Forest...

  Results  (n_test=2,447):
    Accuracy:  0.7842
    Precision: 0.2759
    Recall:    0.4898
    F1 Score:  0.3529

  Confusion Matrix:
    [[1775  378]
 [ 150  144]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Random Forest...

  Results  (n_test=2,633):
    Accuracy:  0.8485
    Precision: 0.1051
    Recall:    0.3679
    F1 Score:  0.1635

  Confusion Matrix:
    [[2195  332]
 [  67   39]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8979
    Precision: 0.5424
    Recall:    0.8285
    F1 Score:  0.6556

  Confusion Matrix:
    [[2110  216]
 [  53  256]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: GDELT only
======================================================================
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...
```

```
 Results  (n_test=1,919):
   Accuracy:  0.7207
   Precision: 0.5187
   Recall:    0.7566
   F1 Score:  0.6155

 Confusion Matrix:
   [[954 398]
[138 429]]

 —— Eval 2: Assessment period 202402 ——
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.6853
   Precision: 0.3341
   Recall:    0.8198
   F1 Score:  0.4748

 Confusion Matrix:
   [[1042  544]
[  60  273]]

 —— Eval 3: Assessment period 202310 ——
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.7452
   Precision: 0.0916
   Recall:    0.7869
   F1 Score:  0.1641

 Confusion Matrix:
   [[1382  476]
[  13   48]]

 —— Eval 4: Assessment period 202306 ——
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.6337
   Precision: 0.4479
   Recall:    0.8711
   F1 Score:  0.5916

 Confusion Matrix:
   [[507 450]
```

```
[ 54 365]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.8067
   Precision: 0.2563
   Recall:    0.5420
   F1 Score:  0.3480

 Confusion Matrix:
   [[1039  206]
[  60   71]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6440
   Precision: 0.2002
   Recall:    0.6167
   F1 Score:  0.3023

 Confusion Matrix:
   [[1360  739]
[ 115  185]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.6594
   Precision: 0.2677
   Recall:    0.7060
   F1 Score:  0.3882

 Confusion Matrix:
   [[1373  736]
[ 112  269]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...
```

```
 Results  (n_test=2,535):
   Accuracy:  0.7467
   Precision: 0.3988
   Recall:    0.7388
   F1 Score:  0.5180

 Confusion Matrix:
   [[1548  520]
[ 122  345]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.6533
   Precision: 0.0513
   Recall:    0.2917
   F1 Score:  0.0872

 Confusion Matrix:
   [[1614  777]
[ 102   42]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7634
   Precision: 0.4842
   Recall:    0.7946
   F1 Score:  0.6018

 Confusion Matrix:
   [[1455  474]
[ 115  445]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7782
   Precision: 0.1508
   Recall:    0.2590
   F1 Score:  0.1906

 Confusion Matrix:
   [[1872  366]
```

```
[ 186   65]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7186
   Precision: 0.1461
   Recall:    0.2776
   F1 Score:  0.1915

 Confusion Matrix:
   [[1707  485]
[ 216   83]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.8067
   Precision: 0.3709
   Recall:    0.6338
   F1 Score:  0.4679

 Confusion Matrix:
   [[1146  229]
[  78  135]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.7554
   Precision: 0.1296
   Recall:    0.6049
   F1 Score:  0.2135

 Confusion Matrix:
   [[1066  329]
[  32   49]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...
```

```
 Results  (n_test=2,400):
   Accuracy:  0.6933
   Precision: 0.1513
   Recall:    0.5583
   F1 Score:  0.2381

 Confusion Matrix:
   [[1549  645]
[  91  115]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7842
   Precision: 0.4338
   Recall:    0.6528
   F1 Score:  0.5213

 Confusion Matrix:
   [[1600  368]
[ 150  282]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8742
   Precision: 0.2628
   Recall:    0.6613
   F1 Score:  0.3761

 Confusion Matrix:
   [[904 115]
[ 21  41]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.4842
   Precision: 0.0651
   Recall:    0.7917
   F1 Score:  0.1203

 Confusion Matrix:
   [[484 546]
```

```
[ 10  38]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.7199
   Precision: 0.3070
   Recall:    0.9180
   F1 Score:  0.4601

 Confusion Matrix:
   [[1464  657]
[  26  291]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8189
   Precision: 0.1996
   Recall:    0.7319
   F1 Score:  0.3137

 Confusion Matrix:
   [[1898  405]
[  37  101]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.8037
   Precision: 0.1948
   Recall:    0.7483
   F1 Score:  0.3092

 Confusion Matrix:
   [[1955  467]
[  38  113]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...
```

```
 Results  (n_test=2,271):
   Accuracy:  0.7508
   Precision: 0.3701
   Recall:    0.4637
   F1 Score:  0.4116

 Confusion Matrix:
   [[1507  337]
[ 229  198]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Random Forest...

 Results  (n_test=2,447):
   Accuracy:  0.8116
   Precision: 0.3212
   Recall:    0.5102
   F1 Score:  0.3942

 Confusion Matrix:
   [[1836  317]
[ 144  150]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Random Forest...

 Results  (n_test=2,633):
   Accuracy:  0.8606
   Precision: 0.1057
   Recall:    0.3302
   F1 Score:  0.1602

 Confusion Matrix:
   [[2231  296]
[  71   35]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Transition target: 309 worsened / 2635 (11.7%)
 Training Random Forest...

 Results  (n_test=2,635):
   Accuracy:  0.8801
   Precision: 0.4932
   Recall:    0.8220
   F1 Score:  0.6165

 Confusion Matrix:
   [[2065  261]
```

```
 [  55  254]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6681
    Precision: 0.4556
    Recall:    0.6332
    F1 Score:  0.5299

  Confusion Matrix:
    [[923 429]
 [208 359]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.5638
    Precision: 0.2402
    Recall:    0.6997
    F1 Score:  0.3576

  Confusion Matrix:
    [[849 737]
 [100 233]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.7134
    Precision: 0.0080
    Recall:    0.0656
    F1 Score:  0.0143

  Confusion Matrix:
    [[1365  493]
 [  57    4]]

  ── Eval 4: Assessment period 202306 ──
```

```
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7682
   Precision: 0.5862
   Recall:    0.8115
   F1 Score:  0.6807

 Confusion Matrix:
   [[717 240]
[ 79 340]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.6926
   Precision: 0.1958
   Recall:    0.7176
   F1 Score:  0.3077

 Confusion Matrix:
   [[859 386]
[ 37  94]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6807
   Precision: 0.2109
   Recall:    0.5667
   F1 Score:  0.3074

 Confusion Matrix:
   [[1463  636]
[ 130  170]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.6506
   Precision: 0.2237
```

```
  Recall:    0.5197
  F1 Score:  0.3128

 Confusion Matrix:
   [[1422  687]
[ 183  198]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.5893
   Precision: 0.2433
   Recall:    0.5824
   F1 Score:  0.3432

 Confusion Matrix:
   [[1222  846]
[ 195  272]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.6517
   Precision: 0.0796
   Recall:    0.4861
   F1 Score:  0.1369

 Confusion Matrix:
   [[1582  809]
[  74   70]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.6472
   Precision: 0.3709
   Recall:    0.8161
   F1 Score:  0.5100

 Confusion Matrix:
   [[1154  775]
[ 103  457]]

 —— Eval 11: Assessment period 202102 ——
```

```
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.2812
   Precision: 0.1052
   Recall:    0.8167
   F1 Score:  0.1864

 Confusion Matrix:
   [[ 495 1743]
[  46  205]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.3452
   Precision: 0.1546
   Recall:    0.9967
   F1 Score:  0.2676

 Confusion Matrix:
   [[ 562 1630]
[   1  298]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.3992
   Precision: 0.1718
   Recall:    0.9108
   F1 Score:  0.2891

 Confusion Matrix:
   [[440 935]
[ 19 194]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.4539
   Precision: 0.0780
```

```
  Recall:    0.8272
  F1 Score:  0.1426

 Confusion Matrix:
   [[603 792]
[ 14  67]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.3554
   Precision: 0.1122
   Recall:    0.9417
   F1 Score:  0.2005

 Confusion Matrix:
   [[ 659 1535]
[  12  194]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.3958
   Precision: 0.2234
   Recall:    0.9514
   F1 Score:  0.3618

 Confusion Matrix:
   [[ 539 1429]
[  21  411]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.1684
   Precision: 0.0485
   Recall:    0.7258
   F1 Score:  0.0910

 Confusion Matrix:
   [[137 882]
[ 17  45]]

 ── Eval 18: Assessment period 201810 ──
```

```
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.1920
   Precision: 0.0453
   Recall:    0.8542
   F1 Score:  0.0860

 Confusion Matrix:
   [[166 864]
[  7  41]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.7818
   Precision: 0.1989
   Recall:    0.2240
   F1 Score:  0.2107

 Confusion Matrix:
   [[1835  286]
[ 246   71]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8370
   Precision: 0.1467
   Recall:    0.3913
   F1 Score:  0.2134

 Confusion Matrix:
   [[1989  314]
[  84   54]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.7594
   Precision: 0.0320
```

```
  Recall:    0.1060
  F1 Score:  0.0492

 Confusion Matrix:
   [[1938  484]
[ 135   16]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...

 Results  (n_test=2,271):
   Accuracy:  0.7274
   Precision: 0.2736
   Recall:    0.2717
   F1 Score:  0.2726

 Confusion Matrix:
   [[1536  308]
[ 311  116]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Random Forest...

 Results  (n_test=2,447):
   Accuracy:  0.8022
   Precision: 0.2780
   Recall:    0.4048
   F1 Score:  0.3296

 Confusion Matrix:
   [[1844  309]
[ 175  119]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training Random Forest...

 Results  (n_test=2,633):
   Accuracy:  0.8401
   Precision: 0.0661
   Recall:    0.2264
   F1 Score:  0.1023

 Confusion Matrix:
   [[2188  339]
[  82   24]]

 —— Eval 25: Assessment period 201606 ——
```

```
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8364
    Precision: 0.3498
    Recall:    0.4595
    F1 Score:  0.3972

  Confusion Matrix:
    [[2062  264]
 [ 167  142]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6759
    Precision: 0.4634
    Recall:    0.6138
    F1 Score:  0.5281

  Confusion Matrix:
    [[949 403]
 [219 348]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.5894
    Precision: 0.2481
    Recall:    0.6727
    F1 Score:  0.3625

  Confusion Matrix:
    [[907 679]
 [109 224]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
```

```
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.7212
   Precision: 0.0083
   Recall:    0.0656
   F1 Score:  0.0147

 Confusion Matrix:
   [[1380  478]
[  57    4]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7609
   Precision: 0.5771
   Recall:    0.8043
   F1 Score:  0.6720

 Confusion Matrix:
   [[710 247]
[ 82 337]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.6751
   Precision: 0.1802
   Recall:    0.6794
   F1 Score:  0.2848

 Confusion Matrix:
   [[840 405]
[ 42  89]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6836
   Precision: 0.2120
   Recall:    0.5633
   F1 Score:  0.3081
```

```
 Confusion Matrix:
   [[1471  628]
[ 131  169]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.6570
   Precision: 0.2227
   Recall:    0.4987
   F1 Score:  0.3079

 Confusion Matrix:
   [[1446  663]
[ 191  190]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.5964
   Precision: 0.2459
   Recall:    0.5760
   F1 Score:  0.3447

 Confusion Matrix:
   [[1243  825]
[ 198  269]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.6607
   Precision: 0.0738
   Recall:    0.4306
   F1 Score:  0.1260

 Confusion Matrix:
   [[1613  778]
[  82   62]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
```

```
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.6525
   Precision: 0.3715
   Recall:    0.7875
   F1 Score:  0.5049

 Confusion Matrix:
   [[1183  746]
[ 119  441]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.3210
   Precision: 0.1036
   Recall:    0.7490
   F1 Score:  0.1820

 Confusion Matrix:
   [[ 611 1627]
[  63  188]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.3786
   Precision: 0.1532
   Recall:    0.9231
   F1 Score:  0.2629

 Confusion Matrix:
   [[ 667 1525]
[  23  276]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.4257
   Precision: 0.1675
   Recall:    0.8263
   F1 Score:  0.2785
```

```
 Confusion Matrix:
   [[500 875]
[ 37 176]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.4898
   Precision: 0.0789
   Recall:    0.7778
   F1 Score:  0.1433

 Confusion Matrix:
   [[660 735]
[ 18  63]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.4008
   Precision: 0.1111
   Recall:    0.8544
   F1 Score:  0.1966

 Confusion Matrix:
   [[ 786 1408]
[  30  176]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.4392
   Precision: 0.2230
   Recall:    0.8519
   F1 Score:  0.3535

 Confusion Matrix:
   [[ 686 1282]
[  64  368]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
```

```
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.2211
   Precision: 0.0423
   Recall:    0.5806
   F1 Score:  0.0788

 Confusion Matrix:
   [[203 816]
[ 26  36]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.2449
   Precision: 0.0419
   Recall:    0.7292
   F1 Score:  0.0792

 Confusion Matrix:
   [[229 801]
[ 13  35]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.4114
   Precision: 0.1665
   Recall:    0.8801
   F1 Score:  0.2800

 Confusion Matrix:
   [[ 724 1397]
[  38  279]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.3458
   Precision: 0.0701
   Recall:    0.8623
   F1 Score:  0.1297
```

```
 Confusion Matrix:
   [[ 725 1578]
[  19  119]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.7726
   Precision: 0.0323
   Recall:    0.0993
   F1 Score:  0.0488

 Confusion Matrix:
   [[1973  449]
[ 136   15]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...

 Results  (n_test=2,271):
   Accuracy:  0.7340
   Precision: 0.2677
   Recall:    0.2389
   F1 Score:  0.2525

 Confusion Matrix:
   [[1565  279]
[ 325  102]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training Random Forest...

 Results  (n_test=2,447):
   Accuracy:  0.8141
   Precision: 0.2876
   Recall:    0.3707
   F1 Score:  0.3239

 Confusion Matrix:
   [[1883  270]
[ 185  109]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
```

```
  Training Random Forest...

  Results  (n_test=2,633):
    Accuracy:  0.8504
    Precision: 0.0689
    Recall:    0.2170
    F1 Score:  0.1045

  Confusion Matrix:
    [[2216  311]
 [  83   23]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8383
    Precision: 0.3496
    Recall:    0.4401
    F1 Score:  0.3897

  Confusion Matrix:
    [[2073  253]
 [ 173  136]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6863
    Precision: 0.4747
    Recall:    0.5785
    F1 Score:  0.5215

  Confusion Matrix:
    [[989 363]
 [239 328]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training Random Forest...

  Results  (n_test=1,919):
```

```
  Accuracy:  0.6102
  Precision: 0.2544
  Recall:    0.6456
  F1 Score:  0.3650

 Confusion Matrix:
   [[956 630]
[118 215]]

 —— Eval 3: Assessment period 202310 ——
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Random Forest...

 Results  (n_test=1,919):
  Accuracy:  0.7530
  Precision: 0.0072
  Recall:    0.0492
  F1 Score:  0.0125

 Confusion Matrix:
   [[1442  416]
[  58    3]]

 —— Eval 4: Assessment period 202306 ——
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
  Accuracy:  0.7624
  Precision: 0.5827
  Recall:    0.7733
  F1 Score:  0.6646

 Confusion Matrix:
   [[725 232]
[ 95 324]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
  Accuracy:  0.6839
  Precision: 0.1780
  Recall:    0.6412
  F1 Score:  0.2786

 Confusion Matrix:
   [[857 388]
[ 47  84]]
```

```
 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6928
   Precision: 0.2166
   Recall:    0.5567
   F1 Score:  0.3119

 Confusion Matrix:
   [[1495  604]
[ 133  167]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.6635
   Precision: 0.2263
   Recall:    0.4961
   F1 Score:  0.3109

 Confusion Matrix:
   [[1463  646]
[ 192  189]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.5964
   Precision: 0.2518
   Recall:    0.6039
   F1 Score:  0.3554

 Confusion Matrix:
   [[1230  838]
[ 185  282]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
```

```
  Accuracy:  0.6584
  Precision: 0.0753
  Recall:    0.4444
  F1 Score:  0.1288

Confusion Matrix:
  [[1605  786]
[  80   64]]

── Eval 10: Assessment period 202106 ──
Training samples:   12,859
Test samples:       2,536  (admin2 areas in 202106)
Transition target: 560 worsened / 2489 (22.5%)
Training Random Forest...

Results  (n_test=2,489):
  Accuracy:  0.6533
  Precision: 0.3706
  Recall:    0.7750
  F1 Score:  0.5014

Confusion Matrix:
  [[1192  737]
[ 126  434]]

── Eval 11: Assessment period 202102 ──
Training samples:   12,839
Test samples:       2,489  (admin2 areas in 202102)
Transition target: 251 worsened / 2489 (10.1%)
Training Random Forest...

Results  (n_test=2,489):
  Accuracy:  0.3712
  Precision: 0.1047
  Recall:    0.6932
  F1 Score:  0.1819

Confusion Matrix:
  [[ 750 1488]
[  77  174]]

── Eval 12: Assessment period 202010 ──
Training samples:   12,858
Test samples:       2,491  (admin2 areas in 202010)
Transition target: 299 worsened / 2491 (12.0%)
Training Random Forest...

Results  (n_test=2,491):
  Accuracy:  0.4155
  Precision: 0.1521
  Recall:    0.8462
  F1 Score:  0.2579

Confusion Matrix:
  [[ 782 1410]
[  46  253]]
```

```
── Eval 13: Assessment period 202006 ──
Training samples:   11,506
Test samples:       2,570  (admin2 areas in 202006)
Transition target: 213 worsened / 1588 (13.4%)
Training Random Forest...

Results  (n_test=1,588):
  Accuracy:  0.4868
  Precision: 0.1714
  Recall:    0.7371
  F1 Score:  0.2781

Confusion Matrix:
  [[616 759]
[ 56 157]]

── Eval 14: Assessment period 202002 ──
Training samples:   12,388
Test samples:       1,588  (admin2 areas in 202002)
Transition target: 81 worsened / 1476 (5.5%)
Training Random Forest...

Results  (n_test=1,476):
  Accuracy:  0.5495
  Precision: 0.0805
  Recall:    0.6914
  F1 Score:  0.1441

Confusion Matrix:
  [[755 640]
[ 25  56]]

── Eval 15: Assessment period 201910 ──
Training samples:   12,533
Test samples:       2,400  (admin2 areas in 201910)
Transition target: 206 worsened / 2400 (8.6%)
Training Random Forest...

Results  (n_test=2,400):
  Accuracy:  0.4567
  Precision: 0.1161
  Recall:    0.8058
  F1 Score:  0.2029

Confusion Matrix:
  [[ 930 1264]
[  40  166]]

── Eval 16: Assessment period 201906 ──
Training samples:   12,605
Test samples:       2,400  (admin2 areas in 201906)
Transition target: 432 worsened / 2400 (18.0%)
Training Random Forest...

Results  (n_test=2,400):
```

```
  Accuracy:  0.4975
  Precision: 0.2331
  Recall:    0.7824
  F1 Score:  0.3592

Confusion Matrix:
  [[ 856 1112]
[  94  338]]

── Eval 17: Assessment period 201902 ──
Training samples:   12,785
Test samples:       2,400  (admin2 areas in 201902)
Transition target: 62 worsened / 1081 (5.7%)
Training Random Forest...

Results  (n_test=1,081):
  Accuracy:  0.7327
  Precision: 0.0842
  Recall:    0.3710
  F1 Score:  0.1373

Confusion Matrix:
  [[769 250]
[ 39  23]]

── Eval 18: Assessment period 201810 ──
Training samples:   13,957
Test samples:       1,081  (admin2 areas in 201810)
Transition target: 48 worsened / 1078 (4.5%)
Training Random Forest...

Results  (n_test=1,078):
  Accuracy:  0.3275
  Precision: 0.0456
  Recall:    0.7083
  F1 Score:  0.0858

Confusion Matrix:
  [[319 711]
[ 14  34]]

── Eval 19: Assessment period 201806 ──
Training samples:   14,374
Test samples:       2,440  (admin2 areas in 201806)
Transition target: 317 worsened / 2438 (13.0%)
Training Random Forest...

Results  (n_test=2,438):
  Accuracy:  0.4914
  Precision: 0.1802
  Recall:    0.8202
  F1 Score:  0.2955

Confusion Matrix:
  [[ 938 1183]
[  57  260]]
```

```
── Eval 20: Assessment period 201802 ──
Training samples:   14,614
Test samples:       2,441  (admin2 areas in 201802)
Transition target: 138 worsened / 2441 (5.7%)
Training Random Forest...

Results  (n_test=2,441):
  Accuracy:  0.4215
  Precision: 0.0753
  Recall:    0.8188
  F1 Score:  0.1380

Confusion Matrix:
  [[ 916 1387]
[  25  113]]

── Eval 21: Assessment period 201710 ──
Training samples:   14,902
Test samples:       2,577  (admin2 areas in 201710)
Transition target: 151 worsened / 2573 (5.9%)
Training Random Forest...

Results  (n_test=2,573):
  Accuracy:  0.7680
  Precision: 0.0522
  Recall:    0.1722
  F1 Score:  0.0801

Confusion Matrix:
  [[1950  472]
[ 125   26]]

── Eval 22: Assessment period 201706 ──
Training samples:   11,953
Test samples:       2,573  (admin2 areas in 201706)
Transition target: 427 worsened / 2271 (18.8%)
Training Random Forest...

Results  (n_test=2,271):
  Accuracy:  0.7525
  Precision: 0.3077
  Recall:    0.2529
  F1 Score:  0.2776

Confusion Matrix:
  [[1601  243]
[ 319  108]]

── Eval 23: Assessment period 201702 ──
Training samples:   9,303
Test samples:       2,449  (admin2 areas in 201702)
Transition target: 294 worsened / 2447 (12.0%)
Training Random Forest...

Results  (n_test=2,447):
```

```
    Accuracy:  0.8165
    Precision: 0.2817
    Recall:    0.3401
    F1 Score:  0.3082

  Confusion Matrix:
    [[1898  255]
 [ 194  100]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Random Forest...

  Results  (n_test=2,633):
    Accuracy:  0.8678
    Precision: 0.0709
    Recall:    0.1887
    F1 Score:  0.1031

  Confusion Matrix:
    [[2265  262]
 [  86   20]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8398
    Precision: 0.3509
    Recall:    0.4304
    F1 Score:  0.3866

  Confusion Matrix:
    [[2080  246]
 [ 176  133]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.7045
    Precision: 0.5000
    Recall:    0.0617
```

```
  F1 Score:  0.1099

 Confusion Matrix:
   [[1317   35]
[ 532   35]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.8103
   Precision: 0.2373
   Recall:    0.0420
   F1 Score:  0.0714

 Confusion Matrix:
   [[1541   45]
[ 319   14]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.9442
   Precision: 0.0741
   Recall:    0.0656
   F1 Score:  0.0696

 Confusion Matrix:
   [[1808   50]
[  57    4]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.6751
   Precision: 0.3372
   Recall:    0.0692
   F1 Score:  0.1149

 Confusion Matrix:
   [[900  57]
[390  29]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
```

```
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.8612
   Precision: 0.0714
   Recall:    0.0382
   F1 Score:  0.0498

 Confusion Matrix:
   [[1180   65]
[ 126    5]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.8554
   Precision: 0.2299
   Recall:    0.0667
   F1 Score:  0.1034

 Confusion Matrix:
   [[2032   67]
[ 280   20]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training Random Forest...

 Results  (n_test=2,490):
   Accuracy:  0.8225
   Precision: 0.1856
   Recall:    0.0472
   F1 Score:  0.0753

 Confusion Matrix:
   [[2030   79]
[ 363   18]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7957
   Precision: 0.3053
   Recall:    0.0857
```

```
  F1 Score:  0.1338

 Confusion Matrix:
   [[1977   91]
[ 427   40]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.9195
   Precision: 0.0946
   Recall:    0.0486
   F1 Score:  0.0642

 Confusion Matrix:
   [[2324   67]
[ 137    7]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7622
   Precision: 0.3095
   Recall:    0.0464
   F1 Score:  0.0807

 Confusion Matrix:
   [[1871   58]
[ 534   26]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.8734
   Precision: 0.1364
   Recall:    0.0478
   F1 Score:  0.0708

 Confusion Matrix:
   [[2162   76]
[ 239   12]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
```

```
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.8483
   Precision: 0.1504
   Recall:    0.0569
   F1 Score:  0.0825

 Confusion Matrix:
   [[2096   96]
[ 282   17]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.8476
   Precision: 0.2364
   Recall:    0.0610
   F1 Score:  0.0970

 Confusion Matrix:
   [[1333   42]
[ 200   13]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training Random Forest...

 Results  (n_test=1,476):
   Accuracy:  0.8970
   Precision: 0.0723
   Recall:    0.0741
   F1 Score:  0.0732

 Confusion Matrix:
   [[1318   77]
[  75    6]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.8746
   Precision: 0.1854
   Recall:    0.1359
```

```
  F1 Score:  0.1569

 Confusion Matrix:
   [[2071  123]
[ 178   28]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7963
   Precision: 0.2791
   Recall:    0.0833
   F1 Score:  0.1283

 Confusion Matrix:
   [[1875   93]
[ 396   36]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8973
   Precision: 0.0984
   Recall:    0.0968
   F1 Score:  0.0976

 Confusion Matrix:
   [[964  55]
[ 56   6]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training Random Forest...

 Results  (n_test=1,078):
   Accuracy:  0.2468
   Precision: 0.0442
   Recall:    0.7708
   F1 Score:  0.0835

 Confusion Matrix:
   [[229 801]
[ 11  37]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
```

```
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training Random Forest...

 Results  (n_test=2,438):
   Accuracy:  0.3257
   Precision: 0.1423
   Recall:    0.8328
   F1 Score:  0.2431

 Confusion Matrix:
   [[ 530 1591]
[  53  264]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8832
   Precision: 0.0847
   Recall:    0.1087
   F1 Score:  0.0952

 Confusion Matrix:
   [[2141  162]
[ 123   15]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.9040
   Precision: 0.1250
   Recall:    0.1060
   F1 Score:  0.1147

 Confusion Matrix:
   [[2310  112]
[ 135   16]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training Random Forest...

 Results  (n_test=2,271):
   Accuracy:  0.7908
   Precision: 0.2333
   Recall:    0.0492
```

```
    F1 Score:  0.0812

  Confusion Matrix:
    [[1775   69]
 [ 406   21]]

  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Transition target: 294 worsened / 2447 (12.0%)
  Training Random Forest...

  Results  (n_test=2,447):
    Accuracy:  0.8619
    Precision: 0.1765
    Recall:    0.0408
    F1 Score:  0.0663

  Confusion Matrix:
    [[2097   56]
 [ 282   12]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training Random Forest...

  Results  (n_test=2,633):
    Accuracy:  0.9556
    Precision: 0.2963
    Recall:    0.0755
    F1 Score:  0.1203

  Confusion Matrix:
    [[2508   19]
 [  98    8]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8748
    Precision: 0.2766
    Recall:    0.0421
    F1 Score:  0.0730

  Confusion Matrix:
    [[2292   34]
 [ 296   13]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC only
```

```
======================================================================
Using 3332 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.8275
    Precision: 0.7138
    Recall:    0.6949
    F1 Score:  0.7042

  Confusion Matrix:
    [[1194  158]
 [ 173  394]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.7066
    Precision: 0.3650
    Recall:    0.9339
    F1 Score:  0.5249

  Confusion Matrix:
    [[1045  541]
 [  22  311]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.7650
    Precision: 0.0903
    Recall:    0.7049
    F1 Score:  0.1601

  Confusion Matrix:
    [[1425  433]
 [  18   43]]

  ── Eval 4: Assessment period 202306 ──
  Training samples:   12,765
  Test samples:       2,613  (admin2 areas in 202306)
  Transition target: 419 worsened / 1376 (30.5%)
  Training CatBoost...
```

```
 Results  (n_test=1,376):
   Accuracy:  0.8328
   Precision: 0.6697
   Recall:    0.8902
   F1 Score:  0.7643

 Confusion Matrix:
   [[773 184]
[ 46 373]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7493
   Precision: 0.2622
   Recall:    0.9008
   F1 Score:  0.4062

 Confusion Matrix:
   [[913 332]
[ 13 118]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.8529
   Precision: 0.4456
   Recall:    0.7233
   F1 Score:  0.5515

 Confusion Matrix:
   [[1829  270]
[  83  217]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training CatBoost...

 Results  (n_test=2,490):
   Accuracy:  0.8558
   Precision: 0.5179
   Recall:    0.8346
   F1 Score:  0.6392

 Confusion Matrix:
```

```
   [[1813  296]
[  63  318]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.8059
   Precision: 0.4835
   Recall:    0.7837
   F1 Score:  0.5980

 Confusion Matrix:
   [[1677  391]
[ 101  366]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.8083
   Precision: 0.1552
   Recall:    0.5347
   F1 Score:  0.2406

 Confusion Matrix:
   [[1972  419]
[  67   77]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.8441
   Precision: 0.6007
   Recall:    0.9161
   F1 Score:  0.7256

 Confusion Matrix:
   [[1588  341]
[  47  513]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...
```

```
 Results  (n_test=2,489):
   Accuracy:  0.7127
   Precision: 0.2428
   Recall:    0.8725
   F1 Score:  0.3799

 Confusion Matrix:
   [[1555  683]
[  32  219]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7591
   Precision: 0.3126
   Recall:    0.8395
   F1 Score:  0.4555

 Confusion Matrix:
   [[1640  552]
[  48  251]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7588
   Precision: 0.3372
   Recall:    0.8263
   F1 Score:  0.4789

 Confusion Matrix:
   [[1029  346]
[  37  176]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.6321
   Precision: 0.1085
   Recall:    0.7901
   F1 Score:  0.1908

 Confusion Matrix:
```

```
   [[869 526]
[ 17  64]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7163
   Precision: 0.1795
   Recall:    0.6456
   F1 Score:  0.2809

 Confusion Matrix:
   [[1586  608]
[  73  133]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7288
   Precision: 0.3865
   Recall:    0.8634
   F1 Score:  0.5340

 Confusion Matrix:
   [[1376  592]
[  59  373]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.7724
   Precision: 0.1913
   Recall:    0.9194
   F1 Score:  0.3167

 Confusion Matrix:
   [[778 241]
[  5  57]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...
```

```
 Results  (n_test=1,078):
   Accuracy:  0.7078
   Precision: 0.0735
   Recall:    0.4792
   F1 Score:  0.1274

 Confusion Matrix:
   [[740 290]
[ 25  23]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.7600
   Precision: 0.3445
   Recall:    0.9369
   F1 Score:  0.5038

 Confusion Matrix:
   [[1556  565]
[  20  297]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.7026
   Precision: 0.1449
   Recall:    0.8696
   F1 Score:  0.2484

 Confusion Matrix:
   [[1595  708]
[  18  120]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.7785
   Precision: 0.1968
   Recall:    0.9007
   F1 Score:  0.3230

 Confusion Matrix:
```

```
   [[1867  555]
[  15  136]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7270
   Precision: 0.3915
   Recall:    0.8150
   F1 Score:  0.5289

 Confusion Matrix:
   [[1303  541]
[  79  348]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training CatBoost...

 Results  (n_test=2,447):
   Accuracy:  0.7160
   Precision: 0.2887
   Recall:    0.9320
   F1 Score:  0.4409

 Confusion Matrix:
   [[1478  675]
[  20  274]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training CatBoost...

 Results  (n_test=2,633):
   Accuracy:  0.8097
   Precision: 0.1429
   Recall:    0.7453
   F1 Score:  0.2398

 Confusion Matrix:
   [[2053  474]
[  27   79]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Transition target: 309 worsened / 2635 (11.7%)
 Training CatBoost...
```

```
  Results  (n_test=2,635):
    Accuracy:  0.9101
    Precision: 0.5684
    Recall:    0.9676
    F1 Score:  0.7162

  Confusion Matrix:
    [[2099  227]
 [  10  299]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT base
======================================================================
Using 3451 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.8525
    Precision: 0.8047
    Recall:    0.6614
    F1 Score:  0.7260

  Confusion Matrix:
    [[1261   91]
 [ 192  375]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6946
    Precision: 0.3541
    Recall:    0.9219
    F1 Score:  0.5117

  Confusion Matrix:
    [[1026  560]
 [  26  307]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.7983
```

```
  Precision: 0.1044
  Recall:    0.7049
  F1 Score:  0.1818

Confusion Matrix:
  [[1489  369]
[  18   43]]

── Eval 4: Assessment period 202306 ──
Training samples:   12,765
Test samples:       2,613  (admin2 areas in 202306)
Transition target: 419 worsened / 1376 (30.5%)
Training CatBoost...

Results  (n_test=1,376):
  Accuracy:  0.8307
  Precision: 0.6667
  Recall:    0.8878
  F1 Score:  0.7615

Confusion Matrix:
  [[771 186]
[ 47 372]]

── Eval 5: Assessment period 202302 ──
Training samples:   13,772
Test samples:       1,376  (admin2 areas in 202302)
Transition target: 131 worsened / 1376 (9.5%)
Training CatBoost...

Results  (n_test=1,376):
  Accuracy:  0.7471
  Precision: 0.2594
  Recall:    0.8931
  F1 Score:  0.4021

Confusion Matrix:
  [[911 334]
[ 14 117]]

── Eval 6: Assessment period 202210 ──
Training samples:   13,866
Test samples:       2,399  (admin2 areas in 202210)
Transition target: 300 worsened / 2399 (12.5%)
Training CatBoost...

Results  (n_test=2,399):
  Accuracy:  0.8458
  Precision: 0.4306
  Recall:    0.7233
  F1 Score:  0.5398

Confusion Matrix:
  [[1812  287]
[  83  217]]
```

```
── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training CatBoost...

Results  (n_test=2,490):
  Accuracy:  0.8590
  Precision: 0.5245
  Recall:    0.8425
  F1 Score:  0.6465

Confusion Matrix:
  [[1818  291]
[  60  321]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.7874
  Precision: 0.4549
  Recall:    0.7773
  F1 Score:  0.5739

Confusion Matrix:
  [[1633  435]
[ 104  363]]

── Eval 9: Assessment period 202110 ──
Training samples:   12,950
Test samples:       2,535  (admin2 areas in 202110)
Transition target: 144 worsened / 2535 (5.7%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.8095
  Precision: 0.1576
  Recall:    0.5417
  F1 Score:  0.2441

Confusion Matrix:
  [[1974  417]
[  66   78]]

── Eval 10: Assessment period 202106 ──
Training samples:   12,859
Test samples:       2,536  (admin2 areas in 202106)
Transition target: 560 worsened / 2489 (22.5%)
Training CatBoost...

Results  (n_test=2,489):
  Accuracy:  0.8505
```

```
  Precision: 0.6133
  Recall:    0.9089
  F1 Score:  0.7324

 Confusion Matrix:
   [[1608  321]
[  51  509]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7674
   Precision: 0.2819
   Recall:    0.8446
   F1 Score:  0.4227

 Confusion Matrix:
   [[1698  540]
[  39  212]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7535
   Precision: 0.3082
   Recall:    0.8462
   F1 Score:  0.4518

 Confusion Matrix:
   [[1624  568]
[  46  253]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7387
   Precision: 0.3282
   Recall:    0.9061
   F1 Score:  0.4819

 Confusion Matrix:
   [[980 395]
[ 20 193]]
```

```
── Eval 14: Assessment period 202002 ──
Training samples:   12,388
Test samples:       1,588  (admin2 areas in 202002)
Transition target: 81 worsened / 1476 (5.5%)
Training CatBoost...

Results  (n_test=1,476):
  Accuracy:  0.6314
  Precision: 0.1083
  Recall:    0.7901
  F1 Score:  0.1905

Confusion Matrix:
  [[868 527]
[ 17  64]]

── Eval 15: Assessment period 201910 ──
Training samples:   12,533
Test samples:       2,400  (admin2 areas in 201910)
Transition target: 206 worsened / 2400 (8.6%)
Training CatBoost...

Results  (n_test=2,400):
  Accuracy:  0.7017
  Precision: 0.1714
  Recall:    0.6456
  F1 Score:  0.2709

Confusion Matrix:
  [[1551  643]
[  73  133]]

── Eval 16: Assessment period 201906 ──
Training samples:   12,605
Test samples:       2,400  (admin2 areas in 201906)
Transition target: 432 worsened / 2400 (18.0%)
Training CatBoost...

Results  (n_test=2,400):
  Accuracy:  0.7421
  Precision: 0.3969
  Recall:    0.8333
  F1 Score:  0.5377

Confusion Matrix:
  [[1421  547]
[  72  360]]

── Eval 17: Assessment period 201902 ──
Training samples:   12,785
Test samples:       2,400  (admin2 areas in 201902)
Transition target: 62 worsened / 1081 (5.7%)
Training CatBoost...

Results  (n_test=1,081):
  Accuracy:  0.7854
```

```
  Precision: 0.2007
  Recall:    0.9194
  F1 Score:  0.3295

 Confusion Matrix:
   [[792 227]
[  5  57]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.7087
   Precision: 0.1065
   Recall:    0.7500
   F1 Score:  0.1865

 Confusion Matrix:
   [[728 302]
[ 12  36]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.7531
   Precision: 0.3322
   Recall:    0.8896
   F1 Score:  0.4837

 Confusion Matrix:
   [[1554  567]
[  35  282]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.7292
   Precision: 0.1461
   Recall:    0.7826
   F1 Score:  0.2463

 Confusion Matrix:
   [[1672  631]
[  30  108]]
```

```
── Eval 21: Assessment period 201710 ──
Training samples:   14,902
Test samples:       2,577  (admin2 areas in 201710)
Transition target: 151 worsened / 2573 (5.9%)
Training CatBoost...

Results  (n_test=2,573):
  Accuracy:  0.7851
  Precision: 0.1889
  Recall:    0.8079
  F1 Score:  0.3061

Confusion Matrix:
  [[1898  524]
[  29  122]]

── Eval 22: Assessment period 201706 ──
Training samples:   11,953
Test samples:       2,573  (admin2 areas in 201706)
Transition target: 427 worsened / 2271 (18.8%)
Training CatBoost...

Results  (n_test=2,271):
  Accuracy:  0.7622
  Precision: 0.4225
  Recall:    0.7213
  F1 Score:  0.5329

Confusion Matrix:
  [[1423  421]
[ 119  308]]

── Eval 23: Assessment period 201702 ──
Training samples:   9,303
Test samples:       2,449  (admin2 areas in 201702)
Transition target: 294 worsened / 2447 (12.0%)
Training CatBoost...

Results  (n_test=2,447):
  Accuracy:  0.7434
  Precision: 0.3085
  Recall:    0.9150
  F1 Score:  0.4614

Confusion Matrix:
  [[1550  603]
[  25  269]]

── Eval 24: Assessment period 201610 ──
Training samples:   6,140
Test samples:       2,748  (admin2 areas in 201610)
Transition target: 106 worsened / 2633 (4.0%)
Training CatBoost...

Results  (n_test=2,633):
  Accuracy:  0.8211
```

```
    Precision: 0.1222
    Recall:    0.5566
    F1 Score:  0.2003

  Confusion Matrix:
    [[2103  424]
 [  47   59]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.9097
    Precision: 0.5681
    Recall:    0.9579
    F1 Score:  0.7133

  Confusion Matrix:
    [[2101  225]
 [  13  296]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.8588
    Precision: 0.8109
    Recall:    0.6808
    F1 Score:  0.7402

  Confusion Matrix:
    [[1262   90]
 [ 181  386]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6957
    Precision: 0.3549
    Recall:    0.9219
    F1 Score:  0.5125
```

```
 Confusion Matrix:
   [[1028  558]
[  26  307]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.7983
   Precision: 0.1063
   Recall:    0.7213
   F1 Score:  0.1853

 Confusion Matrix:
   [[1488  370]
[  17   44]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.8263
   Precision: 0.6679
   Recall:    0.8544
   F1 Score:  0.7497

 Confusion Matrix:
   [[779 178]
[ 61 358]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7406
   Precision: 0.2565
   Recall:    0.9084
   F1 Score:  0.4000

 Confusion Matrix:
   [[900 345]
[ 12 119]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
```

```
Transition target: 300 worsened / 2399 (12.5%)
Training CatBoost...

Results  (n_test=2,399):
  Accuracy:  0.8462
  Precision: 0.4314
  Recall:    0.7233
  F1 Score:  0.5405

Confusion Matrix:
  [[1813  286]
[  83  217]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training CatBoost...

Results  (n_test=2,490):
  Accuracy:  0.8594
  Precision: 0.5261
  Recall:    0.8215
  F1 Score:  0.6414

Confusion Matrix:
  [[1827  282]
[  68  313]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.7992
  Precision: 0.4727
  Recall:    0.7794
  F1 Score:  0.5885

Confusion Matrix:
  [[1662  406]
[ 103  364]]

── Eval 9: Assessment period 202110 ──
Training samples:   12,950
Test samples:       2,535  (admin2 areas in 202110)
Transition target: 144 worsened / 2535 (5.7%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.8024
  Precision: 0.1547
  Recall:    0.5556
  F1 Score:  0.2421
```

```
 Confusion Matrix:
   [[1954  437]
[  64   80]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.8465
   Precision: 0.6060
   Recall:    0.9089
   F1 Score:  0.7271

 Confusion Matrix:
   [[1598  331]
[  51  509]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7557
   Precision: 0.2749
   Recall:    0.8685
   F1 Score:  0.4176

 Confusion Matrix:
   [[1663  575]
[  33  218]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7447
   Precision: 0.2996
   Recall:    0.8428
   F1 Score:  0.4421

 Confusion Matrix:
   [[1603  589]
[  47  252]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
```

```
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7519
   Precision: 0.3327
   Recall:    0.8451
   F1 Score:  0.4775

 Confusion Matrix:
   [[1014  361]
[  33  180]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.6362
   Precision: 0.1096
   Recall:    0.7901
   F1 Score:  0.1925

 Confusion Matrix:
   [[875 520]
[ 17  64]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7021
   Precision: 0.1716
   Recall:    0.6456
   F1 Score:  0.2712

 Confusion Matrix:
   [[1552  642]
[  73  133]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7604
   Precision: 0.4162
   Recall:    0.8218
   F1 Score:  0.5525
```

```
 Confusion Matrix:
   [[1470  498]
[  77  355]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.7641
   Precision: 0.1836
   Recall:    0.9032
   F1 Score:  0.3052

 Confusion Matrix:
   [[770 249]
[  6  56]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.6818
   Precision: 0.0937
   Recall:    0.7083
   F1 Score:  0.1655

 Confusion Matrix:
   [[701 329]
[ 14  34]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.7814
   Precision: 0.3594
   Recall:    0.8707
   F1 Score:  0.5088

 Confusion Matrix:
   [[1629  492]
[  41  276]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
```

```
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.7399
   Precision: 0.1682
   Recall:    0.9130
   F1 Score:  0.2841

 Confusion Matrix:
   [[1680  623]
[  12  126]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.7757
   Precision: 0.1821
   Recall:    0.8079
   F1 Score:  0.2972

 Confusion Matrix:
   [[1874  548]
[  29  122]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7737
   Precision: 0.4408
   Recall:    0.7588
   F1 Score:  0.5577

 Confusion Matrix:
   [[1433  411]
[ 103  324]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training CatBoost...

 Results  (n_test=2,447):
   Accuracy:  0.7425
   Precision: 0.3056
   Recall:    0.8980
   F1 Score:  0.4560
```

```
  Confusion Matrix:
    [[1553  600]
 [  30  264]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training CatBoost...

  Results  (n_test=2,633):
    Accuracy:  0.8040
    Precision: 0.1175
    Recall:    0.5943
    F1 Score:  0.1963

  Confusion Matrix:
    [[2054  473]
 [  43   63]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.9089
    Precision: 0.5660
    Recall:    0.9579
    F1 Score:  0.7115

  Confusion Matrix:
    [[2099  227]
 [  13  296]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: GDELT only
======================================================================
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6983
    Precision: 0.4935
    Recall:    0.8060
    F1 Score:  0.6122

  Confusion Matrix:
    [[883 469]
```

```
[110 457]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.6509
   Precision: 0.3228
   Recall:    0.9219
   F1 Score:  0.4782

 Confusion Matrix:
   [[942 644]
[ 26 307]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.3882
   Precision: 0.0494
   Recall:    1.0000
   F1 Score:  0.0941

 Confusion Matrix:
   [[ 684 1174]
[   0   61]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.5865
   Precision: 0.4195
   Recall:    0.9332
   F1 Score:  0.5788

 Confusion Matrix:
   [[416 541]
[ 28 391]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...
```

```
 Results  (n_test=1,376):
   Accuracy:  0.4273
   Precision: 0.1338
   Recall:    0.9160
   F1 Score:  0.2335

 Confusion Matrix:
   [[468 777]
[ 11 120]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.5498
   Precision: 0.1793
   Recall:    0.7267
   F1 Score:  0.2876

 Confusion Matrix:
   [[1101  998]
[  82  218]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training CatBoost...

 Results  (n_test=2,490):
   Accuracy:  0.6141
   Precision: 0.2684
   Recall:    0.8819
   F1 Score:  0.4115

 Confusion Matrix:
   [[1193  916]
[  45  336]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7645
   Precision: 0.4281
   Recall:    0.8287
   F1 Score:  0.5646

 Confusion Matrix:
   [[1551  517]
```

```
[  80  387]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.5748
   Precision: 0.0619
   Recall:    0.4583
   F1 Score:  0.1091

 Confusion Matrix:
   [[1391 1000]
[  78   66]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.6501
   Precision: 0.3859
   Recall:    0.9393
   F1 Score:  0.5471

 Confusion Matrix:
   [[1092  837]
[  34  526]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.5440
   Precision: 0.1441
   Recall:    0.7131
   F1 Score:  0.2398

 Confusion Matrix:
   [[1175 1063]
[  72  179]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...
```

```
 Results  (n_test=2,491):
   Accuracy:  0.6042
   Precision: 0.1936
   Recall:    0.7258
   F1 Score:  0.3056

 Confusion Matrix:
   [[1288  904]
[  82  217]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7053
   Precision: 0.3005
   Recall:    0.9014
   F1 Score:  0.4507

 Confusion Matrix:
   [[928 447]
[ 21 192]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.5474
   Precision: 0.0929
   Recall:    0.8272
   F1 Score:  0.1671

 Confusion Matrix:
   [[741 654]
[ 14  67]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.5546
   Precision: 0.1164
   Recall:    0.6359
   F1 Score:  0.1968

 Confusion Matrix:
   [[1200  994]
```

```
[  75  131]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6979
   Precision: 0.3548
   Recall:    0.8287
   F1 Score:  0.4969

 Confusion Matrix:
   [[1317  651]
[  74  358]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.6947
   Precision: 0.1378
   Recall:    0.8226
   F1 Score:  0.2361

 Confusion Matrix:
   [[700 319]
[ 11  51]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.5028
   Precision: 0.0448
   Recall:    0.5000
   F1 Score:  0.0822

 Confusion Matrix:
   [[518 512]
[ 24  24]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...
```

```
 Results  (n_test=2,438):
   Accuracy:  0.7034
   Precision: 0.2894
   Recall:    0.8801
   F1 Score:  0.4356

 Confusion Matrix:
   [[1436  685]
[  38  279]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.7124
   Precision: 0.1457
   Recall:    0.8406
   F1 Score:  0.2484

 Confusion Matrix:
   [[1623  680]
[  22  116]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.7159
   Precision: 0.1556
   Recall:    0.8675
   F1 Score:  0.2638

 Confusion Matrix:
   [[1711  711]
[  20  131]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7486
   Precision: 0.4133
   Recall:    0.8033
   F1 Score:  0.5457

 Confusion Matrix:
   [[1357  487]
```

```
 [  84  343]]

  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Transition target: 294 worsened / 2447 (12.0%)
  Training CatBoost...

  Results  (n_test=2,447):
    Accuracy:  0.6257
    Precision: 0.2328
    Recall:    0.9218
    F1 Score:  0.3717

  Confusion Matrix:
    [[1260  893]
 [  23  271]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training CatBoost...

  Results  (n_test=2,633):
    Accuracy:  0.7520
    Precision: 0.0900
    Recall:    0.5660
    F1 Score:  0.1552

  Confusion Matrix:
    [[1920  607]
 [  46   60]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8763
    Precision: 0.4859
    Recall:    0.9482
    F1 Score:  0.6425

  Confusion Matrix:
    [[2016  310]
 [  16  293]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  ── Eval 1: Assessment period 202406 ──
```

```
 Training samples:   11,437
 Test samples:       1,919  (admin2 areas in 202406)
 Transition target: 567 worsened / 1919 (29.5%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.6681
   Precision: 0.4556
   Recall:    0.6332
   F1 Score:  0.5299

 Confusion Matrix:
   [[923 429]
[208 359]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.5638
   Precision: 0.2402
   Recall:    0.6997
   F1 Score:  0.3576

 Confusion Matrix:
   [[849 737]
[100 233]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.7134
   Precision: 0.0080
   Recall:    0.0656
   F1 Score:  0.0143

 Confusion Matrix:
   [[1365  493]
[  57    4]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7682
   Precision: 0.5862
```

```
  Recall:    0.8115
  F1 Score:  0.6807

 Confusion Matrix:
   [[717 240]
[ 79 340]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.6926
   Precision: 0.1958
   Recall:    0.7176
   F1 Score:  0.3077

 Confusion Matrix:
   [[859 386]
[ 37  94]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.6807
   Precision: 0.2109
   Recall:    0.5667
   F1 Score:  0.3074

 Confusion Matrix:
   [[1463  636]
[ 130  170]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training CatBoost...

 Results  (n_test=2,490):
   Accuracy:  0.6506
   Precision: 0.2237
   Recall:    0.5197
   F1 Score:  0.3128

 Confusion Matrix:
   [[1422  687]
[ 183  198]]

 ── Eval 8: Assessment period 202202 ──
```

```
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.5893
   Precision: 0.2433
   Recall:    0.5824
   F1 Score:  0.3432

 Confusion Matrix:
   [[1222  846]
[ 195  272]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.6517
   Precision: 0.0796
   Recall:    0.4861
   F1 Score:  0.1369

 Confusion Matrix:
   [[1582  809]
[  74   70]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.6472
   Precision: 0.3709
   Recall:    0.8161
   F1 Score:  0.5100

 Confusion Matrix:
   [[1154  775]
[ 103  457]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.2812
   Precision: 0.1052
```

```
  Recall:    0.8167
  F1 Score:  0.1864

 Confusion Matrix:
   [[ 495 1743]
[  46  205]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.3452
   Precision: 0.1546
   Recall:    0.9967
   F1 Score:  0.2676

 Confusion Matrix:
   [[ 562 1630]
[   1  298]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.3992
   Precision: 0.1718
   Recall:    0.9108
   F1 Score:  0.2891

 Confusion Matrix:
   [[440 935]
[ 19 194]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.4539
   Precision: 0.0780
   Recall:    0.8272
   F1 Score:  0.1426

 Confusion Matrix:
   [[603 792]
[ 14  67]]

 —— Eval 15: Assessment period 201910 ——
```

```
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.3554
   Precision: 0.1122
   Recall:    0.9417
   F1 Score:  0.2005

 Confusion Matrix:
   [[ 659 1535]
[  12  194]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.3958
   Precision: 0.2234
   Recall:    0.9514
   F1 Score:  0.3618

 Confusion Matrix:
   [[ 539 1429]
[  21  411]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.1684
   Precision: 0.0485
   Recall:    0.7258
   F1 Score:  0.0910

 Confusion Matrix:
   [[137 882]
[ 17  45]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.1920
   Precision: 0.0453
```

```
  Recall:    0.8542
  F1 Score:  0.0860

 Confusion Matrix:
   [[166 864]
[  7  41]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.7818
   Precision: 0.1989
   Recall:    0.2240
   F1 Score:  0.2107

 Confusion Matrix:
   [[1835  286]
[ 246   71]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8370
   Precision: 0.1467
   Recall:    0.3913
   F1 Score:  0.2134

 Confusion Matrix:
   [[1989  314]
[  84   54]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.7594
   Precision: 0.0320
   Recall:    0.1060
   F1 Score:  0.0492

 Confusion Matrix:
   [[1938  484]
[ 135   16]]

 —— Eval 22: Assessment period 201706 ——
```

```
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7274
   Precision: 0.2736
   Recall:    0.2717
   F1 Score:  0.2726

 Confusion Matrix:
   [[1536  308]
[ 311  116]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training CatBoost...

 Results  (n_test=2,447):
   Accuracy:  0.8022
   Precision: 0.2780
   Recall:    0.4048
   F1 Score:  0.3296

 Confusion Matrix:
   [[1844  309]
[ 175  119]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training CatBoost...

 Results  (n_test=2,633):
   Accuracy:  0.8401
   Precision: 0.0661
   Recall:    0.2264
   F1 Score:  0.1023

 Confusion Matrix:
   [[2188  339]
[  82   24]]

 —— Eval 25: Assessment period 201606 ——
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Transition target: 309 worsened / 2635 (11.7%)
 Training CatBoost...

 Results  (n_test=2,635):
   Accuracy:  0.8364
   Precision: 0.3498
```

```
    Recall:    0.4595
    F1 Score:  0.3972

  Confusion Matrix:
    [[2062  264]
 [ 167  142]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6644
    Precision: 0.4517
    Recall:    0.6349
    F1 Score:  0.5279

  Confusion Matrix:
    [[915 437]
 [207 360]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.5586
    Precision: 0.2404
    Recall:    0.7147
    F1 Score:  0.3598

  Confusion Matrix:
    [[834 752]
 [ 95 238]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Transition target: 61 worsened / 1919 (3.2%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.7024
    Precision: 0.0115
    Recall:    0.0984
    F1 Score:  0.0206
```

```
 Confusion Matrix:
   [[1342  516]
[  55    6]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7398
   Precision: 0.5482
   Recall:    0.8282
   F1 Score:  0.6597

 Confusion Matrix:
   [[671 286]
[ 72 347]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.6548
   Precision: 0.1767
   Recall:    0.7176
   F1 Score:  0.2836

 Confusion Matrix:
   [[807 438]
[ 37  94]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.6728
   Precision: 0.2150
   Recall:    0.6100
   F1 Score:  0.3180

 Confusion Matrix:
   [[1431  668]
[ 117  183]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
```

```
 Training CatBoost...

 Results  (n_test=2,490):
   Accuracy:  0.6466
   Precision: 0.2237
   Recall:    0.5302
   F1 Score:  0.3146

 Confusion Matrix:
   [[1408  701]
[ 179  202]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.5799
   Precision: 0.2427
   Recall:    0.6039
   F1 Score:  0.3462

 Confusion Matrix:
   [[1188  880]
[ 185  282]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.6454
   Precision: 0.0773
   Recall:    0.4792
   F1 Score:  0.1331

 Confusion Matrix:
   [[1567  824]
[  75   69]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.6436
   Precision: 0.3691
   Recall:    0.8232
   F1 Score:  0.5097
```

```
 Confusion Matrix:
   [[1141  788]
[  99  461]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.2981
   Precision: 0.1051
   Recall:    0.7928
   F1 Score:  0.1855

 Confusion Matrix:
   [[ 543 1695]
[  52  199]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.3593
   Precision: 0.1556
   Recall:    0.9799
   F1 Score:  0.2686

 Confusion Matrix:
   [[ 602 1590]
[   6  293]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.4049
   Precision: 0.1685
   Recall:    0.8732
   F1 Score:  0.2825

 Confusion Matrix:
   [[457 918]
[ 27 186]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
```

```
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.4627
   Precision: 0.0772
   Recall:    0.8025
   F1 Score:  0.1408

 Confusion Matrix:
   [[618 777]
[ 16  65]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.3783
   Precision: 0.1140
   Recall:    0.9223
   F1 Score:  0.2030

 Confusion Matrix:
   [[ 718 1476]
[  16  190]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.4250
   Precision: 0.2270
   Recall:    0.9120
   F1 Score:  0.3635

 Confusion Matrix:
   [[ 626 1342]
[  38  394]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.2054
   Precision: 0.0446
   Recall:    0.6290
   F1 Score:  0.0832
```

```
 Confusion Matrix:
   [[183 836]
[ 23  39]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.2236
   Precision: 0.0418
   Recall:    0.7500
   F1 Score:  0.0792

 Confusion Matrix:
   [[205 825]
[ 12  36]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.4020
   Precision: 0.1689
   Recall:    0.9180
   F1 Score:  0.2853

 Confusion Matrix:
   [[ 689 1432]
[  26  291]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8320
   Precision: 0.1402
   Recall:    0.3841
   F1 Score:  0.2054

 Confusion Matrix:
   [[1978  325]
[  85   53]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
```

```
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.7474
   Precision: 0.0354
   Recall:    0.1258
   F1 Score:  0.0552

 Confusion Matrix:
   [[1904  518]
[ 132   19]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7195
   Precision: 0.2677
   Recall:    0.2834
   F1 Score:  0.2753

 Confusion Matrix:
   [[1513  331]
[ 306  121]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training CatBoost...

 Results  (n_test=2,447):
   Accuracy:  0.7924
   Precision: 0.2694
   Recall:    0.4252
   F1 Score:  0.3298

 Confusion Matrix:
   [[1814  339]
[ 169  125]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Transition target: 106 worsened / 2633 (4.0%)
 Training CatBoost...

 Results  (n_test=2,633):
   Accuracy:  0.8283
   Precision: 0.0697
   Recall:    0.2642
   F1 Score:  0.1102
```

```
  Confusion Matrix:
    [[2153  374]
 [  78   28]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8319
    Precision: 0.3442
    Recall:    0.4790
    F1 Score:  0.4005

  Confusion Matrix:
    [[2044  282]
 [ 161  148]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6686
    Precision: 0.4568
    Recall:    0.6437
    F1 Score:  0.5344

  Confusion Matrix:
    [[918 434]
 [202 365]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Transition target: 333 worsened / 1919 (17.4%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.5586
    Precision: 0.2414
    Recall:    0.7207
    F1 Score:  0.3617

  Confusion Matrix:
    [[832 754]
 [ 93 240]]
```

```
── Eval 3: Assessment period 202310 ──
Training samples:   12,645
Test samples:       1,919  (admin2 areas in 202310)
Transition target: 61 worsened / 1919 (3.2%)
Training CatBoost...

Results  (n_test=1,919):
  Accuracy:  0.6905
  Precision: 0.0181
  Recall:    0.1639
  F1 Score:  0.0326

Confusion Matrix:
  [[1315  543]
[  51   10]]

── Eval 4: Assessment period 202306 ──
Training samples:   12,765
Test samples:       2,613  (admin2 areas in 202306)
Transition target: 419 worsened / 1376 (30.5%)
Training CatBoost...

Results  (n_test=1,376):
  Accuracy:  0.7362
  Precision: 0.5439
  Recall:    0.8282
  F1 Score:  0.6566

Confusion Matrix:
  [[666 291]
[ 72 347]]

── Eval 5: Assessment period 202302 ──
Training samples:   13,772
Test samples:       1,376  (admin2 areas in 202302)
Transition target: 131 worsened / 1376 (9.5%)
Training CatBoost...

Results  (n_test=1,376):
  Accuracy:  0.6403
  Precision: 0.1807
  Recall:    0.7863
  F1 Score:  0.2939

Confusion Matrix:
  [[778 467]
[ 28 103]]

── Eval 6: Assessment period 202210 ──
Training samples:   13,866
Test samples:       2,399  (admin2 areas in 202210)
Transition target: 300 worsened / 2399 (12.5%)
Training CatBoost...

Results  (n_test=2,399):
```

```
  Accuracy:  0.6678
  Precision: 0.2173
  Recall:    0.6367
  F1 Score:  0.3240

Confusion Matrix:
  [[1411  688]
[ 109  191]]

── Eval 7: Assessment period 202206 ──
Training samples:   13,795
Test samples:       2,568  (admin2 areas in 202206)
Transition target: 381 worsened / 2490 (15.3%)
Training CatBoost...

Results  (n_test=2,490):
  Accuracy:  0.6414
  Precision: 0.2247
  Recall:    0.5486
  F1 Score:  0.3188

Confusion Matrix:
  [[1388  721]
[ 172  209]]

── Eval 8: Assessment period 202202 ──
Training samples:   13,862
Test samples:       2,535  (admin2 areas in 202202)
Transition target: 467 worsened / 2535 (18.4%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.5775
  Precision: 0.2483
  Recall:    0.6381
  F1 Score:  0.3575

Confusion Matrix:
  [[1166  902]
[ 169  298]]

── Eval 9: Assessment period 202110 ──
Training samples:   12,950
Test samples:       2,535  (admin2 areas in 202110)
Transition target: 144 worsened / 2535 (5.7%)
Training CatBoost...

Results  (n_test=2,535):
  Accuracy:  0.6308
  Precision: 0.0778
  Recall:    0.5069
  F1 Score:  0.1349

Confusion Matrix:
  [[1526  865]
[  71   73]]
```

```
── Eval 10: Assessment period 202106 ──
Training samples:   12,859
Test samples:       2,536  (admin2 areas in 202106)
Transition target: 560 worsened / 2489 (22.5%)
Training CatBoost...

Results  (n_test=2,489):
  Accuracy:  0.6388
  Precision: 0.3652
  Recall:    0.8196
  F1 Score:  0.5052

Confusion Matrix:
  [[1131  798]
[ 101  459]]

── Eval 11: Assessment period 202102 ──
Training samples:   12,839
Test samples:       2,489  (admin2 areas in 202102)
Transition target: 251 worsened / 2489 (10.1%)
Training CatBoost...

Results  (n_test=2,489):
  Accuracy:  0.3122
  Precision: 0.1062
  Recall:    0.7849
  F1 Score:  0.1871

Confusion Matrix:
  [[ 580 1658]
[  54  197]]

── Eval 12: Assessment period 202010 ──
Training samples:   12,858
Test samples:       2,491  (admin2 areas in 202010)
Transition target: 299 worsened / 2491 (12.0%)
Training CatBoost...

Results  (n_test=2,491):
  Accuracy:  0.3737
  Precision: 0.1541
  Recall:    0.9398
  F1 Score:  0.2648

Confusion Matrix:
  [[ 650 1542]
[  18  281]]

── Eval 13: Assessment period 202006 ──
Training samples:   11,506
Test samples:       2,570  (admin2 areas in 202006)
Transition target: 213 worsened / 1588 (13.4%)
Training CatBoost...

Results  (n_test=1,588):
```

```
  Accuracy:  0.4427
  Precision: 0.1788
  Recall:    0.8779
  F1 Score:  0.2971

 Confusion Matrix:
   [[516 859]
[ 26 187]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.4837
   Precision: 0.0781
   Recall:    0.7778
   F1 Score:  0.1419

 Confusion Matrix:
   [[651 744]
[ 18  63]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.4188
   Precision: 0.1157
   Recall:    0.8689
   F1 Score:  0.2042

 Confusion Matrix:
   [[ 826 1368]
[  27  179]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.4608
   Precision: 0.2353
   Recall:    0.8866
   F1 Score:  0.3718

 Confusion Matrix:
   [[ 723 1245]
[  49  383]]
```

```
── Eval 17: Assessment period 201902 ──
Training samples:   12,785
Test samples:       2,400  (admin2 areas in 201902)
Transition target: 62 worsened / 1081 (5.7%)
Training CatBoost...

Results  (n_test=1,081):
  Accuracy:  0.2266
  Precision: 0.0458
  Recall:    0.6290
  F1 Score:  0.0853

Confusion Matrix:
  [[206 813]
[ 23  39]]

── Eval 18: Assessment period 201810 ──
Training samples:   13,957
Test samples:       1,081  (admin2 areas in 201810)
Transition target: 48 worsened / 1078 (4.5%)
Training CatBoost...

Results  (n_test=1,078):
  Accuracy:  0.2681
  Precision: 0.0432
  Recall:    0.7292
  F1 Score:  0.0815

Confusion Matrix:
  [[254 776]
[ 13  35]]

── Eval 19: Assessment period 201806 ──
Training samples:   14,374
Test samples:       2,440  (admin2 areas in 201806)
Transition target: 317 worsened / 2438 (13.0%)
Training CatBoost...

Results  (n_test=2,438):
  Accuracy:  0.4586
  Precision: 0.1791
  Recall:    0.8833
  F1 Score:  0.2979

Confusion Matrix:
  [[ 838 1283]
[  37  280]]

── Eval 20: Assessment period 201802 ──
Training samples:   14,614
Test samples:       2,441  (admin2 areas in 201802)
Transition target: 138 worsened / 2441 (5.7%)
Training CatBoost...

Results  (n_test=2,441):
```

```
  Accuracy:  0.7866
  Precision: 0.1147
  Recall:    0.4130
  F1 Score:  0.1795

 Confusion Matrix:
  [[1863  440]
[  81   57]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
  Accuracy:  0.7311
  Precision: 0.0529
  Recall:    0.2119
  F1 Score:  0.0847

 Confusion Matrix:
  [[1849  573]
[ 119   32]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
  Accuracy:  0.7160
  Precision: 0.2739
  Recall:    0.3091
  F1 Score:  0.2904

 Confusion Matrix:
  [[1494  350]
[ 295  132]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Transition target: 294 worsened / 2447 (12.0%)
 Training CatBoost...

 Results  (n_test=2,447):
  Accuracy:  0.7765
  Precision: 0.2544
  Recall:    0.4456
  F1 Score:  0.3239

 Confusion Matrix:
  [[1769  384]
[ 163  131]]
```

```
  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training CatBoost...

  Results  (n_test=2,633):
    Accuracy:  0.8299
    Precision: 0.0704
    Recall:    0.2642
    F1 Score:  0.1111

  Confusion Matrix:
    [[2157  370]
 [  78   28]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8315
    Precision: 0.3441
    Recall:    0.4822
    F1 Score:  0.4016

  Confusion Matrix:
    [[2042  284]
 [ 160  149]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Transition target: 567 worsened / 1919 (29.5%)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6727
    Precision: 0.3082
    Recall:    0.0864
    F1 Score:  0.1350

  Confusion Matrix:
    [[1242  110]
 [ 518   49]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
```

```
 Test samples:       2,376  (admin2 areas in 202402)
 Transition target: 333 worsened / 1919 (17.4%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.7556
   Precision: 0.1344
   Recall:    0.0751
   F1 Score:  0.0963

 Confusion Matrix:
   [[1425  161]
[ 308   25]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Transition target: 61 worsened / 1919 (3.2%)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.8953
   Precision: 0.0732
   Recall:    0.1967
   F1 Score:  0.1067

 Confusion Matrix:
   [[1706  152]
[  49   12]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Transition target: 419 worsened / 1376 (30.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.6519
   Precision: 0.3235
   Recall:    0.1313
   F1 Score:  0.1868

 Confusion Matrix:
   [[842 115]
[364  55]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Transition target: 131 worsened / 1376 (9.5%)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.8060
   Precision: 0.1000
   Recall:    0.1298
```

```
  F1 Score:  0.1130

 Confusion Matrix:
   [[1092  153]
[ 114   17]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Transition target: 300 worsened / 2399 (12.5%)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.8195
   Precision: 0.2308
   Recall:    0.1900
   F1 Score:  0.2084

 Confusion Matrix:
   [[1909  190]
[ 243   57]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Transition target: 381 worsened / 2490 (15.3%)
 Training CatBoost...

 Results  (n_test=2,490):
   Accuracy:  0.7791
   Precision: 0.1712
   Recall:    0.1155
   F1 Score:  0.1379

 Confusion Matrix:
   [[1896  213]
[ 337   44]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Transition target: 467 worsened / 2535 (18.4%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7594
   Precision: 0.2671
   Recall:    0.1756
   F1 Score:  0.2119

 Confusion Matrix:
   [[1843  225]
[ 385   82]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
```

```
 Test samples:       2,535  (admin2 areas in 202110)
 Transition target: 144 worsened / 2535 (5.7%)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.8667
   Precision: 0.0855
   Recall:    0.1389
   F1 Score:  0.1058

 Confusion Matrix:
   [[2177  214]
[ 124   20]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Transition target: 560 worsened / 2489 (22.5%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7192
   Precision: 0.2473
   Recall:    0.1214
   F1 Score:  0.1629

 Confusion Matrix:
   [[1722  207]
[ 492   68]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Transition target: 251 worsened / 2489 (10.1%)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.8216
   Precision: 0.1303
   Recall:    0.1355
   F1 Score:  0.1328

 Confusion Matrix:
   [[2011  227]
[ 217   34]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Transition target: 299 worsened / 2491 (12.0%)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.8009
   Precision: 0.1365
   Recall:    0.1237
```

```
  F1 Score:  0.1298

 Confusion Matrix:
   [[1958  234]
[ 262   37]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Transition target: 213 worsened / 1588 (13.4%)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.8079
   Precision: 0.2051
   Recall:    0.1502
   F1 Score:  0.1734

 Confusion Matrix:
   [[1251  124]
[ 181   32]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Transition target: 81 worsened / 1476 (5.5%)
 Training CatBoost...

 Results  (n_test=1,476):
   Accuracy:  0.7988
   Precision: 0.0680
   Recall:    0.2099
   F1 Score:  0.1027

 Confusion Matrix:
   [[1162  233]
[  64   17]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Transition target: 206 worsened / 2400 (8.6%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7992
   Precision: 0.1310
   Recall:    0.2379
   F1 Score:  0.1690

 Confusion Matrix:
   [[1869  325]
[ 157   49]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
```

```
 Test samples:       2,400  (admin2 areas in 201906)
 Transition target: 432 worsened / 2400 (18.0%)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7500
   Precision: 0.2290
   Recall:    0.1644
   F1 Score:  0.1914

 Confusion Matrix:
   [[1729  239]
[ 361   71]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Transition target: 62 worsened / 1081 (5.7%)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8150
   Precision: 0.0577
   Recall:    0.1452
   F1 Score:  0.0826

 Confusion Matrix:
   [[872 147]
[ 53   9]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Transition target: 48 worsened / 1078 (4.5%)
 Training CatBoost...

 Results  (n_test=1,078):
   Accuracy:  0.1976
   Precision: 0.0446
   Recall:    0.8333
   F1 Score:  0.0847

 Confusion Matrix:
   [[173 857]
[  8  40]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Transition target: 317 worsened / 2438 (13.0%)
 Training CatBoost...

 Results  (n_test=2,438):
   Accuracy:  0.3019
   Precision: 0.1447
   Recall:    0.8896
```

```
  F1 Score:  0.2489

 Confusion Matrix:
   [[ 454 1667]
[  35  282]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Transition target: 138 worsened / 2441 (5.7%)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8365
   Precision: 0.0721
   Recall:    0.1594
   F1 Score:  0.0993

 Confusion Matrix:
   [[2020  283]
[ 116   22]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Transition target: 151 worsened / 2573 (5.9%)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.8263
   Precision: 0.0866
   Recall:    0.2053
   F1 Score:  0.1218

 Confusion Matrix:
   [[2095  327]
[ 120   31]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Transition target: 427 worsened / 2271 (18.8%)
 Training CatBoost...

 Results  (n_test=2,271):
   Accuracy:  0.7428
   Precision: 0.2060
   Recall:    0.1288
   F1 Score:  0.1585

 Confusion Matrix:
   [[1632  212]
[ 372   55]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
```

```
  Test samples:       2,449  (admin2 areas in 201702)
  Transition target: 294 worsened / 2447 (12.0%)
  Training CatBoost...

  Results  (n_test=2,447):
    Accuracy:  0.8124
    Precision: 0.1687
    Recall:    0.1429
    F1 Score:  0.1547

  Confusion Matrix:
    [[1946  207]
 [ 252   42]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Transition target: 106 worsened / 2633 (4.0%)
  Training CatBoost...

  Results  (n_test=2,633):
    Accuracy:  0.9081
    Precision: 0.0952
    Recall:    0.1509
    F1 Score:  0.1168

  Confusion Matrix:
    [[2375  152]
 [  90   16]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Transition target: 309 worsened / 2635 (11.7%)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8573
    Precision: 0.2403
    Recall:    0.1003
    F1 Score:  0.1416

  Confusion Matrix:
    [[2228   98]
 [ 278   31]]

======================================================================
SUMMARY: TRANSITION PREDICTION — AVERAGE RESULTS BY MODEL & FEATURE SET
======================================================================
(Precision/Recall/F1 are for the WORSENED class)

Logistic Regression — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.9135
    Precision (W):  0.5923
```

```
    Recall (W):     0.9952
    F1 (worsen):    0.7181

  Per assessment:
    202406:  n=1,919  acc=0.965  f1=0.943
    202402:  n=1,919  acc=0.796  f1=0.629
    202310:  n=1,919  acc=0.929  f1=0.473
    202306:  n=1,376  acc=0.903  f1=0.862
    202302:  n=1,376  acc=0.823  f1=0.518
    202210:  n=2,399  acc=0.940  f1=0.803
    202206:  n=2,490  acc=0.938  f1=0.831
    202202:  n=2,535  acc=0.886  f1=0.763
    202110:  n=2,535  acc=0.910  f1=0.559
    202106:  n=2,489  acc=0.929  f1=0.863
    202102:  n=2,489  acc=0.894  f1=0.653
    202010:  n=2,491  acc=0.943  f1=0.806
    202006:  n=1,588  acc=0.872  f1=0.675
    202002:  n=1,476  acc=0.911  f1=0.550
    201910:  n=2,400  acc=0.951  f1=0.778
    201906:  n=2,400  acc=0.945  f1=0.868
    201902:  n=1,081  acc=0.929  f1=0.617
    201810:  n=1,078  acc=0.843  f1=0.362
    201806:  n=2,438  acc=0.873  f1=0.672
    201802:  n=2,441  acc=0.821  f1=0.387
    201710:  n=2,573  acc=0.925  f1=0.609
    201706:  n=2,271  acc=0.967  f1=0.918
    201702:  n=2,447  acc=0.945  f1=0.814
    201610:  n=2,633  acc=1.000  f1=1.000
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC + GDELT base:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.9124
    Precision (W):  0.5860
    Recall (W):     0.9711
    F1 (worsen):    0.7076

  Per assessment:
    202406:  n=1,919  acc=0.964  f1=0.940
    202402:  n=1,919  acc=0.798  f1=0.631
    202310:  n=1,919  acc=0.929  f1=0.469
    202306:  n=1,376  acc=0.902  f1=0.860
    202302:  n=1,376  acc=0.824  f1=0.520
    202210:  n=2,399  acc=0.939  f1=0.798
    202206:  n=2,490  acc=0.936  f1=0.823
    202202:  n=2,535  acc=0.886  f1=0.760
    202110:  n=2,535  acc=0.910  f1=0.550
    202106:  n=2,489  acc=0.927  f1=0.858
    202102:  n=2,489  acc=0.896  f1=0.656
    202010:  n=2,491  acc=0.941  f1=0.799
    202006:  n=1,588  acc=0.870  f1=0.668
    202002:  n=1,476  acc=0.913  f1=0.549
    201910:  n=2,400  acc=0.951  f1=0.775
    201906:  n=2,400  acc=0.943  f1=0.858
    201902:  n=1,081  acc=0.925  f1=0.593
```

```
    201810:  n=1,078  acc=0.841  f1=0.335
    201806:  n=2,438  acc=0.875  f1=0.673
    201802:  n=2,441  acc=0.823  f1=0.386
    201710:  n=2,573  acc=0.925  f1=0.604
    201706:  n=2,271  acc=0.961  f1=0.903
    201702:  n=2,447  acc=0.940  f1=0.794
    201610:  n=2,633  acc=0.991  f1=0.888
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC + GDELT + temporal:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.9130
    Precision (W):  0.5834
    Recall (W):     0.9590
    F1 (worsen):    0.7036

  Per assessment:
    202406:  n=1,919  acc=0.964  f1=0.941
    202402:  n=1,919  acc=0.807  f1=0.639
    202310:  n=1,919  acc=0.930  f1=0.460
    202306:  n=1,376  acc=0.903  f1=0.859
    202302:  n=1,376  acc=0.830  f1=0.526
    202210:  n=2,399  acc=0.935  f1=0.783
    202206:  n=2,490  acc=0.934  f1=0.816
    202202:  n=2,535  acc=0.887  f1=0.759
    202110:  n=2,535  acc=0.913  f1=0.561
    202106:  n=2,489  acc=0.926  f1=0.854
    202102:  n=2,489  acc=0.898  f1=0.658
    202010:  n=2,491  acc=0.942  f1=0.799
    202006:  n=1,588  acc=0.872  f1=0.670
    202002:  n=1,476  acc=0.912  f1=0.545
    201910:  n=2,400  acc=0.950  f1=0.765
    201906:  n=2,400  acc=0.943  f1=0.858
    201902:  n=1,081  acc=0.926  f1=0.596
    201810:  n=1,078  acc=0.845  f1=0.335
    201806:  n=2,438  acc=0.877  f1=0.677
    201802:  n=2,441  acc=0.830  f1=0.392
    201710:  n=2,573  acc=0.927  f1=0.610
    201706:  n=2,271  acc=0.959  f1=0.901
    201702:  n=2,447  acc=0.932  f1=0.767
    201610:  n=2,633  acc=0.985  f1=0.820
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — GDELT only:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7108
    Precision (W):  0.3148
    Recall (W):     0.9520
    F1 (worsen):    0.4448

  Per assessment:
    202406:  n=1,919  acc=0.718  f1=0.662
    202402:  n=1,919  acc=0.679  f1=0.512
    202310:  n=1,919  acc=0.525  f1=0.115
```

```
    202306:  n=1,376  acc=0.685  f1=0.652
    202302:  n=1,376  acc=0.534  f1=0.287
    202210:  n=2,399  acc=0.640  f1=0.395
    202206:  n=2,490  acc=0.653  f1=0.458
    202202:  n=2,535  acc=0.690  f1=0.537
    202110:  n=2,535  acc=0.592  f1=0.208
    202106:  n=2,489  acc=0.726  f1=0.613
    202102:  n=2,489  acc=0.675  f1=0.374
    202010:  n=2,491  acc=0.684  f1=0.420
    202006:  n=1,588  acc=0.758  f1=0.511
    202002:  n=1,476  acc=0.644  f1=0.229
    201910:  n=2,400  acc=0.701  f1=0.360
    201906:  n=2,400  acc=0.809  f1=0.643
    201902:  n=1,081  acc=0.771  f1=0.303
    201810:  n=1,078  acc=0.618  f1=0.179
    201806:  n=2,438  acc=0.740  f1=0.497
    201802:  n=2,441  acc=0.694  f1=0.261
    201710:  n=2,573  acc=0.679  f1=0.259
    201706:  n=2,271  acc=0.803  f1=0.650
    201702:  n=2,447  acc=0.867  f1=0.624
    201610:  n=2,633  acc=0.886  f1=0.372
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC only (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5891
    Precision (W):  0.1766
    Recall (W):     0.5491
    F1 (worsen):    0.2516

  Per assessment:
    202406:  n=1,919  acc=0.668  f1=0.530
    202402:  n=1,919  acc=0.564  f1=0.358
    202310:  n=1,919  acc=0.713  f1=0.014
    202306:  n=1,376  acc=0.768  f1=0.681
    202302:  n=1,376  acc=0.693  f1=0.308
    202210:  n=2,399  acc=0.681  f1=0.307
    202206:  n=2,490  acc=0.651  f1=0.313
    202202:  n=2,535  acc=0.589  f1=0.343
    202110:  n=2,535  acc=0.652  f1=0.137
    202106:  n=2,489  acc=0.647  f1=0.510
    202102:  n=2,489  acc=0.546  f1=0.186
    202010:  n=2,491  acc=0.581  f1=0.269
    202006:  n=1,588  acc=0.565  f1=0.216
    202002:  n=1,476  acc=0.664  f1=0.145
    201910:  n=2,400  acc=0.500  f1=0.134
    201906:  n=2,400  acc=0.450  f1=0.290
    201902:  n=1,081  acc=0.386  f1=0.059
    201810:  n=1,078  acc=0.442  f1=0.082
    201806:  n=2,438  acc=0.440  f1=0.253
    201802:  n=2,441  acc=0.381  f1=0.090
    201710:  n=2,573  acc=0.460  f1=0.141
    201706:  n=2,271  acc=0.627  f1=0.266
    201702:  n=2,447  acc=0.733  f1=0.288
    201610:  n=2,633  acc=0.631  f1=0.069
```

```
    201606:  n=2,635  acc=0.695  f1=0.300

Logistic Regression — IPC + GDELT base (no geo):
--------------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.6235
    Precision (W):  0.1719
    Recall (W):     0.4925
    F1 (worsen):    0.2342

  Per assessment:
    202406:  n=1,919  acc=0.658  f1=0.525
    202402:  n=1,919  acc=0.549  f1=0.355
    202310:  n=1,919  acc=0.689  f1=0.020
    202306:  n=1,376  acc=0.738  f1=0.657
    202302:  n=1,376  acc=0.657  f1=0.285
    202210:  n=2,399  acc=0.670  f1=0.315
    202206:  n=2,490  acc=0.643  f1=0.309
    202202:  n=2,535  acc=0.578  f1=0.342
    202110:  n=2,535  acc=0.642  f1=0.132
    202106:  n=2,489  acc=0.645  f1=0.511
    202102:  n=2,489  acc=0.543  f1=0.188
    202010:  n=2,491  acc=0.579  f1=0.273
    202006:  n=1,588  acc=0.559  f1=0.219
    202002:  n=1,476  acc=0.650  f1=0.134
    201910:  n=2,400  acc=0.507  f1=0.147
    201906:  n=2,400  acc=0.466  f1=0.300
    201902:  n=1,081  acc=0.389  f1=0.062
    201810:  n=1,078  acc=0.448  f1=0.075
    201806:  n=2,438  acc=0.466  f1=0.258
    201802:  n=2,441  acc=0.402  f1=0.100
    201710:  n=2,573  acc=0.896  f1=0.057
    201706:  n=2,271  acc=0.779  f1=0.084
    201702:  n=2,447  acc=0.829  f1=0.122
    201610:  n=2,633  acc=0.911  f1=0.086
    201606:  n=2,635  acc=0.696  f1=0.299

Logistic Regression — IPC + GDELT + temporal (no geo):
--------------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.6184
    Precision (W):  0.1746
    Recall (W):     0.5178
    F1 (worsen):    0.2451

  Per assessment:
    202406:  n=1,919  acc=0.661  f1=0.533
    202402:  n=1,919  acc=0.548  f1=0.358
    202310:  n=1,919  acc=0.683  f1=0.032
    202306:  n=1,376  acc=0.733  f1=0.654
    202302:  n=1,376  acc=0.643  f1=0.283
    202210:  n=2,399  acc=0.666  f1=0.317
    202206:  n=2,490  acc=0.637  f1=0.308
    202202:  n=2,535  acc=0.576  f1=0.347
    202110:  n=2,535  acc=0.639  f1=0.134
    202106:  n=2,489  acc=0.645  f1=0.512
```

```
    202102:  n=2,489  acc=0.538  f1=0.192
    202010:  n=2,491  acc=0.577  f1=0.269
    202006:  n=1,588  acc=0.566  f1=0.239
    202002:  n=1,476  acc=0.653  f1=0.144
    201910:  n=2,400  acc=0.511  f1=0.157
    201906:  n=2,400  acc=0.478  f1=0.305
    201902:  n=1,081  acc=0.392  f1=0.063
    201810:  n=1,078  acc=0.456  f1=0.070
    201806:  n=2,438  acc=0.494  f1=0.263
    201802:  n=2,441  acc=0.437  f1=0.115
    201710:  n=2,573  acc=0.837  f1=0.132
    201706:  n=2,271  acc=0.729  f1=0.163
    201702:  n=2,447  acc=0.796  f1=0.167
    201610:  n=2,633  acc=0.880  f1=0.086
    201606:  n=2,635  acc=0.683  f1=0.284

Logistic Regression — GDELT only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7361
    Precision (W):  0.1411
    Recall (W):     0.2123
    F1 (worsen):    0.1405

  Per assessment:
    202406:  n=1,919  acc=0.640  f1=0.198
    202402:  n=1,919  acc=0.738  f1=0.097
    202310:  n=1,919  acc=0.859  f1=0.081
    202306:  n=1,376  acc=0.630  f1=0.172
    202302:  n=1,376  acc=0.812  f1=0.116
    202210:  n=2,399  acc=0.822  f1=0.168
    202206:  n=2,490  acc=0.774  f1=0.099
    202202:  n=2,535  acc=0.743  f1=0.222
    202110:  n=2,535  acc=0.839  f1=0.102
    202106:  n=2,489  acc=0.693  f1=0.186
    202102:  n=2,489  acc=0.784  f1=0.118
    202010:  n=2,491  acc=0.770  f1=0.125
    202006:  n=1,588  acc=0.788  f1=0.184
    202002:  n=1,476  acc=0.796  f1=0.107
    201910:  n=2,400  acc=0.810  f1=0.139
    201906:  n=2,400  acc=0.733  f1=0.173
    201902:  n=1,081  acc=0.811  f1=0.105
    201810:  n=1,078  acc=0.182  f1=0.081
    201806:  n=2,438  acc=0.282  f1=0.245
    201802:  n=2,441  acc=0.819  f1=0.094
    201710:  n=2,573  acc=0.834  f1=0.138
    201706:  n=2,271  acc=0.731  f1=0.166
    201702:  n=2,447  acc=0.805  f1=0.176
    201610:  n=2,633  acc=0.880  f1=0.092
    201606:  n=2,635  acc=0.826  f1=0.126

Random Forest — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.6858
    Precision (W):  0.2602
```

```
    Recall (W):     0.8125
    F1 (worsen):    0.3756

  Per assessment:
    202406:  n=1,919  acc=0.761  f1=0.654
    202402:  n=1,919  acc=0.695  f1=0.501
    202310:  n=1,919  acc=0.734  f1=0.118
    202306:  n=1,376  acc=0.770  f1=0.690
    202302:  n=1,376  acc=0.517  f1=0.267
    202210:  n=2,399  acc=0.641  f1=0.324
    202206:  n=2,490  acc=0.663  f1=0.452
    202202:  n=2,535  acc=0.739  f1=0.554
    202110:  n=2,535  acc=0.536  f1=0.126
    202106:  n=2,489  acc=0.708  f1=0.591
    202102:  n=2,489  acc=0.575  f1=0.261
    202010:  n=2,491  acc=0.630  f1=0.343
    202006:  n=1,588  acc=0.720  f1=0.450
    202002:  n=1,476  acc=0.642  f1=0.177
    201910:  n=2,400  acc=0.655  f1=0.236
    201906:  n=2,400  acc=0.727  f1=0.530
    201902:  n=1,081  acc=0.810  f1=0.357
    201810:  n=1,078  acc=0.576  f1=0.152
    201806:  n=2,438  acc=0.744  f1=0.484
    201802:  n=2,441  acc=0.682  f1=0.260
    201710:  n=2,573  acc=0.639  f1=0.228
    201706:  n=2,271  acc=0.681  f1=0.468
    201702:  n=2,447  acc=0.704  f1=0.407
    201610:  n=2,633  acc=0.750  f1=0.163
    201606:  n=2,635  acc=0.846  f1=0.596

Random Forest — IPC + GDELT base:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7707
    Precision (W):  0.3090
    Recall (W):     0.6859
    F1 (worsen):    0.4039

  Per assessment:
    202406:  n=1,919  acc=0.822  f1=0.663
    202402:  n=1,919  acc=0.754  f1=0.551
    202310:  n=1,919  acc=0.745  f1=0.109
    202306:  n=1,376  acc=0.809  f1=0.724
    202302:  n=1,376  acc=0.737  f1=0.372
    202210:  n=2,399  acc=0.687  f1=0.350
    202206:  n=2,490  acc=0.707  f1=0.451
    202202:  n=2,535  acc=0.778  f1=0.576
    202110:  n=2,535  acc=0.756  f1=0.153
    202106:  n=2,489  acc=0.789  f1=0.642
    202102:  n=2,489  acc=0.754  f1=0.288
    202010:  n=2,491  acc=0.736  f1=0.296
    202006:  n=1,588  acc=0.858  f1=0.556
    202002:  n=1,476  acc=0.800  f1=0.230
    201910:  n=2,400  acc=0.728  f1=0.275
    201906:  n=2,400  acc=0.784  f1=0.561
    201902:  n=1,081  acc=0.886  f1=0.453
```

```
    201810:  n=1,078  acc=0.594  f1=0.131
    201806:  n=2,438  acc=0.736  f1=0.480
    201802:  n=2,441  acc=0.772  f1=0.322
    201710:  n=2,573  acc=0.779  f1=0.302
    201706:  n=2,271  acc=0.749  f1=0.501
    201702:  n=2,447  acc=0.776  f1=0.387
    201610:  n=2,633  acc=0.852  f1=0.118
    201606:  n=2,635  acc=0.880  f1=0.607

Random Forest — IPC + GDELT + temporal:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7786
    Precision (W):  0.3104
    Recall (W):     0.6426
    F1 (worsen):    0.3955

  Per assessment:
    202406:  n=1,919  acc=0.838  f1=0.688
    202402:  n=1,919  acc=0.760  f1=0.552
    202310:  n=1,919  acc=0.803  f1=0.108
    202306:  n=1,376  acc=0.823  f1=0.739
    202302:  n=1,376  acc=0.721  f1=0.364
    202210:  n=2,399  acc=0.709  f1=0.358
    202206:  n=2,490  acc=0.779  f1=0.456
    202202:  n=2,535  acc=0.782  f1=0.576
    202110:  n=2,535  acc=0.737  f1=0.128
    202106:  n=2,489  acc=0.789  f1=0.643
    202102:  n=2,489  acc=0.742  f1=0.228
    202010:  n=2,491  acc=0.730  f1=0.229
    202006:  n=1,588  acc=0.843  f1=0.497
    202002:  n=1,476  acc=0.809  f1=0.250
    201910:  n=2,400  acc=0.714  f1=0.268
    201906:  n=2,400  acc=0.798  f1=0.564
    201902:  n=1,081  acc=0.872  f1=0.378
    201810:  n=1,078  acc=0.590  f1=0.133
    201806:  n=2,438  acc=0.708  f1=0.455
    201802:  n=2,441  acc=0.821  f1=0.350
    201710:  n=2,573  acc=0.823  f1=0.345
    201706:  n=2,271  acc=0.742  f1=0.403
    201702:  n=2,447  acc=0.784  f1=0.353
    201610:  n=2,633  acc=0.848  f1=0.164
    201606:  n=2,635  acc=0.898  f1=0.656

Random Forest — GDELT only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7439
    Precision (W):  0.2701
    Recall (W):     0.6355
    F1 (worsen):    0.3630

  Per assessment:
    202406:  n=1,919  acc=0.721  f1=0.615
    202402:  n=1,919  acc=0.685  f1=0.475
    202310:  n=1,919  acc=0.745  f1=0.164
```

```
    202306:  n=1,376  acc=0.634  f1=0.592
    202302:  n=1,376  acc=0.807  f1=0.348
    202210:  n=2,399  acc=0.644  f1=0.302
    202206:  n=2,490  acc=0.659  f1=0.388
    202202:  n=2,535  acc=0.747  f1=0.518
    202110:  n=2,535  acc=0.653  f1=0.087
    202106:  n=2,489  acc=0.763  f1=0.602
    202102:  n=2,489  acc=0.778  f1=0.191
    202010:  n=2,491  acc=0.719  f1=0.191
    202006:  n=1,588  acc=0.807  f1=0.468
    202002:  n=1,476  acc=0.755  f1=0.214
    201910:  n=2,400  acc=0.693  f1=0.238
    201906:  n=2,400  acc=0.784  f1=0.521
    201902:  n=1,081  acc=0.874  f1=0.376
    201810:  n=1,078  acc=0.484  f1=0.120
    201806:  n=2,438  acc=0.720  f1=0.460
    201802:  n=2,441  acc=0.819  f1=0.314
    201710:  n=2,573  acc=0.804  f1=0.309
    201706:  n=2,271  acc=0.751  f1=0.412
    201702:  n=2,447  acc=0.812  f1=0.394
    201610:  n=2,633  acc=0.861  f1=0.160
    201606:  n=2,635  acc=0.880  f1=0.617

Random Forest — IPC only (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5920
    Precision (W):  0.1959
    Recall (W):     0.6003
    F1 (worsen):    0.2680

  Per assessment:
    202406:  n=1,919  acc=0.668  f1=0.530
    202402:  n=1,919  acc=0.564  f1=0.358
    202310:  n=1,919  acc=0.713  f1=0.014
    202306:  n=1,376  acc=0.768  f1=0.681
    202302:  n=1,376  acc=0.693  f1=0.308
    202210:  n=2,399  acc=0.681  f1=0.307
    202206:  n=2,490  acc=0.651  f1=0.313
    202202:  n=2,535  acc=0.589  f1=0.343
    202110:  n=2,535  acc=0.652  f1=0.137
    202106:  n=2,489  acc=0.647  f1=0.510
    202102:  n=2,489  acc=0.281  f1=0.186
    202010:  n=2,491  acc=0.345  f1=0.268
    202006:  n=1,588  acc=0.399  f1=0.289
    202002:  n=1,476  acc=0.454  f1=0.143
    201910:  n=2,400  acc=0.355  f1=0.201
    201906:  n=2,400  acc=0.396  f1=0.362
    201902:  n=1,081  acc=0.168  f1=0.091
    201810:  n=1,078  acc=0.192  f1=0.086
    201806:  n=2,438  acc=0.782  f1=0.211
    201802:  n=2,441  acc=0.837  f1=0.213
    201710:  n=2,573  acc=0.759  f1=0.049
    201706:  n=2,271  acc=0.727  f1=0.273
    201702:  n=2,447  acc=0.802  f1=0.330
    201610:  n=2,633  acc=0.840  f1=0.102
```

```
    201606:  n=2,635  acc=0.836  f1=0.397

Random Forest — IPC + GDELT base (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5744
    Precision (W):  0.1907
    Recall (W):     0.6037
    F1 (worsen):    0.2623

  Per assessment:
    202406:  n=1,919  acc=0.676  f1=0.528
    202402:  n=1,919  acc=0.589  f1=0.362
    202310:  n=1,919  acc=0.721  f1=0.015
    202306:  n=1,376  acc=0.761  f1=0.672
    202302:  n=1,376  acc=0.675  f1=0.285
    202210:  n=2,399  acc=0.684  f1=0.308
    202206:  n=2,490  acc=0.657  f1=0.308
    202202:  n=2,535  acc=0.596  f1=0.345
    202110:  n=2,535  acc=0.661  f1=0.126
    202106:  n=2,489  acc=0.652  f1=0.505
    202102:  n=2,489  acc=0.321  f1=0.182
    202010:  n=2,491  acc=0.379  f1=0.263
    202006:  n=1,588  acc=0.426  f1=0.278
    202002:  n=1,476  acc=0.490  f1=0.143
    201910:  n=2,400  acc=0.401  f1=0.197
    201906:  n=2,400  acc=0.439  f1=0.354
    201902:  n=1,081  acc=0.221  f1=0.079
    201810:  n=1,078  acc=0.245  f1=0.079
    201806:  n=2,438  acc=0.411  f1=0.280
    201802:  n=2,441  acc=0.346  f1=0.130
    201710:  n=2,573  acc=0.773  f1=0.049
    201706:  n=2,271  acc=0.734  f1=0.252
    201702:  n=2,447  acc=0.814  f1=0.324
    201610:  n=2,633  acc=0.850  f1=0.105
    201606:  n=2,635  acc=0.838  f1=0.390

Random Forest — IPC + GDELT + temporal (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.6222
    Precision (W):  0.1978
    Recall (W):     0.5689
    F1 (worsen):    0.2675

  Per assessment:
    202406:  n=1,919  acc=0.686  f1=0.521
    202402:  n=1,919  acc=0.610  f1=0.365
    202310:  n=1,919  acc=0.753  f1=0.013
    202306:  n=1,376  acc=0.762  f1=0.665
    202302:  n=1,376  acc=0.684  f1=0.279
    202210:  n=2,399  acc=0.693  f1=0.312
    202206:  n=2,490  acc=0.663  f1=0.311
    202202:  n=2,535  acc=0.596  f1=0.355
    202110:  n=2,535  acc=0.658  f1=0.129
    202106:  n=2,489  acc=0.653  f1=0.501
```

```
    202102:  n=2,489  acc=0.371  f1=0.182
    202010:  n=2,491  acc=0.415  f1=0.258
    202006:  n=1,588  acc=0.487  f1=0.278
    202002:  n=1,476  acc=0.549  f1=0.144
    201910:  n=2,400  acc=0.457  f1=0.203
    201906:  n=2,400  acc=0.497  f1=0.359
    201902:  n=1,081  acc=0.733  f1=0.137
    201810:  n=1,078  acc=0.327  f1=0.086
    201806:  n=2,438  acc=0.491  f1=0.295
    201802:  n=2,441  acc=0.422  f1=0.138
    201710:  n=2,573  acc=0.768  f1=0.080
    201706:  n=2,271  acc=0.753  f1=0.278
    201702:  n=2,447  acc=0.817  f1=0.308
    201610:  n=2,633  acc=0.868  f1=0.103
    201606:  n=2,635  acc=0.840  f1=0.387

Random Forest — GDELT only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.8011
    Precision (W):  0.1953
    Recall (W):     0.1261
    F1 (worsen):    0.0983

  Per assessment:
    202406:  n=1,919  acc=0.705  f1=0.110
    202402:  n=1,919  acc=0.810  f1=0.071
    202310:  n=1,919  acc=0.944  f1=0.070
    202306:  n=1,376  acc=0.675  f1=0.115
    202302:  n=1,376  acc=0.861  f1=0.050
    202210:  n=2,399  acc=0.855  f1=0.103
    202206:  n=2,490  acc=0.822  f1=0.075
    202202:  n=2,535  acc=0.796  f1=0.134
    202110:  n=2,535  acc=0.920  f1=0.064
    202106:  n=2,489  acc=0.762  f1=0.081
    202102:  n=2,489  acc=0.873  f1=0.071
    202010:  n=2,491  acc=0.848  f1=0.083
    202006:  n=1,588  acc=0.848  f1=0.097
    202002:  n=1,476  acc=0.897  f1=0.073
    201910:  n=2,400  acc=0.875  f1=0.157
    201906:  n=2,400  acc=0.796  f1=0.128
    201902:  n=1,081  acc=0.897  f1=0.098
    201810:  n=1,078  acc=0.247  f1=0.084
    201806:  n=2,438  acc=0.326  f1=0.243
    201802:  n=2,441  acc=0.883  f1=0.095
    201710:  n=2,573  acc=0.904  f1=0.115
    201706:  n=2,271  acc=0.791  f1=0.081
    201702:  n=2,447  acc=0.862  f1=0.066
    201610:  n=2,633  acc=0.956  f1=0.120
    201606:  n=2,635  acc=0.875  f1=0.073

CatBoost — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7696
    Precision (W):  0.3285
```

```
    Recall (W):     0.8128
    F1 (worsen):    0.4432

  Per assessment:
    202406:  n=1,919  acc=0.828  f1=0.704
    202402:  n=1,919  acc=0.707  f1=0.525
    202310:  n=1,919  acc=0.765  f1=0.160
    202306:  n=1,376  acc=0.833  f1=0.764
    202302:  n=1,376  acc=0.749  f1=0.406
    202210:  n=2,399  acc=0.853  f1=0.551
    202206:  n=2,490  acc=0.856  f1=0.639
    202202:  n=2,535  acc=0.806  f1=0.598
    202110:  n=2,535  acc=0.808  f1=0.241
    202106:  n=2,489  acc=0.844  f1=0.726
    202102:  n=2,489  acc=0.713  f1=0.380
    202010:  n=2,491  acc=0.759  f1=0.456
    202006:  n=1,588  acc=0.759  f1=0.479
    202002:  n=1,476  acc=0.632  f1=0.191
    201910:  n=2,400  acc=0.716  f1=0.281
    201906:  n=2,400  acc=0.729  f1=0.534
    201902:  n=1,081  acc=0.772  f1=0.317
    201810:  n=1,078  acc=0.708  f1=0.127
    201806:  n=2,438  acc=0.760  f1=0.504
    201802:  n=2,441  acc=0.703  f1=0.248
    201710:  n=2,573  acc=0.778  f1=0.323
    201706:  n=2,271  acc=0.727  f1=0.529
    201702:  n=2,447  acc=0.716  f1=0.441
    201610:  n=2,633  acc=0.810  f1=0.240
    201606:  n=2,635  acc=0.910  f1=0.716

CatBoost — IPC + GDELT base:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7763
    Precision (W):  0.3344
    Recall (W):     0.8012
    F1 (worsen):    0.4454

  Per assessment:
    202406:  n=1,919  acc=0.853  f1=0.726
    202402:  n=1,919  acc=0.695  f1=0.512
    202310:  n=1,919  acc=0.798  f1=0.182
    202306:  n=1,376  acc=0.831  f1=0.762
    202302:  n=1,376  acc=0.747  f1=0.402
    202210:  n=2,399  acc=0.846  f1=0.540
    202206:  n=2,490  acc=0.859  f1=0.647
    202202:  n=2,535  acc=0.787  f1=0.574
    202110:  n=2,535  acc=0.809  f1=0.244
    202106:  n=2,489  acc=0.851  f1=0.732
    202102:  n=2,489  acc=0.767  f1=0.423
    202010:  n=2,491  acc=0.754  f1=0.452
    202006:  n=1,588  acc=0.739  f1=0.482
    202002:  n=1,476  acc=0.631  f1=0.190
    201910:  n=2,400  acc=0.702  f1=0.271
    201906:  n=2,400  acc=0.742  f1=0.538
    201902:  n=1,081  acc=0.785  f1=0.329
```

```
    201810:  n=1,078  acc=0.709  f1=0.187
    201806:  n=2,438  acc=0.753  f1=0.484
    201802:  n=2,441  acc=0.729  f1=0.246
    201710:  n=2,573  acc=0.785  f1=0.306
    201706:  n=2,271  acc=0.762  f1=0.533
    201702:  n=2,447  acc=0.743  f1=0.461
    201610:  n=2,633  acc=0.821  f1=0.200
    201606:  n=2,635  acc=0.910  f1=0.713

CatBoost — IPC + GDELT + temporal:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7759
    Precision (W):  0.3364
    Recall (W):     0.8041
    F1 (worsen):    0.4465

  Per assessment:
    202406:  n=1,919  acc=0.859  f1=0.740
    202402:  n=1,919  acc=0.696  f1=0.513
    202310:  n=1,919  acc=0.798  f1=0.185
    202306:  n=1,376  acc=0.826  f1=0.750
    202302:  n=1,376  acc=0.741  f1=0.400
    202210:  n=2,399  acc=0.846  f1=0.540
    202206:  n=2,490  acc=0.859  f1=0.641
    202202:  n=2,535  acc=0.799  f1=0.589
    202110:  n=2,535  acc=0.802  f1=0.242
    202106:  n=2,489  acc=0.847  f1=0.727
    202102:  n=2,489  acc=0.756  f1=0.418
    202010:  n=2,491  acc=0.745  f1=0.442
    202006:  n=1,588  acc=0.752  f1=0.477
    202002:  n=1,476  acc=0.636  f1=0.192
    201910:  n=2,400  acc=0.702  f1=0.271
    201906:  n=2,400  acc=0.760  f1=0.553
    201902:  n=1,081  acc=0.764  f1=0.305
    201810:  n=1,078  acc=0.682  f1=0.165
    201806:  n=2,438  acc=0.781  f1=0.509
    201802:  n=2,441  acc=0.740  f1=0.284
    201710:  n=2,573  acc=0.776  f1=0.297
    201706:  n=2,271  acc=0.774  f1=0.558
    201702:  n=2,447  acc=0.743  f1=0.456
    201610:  n=2,633  acc=0.804  f1=0.196
    201606:  n=2,635  acc=0.909  f1=0.712

CatBoost — GDELT only:
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.6356
    Precision (W):  0.2376
    Recall (W):     0.8078
    F1 (worsen):    0.3502

  Per assessment:
    202406:  n=1,919  acc=0.698  f1=0.612
    202402:  n=1,919  acc=0.651  f1=0.478
    202310:  n=1,919  acc=0.388  f1=0.094
```

```
    202306:  n=1,376  acc=0.586  f1=0.579
    202302:  n=1,376  acc=0.427  f1=0.233
    202210:  n=2,399  acc=0.550  f1=0.288
    202206:  n=2,490  acc=0.614  f1=0.412
    202202:  n=2,535  acc=0.764  f1=0.565
    202110:  n=2,535  acc=0.575  f1=0.109
    202106:  n=2,489  acc=0.650  f1=0.547
    202102:  n=2,489  acc=0.544  f1=0.240
    202010:  n=2,491  acc=0.604  f1=0.306
    202006:  n=1,588  acc=0.705  f1=0.451
    202002:  n=1,476  acc=0.547  f1=0.167
    201910:  n=2,400  acc=0.555  f1=0.197
    201906:  n=2,400  acc=0.698  f1=0.497
    201902:  n=1,081  acc=0.695  f1=0.236
    201810:  n=1,078  acc=0.503  f1=0.082
    201806:  n=2,438  acc=0.703  f1=0.436
    201802:  n=2,441  acc=0.712  f1=0.248
    201710:  n=2,573  acc=0.716  f1=0.264
    201706:  n=2,271  acc=0.749  f1=0.546
    201702:  n=2,447  acc=0.626  f1=0.372
    201610:  n=2,633  acc=0.752  f1=0.155
    201606:  n=2,635  acc=0.876  f1=0.643

CatBoost — IPC only (no geo):
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5920
    Precision (W):  0.1959
    Recall (W):     0.6003
    F1 (worsen):    0.2680

  Per assessment:
    202406:  n=1,919  acc=0.668  f1=0.530
    202402:  n=1,919  acc=0.564  f1=0.358
    202310:  n=1,919  acc=0.713  f1=0.014
    202306:  n=1,376  acc=0.768  f1=0.681
    202302:  n=1,376  acc=0.693  f1=0.308
    202210:  n=2,399  acc=0.681  f1=0.307
    202206:  n=2,490  acc=0.651  f1=0.313
    202202:  n=2,535  acc=0.589  f1=0.343
    202110:  n=2,535  acc=0.652  f1=0.137
    202106:  n=2,489  acc=0.647  f1=0.510
    202102:  n=2,489  acc=0.281  f1=0.186
    202010:  n=2,491  acc=0.345  f1=0.268
    202006:  n=1,588  acc=0.399  f1=0.289
    202002:  n=1,476  acc=0.454  f1=0.143
    201910:  n=2,400  acc=0.355  f1=0.201
    201906:  n=2,400  acc=0.396  f1=0.362
    201902:  n=1,081  acc=0.168  f1=0.091
    201810:  n=1,078  acc=0.192  f1=0.086
    201806:  n=2,438  acc=0.782  f1=0.211
    201802:  n=2,441  acc=0.837  f1=0.213
    201710:  n=2,573  acc=0.759  f1=0.049
    201706:  n=2,271  acc=0.727  f1=0.273
    201702:  n=2,447  acc=0.802  f1=0.330
    201610:  n=2,633  acc=0.840  f1=0.102
```

```
    201606:  n=2,635  acc=0.836  f1=0.397

CatBoost — IPC + GDELT base (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5768
    Precision (W):  0.1914
    Recall (W):     0.6233
    F1 (worsen):    0.2697

  Per assessment:
    202406:  n=1,919  acc=0.664  f1=0.528
    202402:  n=1,919  acc=0.559  f1=0.360
    202310:  n=1,919  acc=0.702  f1=0.021
    202306:  n=1,376  acc=0.740  f1=0.660
    202302:  n=1,376  acc=0.655  f1=0.284
    202210:  n=2,399  acc=0.673  f1=0.318
    202206:  n=2,490  acc=0.647  f1=0.315
    202202:  n=2,535  acc=0.580  f1=0.346
    202110:  n=2,535  acc=0.645  f1=0.133
    202106:  n=2,489  acc=0.644  f1=0.510
    202102:  n=2,489  acc=0.298  f1=0.186
    202010:  n=2,491  acc=0.359  f1=0.269
    202006:  n=1,588  acc=0.405  f1=0.282
    202002:  n=1,476  acc=0.463  f1=0.141
    201910:  n=2,400  acc=0.378  f1=0.203
    201906:  n=2,400  acc=0.425  f1=0.363
    201902:  n=1,081  acc=0.205  f1=0.083
    201810:  n=1,078  acc=0.224  f1=0.079
    201806:  n=2,438  acc=0.402  f1=0.285
    201802:  n=2,441  acc=0.832  f1=0.205
    201710:  n=2,573  acc=0.747  f1=0.055
    201706:  n=2,271  acc=0.720  f1=0.275
    201702:  n=2,447  acc=0.792  f1=0.330
    201610:  n=2,633  acc=0.828  f1=0.110
    201606:  n=2,635  acc=0.832  f1=0.401

CatBoost — IPC + GDELT + temporal (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.5827
    Precision (W):  0.1928
    Recall (W):     0.6318
    F1 (worsen):    0.2737

  Per assessment:
    202406:  n=1,919  acc=0.669  f1=0.534
    202402:  n=1,919  acc=0.559  f1=0.362
    202310:  n=1,919  acc=0.690  f1=0.033
    202306:  n=1,376  acc=0.736  f1=0.657
    202302:  n=1,376  acc=0.640  f1=0.294
    202210:  n=2,399  acc=0.668  f1=0.324
    202206:  n=2,490  acc=0.641  f1=0.319
    202202:  n=2,535  acc=0.578  f1=0.358
    202110:  n=2,535  acc=0.631  f1=0.135
    202106:  n=2,489  acc=0.639  f1=0.505
```

```
    202102:  n=2,489  acc=0.312  f1=0.187
    202010:  n=2,491  acc=0.374  f1=0.265
    202006:  n=1,588  acc=0.443  f1=0.297
    202002:  n=1,476  acc=0.484  f1=0.142
    201910:  n=2,400  acc=0.419  f1=0.204
    201906:  n=2,400  acc=0.461  f1=0.372
    201902:  n=1,081  acc=0.227  f1=0.085
    201810:  n=1,078  acc=0.268  f1=0.081
    201806:  n=2,438  acc=0.459  f1=0.298
    201802:  n=2,441  acc=0.787  f1=0.180
    201710:  n=2,573  acc=0.731  f1=0.085
    201706:  n=2,271  acc=0.716  f1=0.290
    201702:  n=2,447  acc=0.776  f1=0.324
    201610:  n=2,633  acc=0.830  f1=0.111
    201606:  n=2,635  acc=0.831  f1=0.402

CatBoost — GDELT only (no geo):
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 53,398 total test samples):
    Accuracy:       0.7521
    Precision (W):  0.1583
    Recall (W):     0.2055
    F1 (worsen):    0.1429

  Per assessment:
    202406:  n=1,919  acc=0.673  f1=0.135
    202402:  n=1,919  acc=0.756  f1=0.096
    202310:  n=1,919  acc=0.895  f1=0.107
    202306:  n=1,376  acc=0.652  f1=0.187
    202302:  n=1,376  acc=0.806  f1=0.113
    202210:  n=2,399  acc=0.820  f1=0.208
    202206:  n=2,490  acc=0.779  f1=0.138
    202202:  n=2,535  acc=0.759  f1=0.212
    202110:  n=2,535  acc=0.867  f1=0.106
    202106:  n=2,489  acc=0.719  f1=0.163
    202102:  n=2,489  acc=0.822  f1=0.133
    202010:  n=2,491  acc=0.801  f1=0.130
    202006:  n=1,588  acc=0.808  f1=0.173
    202002:  n=1,476  acc=0.799  f1=0.103
    201910:  n=2,400  acc=0.799  f1=0.169
    201906:  n=2,400  acc=0.750  f1=0.191
    201902:  n=1,081  acc=0.815  f1=0.083
    201810:  n=1,078  acc=0.198  f1=0.085
    201806:  n=2,438  acc=0.302  f1=0.249
    201802:  n=2,441  acc=0.837  f1=0.099
    201710:  n=2,573  acc=0.826  f1=0.122
    201706:  n=2,271  acc=0.743  f1=0.159
    201702:  n=2,447  acc=0.812  f1=0.155
    201610:  n=2,633  acc=0.908  f1=0.117
    201606:  n=2,635  acc=0.857  f1=0.142

========================================================================
SUMMARY TABLE
========================================================================
              Model                     Feature Set  N Features  IPC  GDELT
Base  GDELT Temporal  Geographic  Accuracy  Prec (W)  Recall (W)  F1 (worse
```

```
n)  Total Test  N Evals
Logistic Regression                      IPC only        3332    1
0               0        3331  0.913540  0.592340    0.995202     0.718133
53398       25
Logistic Regression                IPC + GDELT base        3451    1
119              0        3331  0.912353  0.586028    0.971117     0.707565
53398       25
Logistic Regression          IPC + GDELT + temporal        3486    1
119             35        3331  0.913041  0.583366    0.959005     0.703608
53398       25
          CatBoost          IPC + GDELT + temporal        3486    1
119             35        3331  0.775857  0.336354    0.804070     0.446509
53398       25
          CatBoost                IPC + GDELT base        3451    1
119              0        3331  0.776321  0.334424    0.801168     0.445414
53398       25
Logistic Regression                     GDELT only        3485    0
119             35        3331  0.710816  0.314799    0.951969     0.444828
53398       25
          CatBoost                      IPC only        3332    1
0               0        3331  0.769604  0.328540    0.812808     0.443190
53398       25
      Random Forest                IPC + GDELT base        3451    1
119              0        3331  0.770681  0.309031    0.685851     0.403905
53398       25
      Random Forest          IPC + GDELT + temporal        3486    1
119             35        3331  0.778561  0.310356    0.642611     0.395454
53398       25
      Random Forest                      IPC only        3332    1
0               0        3331  0.685821  0.260219    0.812539     0.375589
53398       25
      Random Forest                     GDELT only        3485    0
119             35        3331  0.743950  0.270121    0.635512     0.363047
53398       25
          CatBoost                     GDELT only        3485    0
119             35        3331  0.635580  0.237609    0.807769     0.350199
53398       25
          CatBoost IPC + GDELT + temporal (no geo)         155    1
119             35           0  0.582680  0.192831    0.631845     0.273701
53398       25
          CatBoost       IPC + GDELT base (no geo)         120    1
119              0           0  0.576767  0.191416    0.623261     0.269651
53398       25
          CatBoost               IPC only (no geo)           1    1
0               0           0  0.592050  0.195942    0.600260     0.268030
53398       25
      Random Forest               IPC only (no geo)           1    1
0               0           0  0.592050  0.195942    0.600260     0.268030
53398       25
      Random Forest IPC + GDELT + temporal (no geo)         155    1
119             35           0  0.622198  0.197773    0.568903     0.267475
53398       25
      Random Forest       IPC + GDELT base (no geo)         120    1
119              0           0  0.574420  0.190687    0.603699     0.262301
53398       25
Logistic Regression               IPC only (no geo)           1    1
```

```
0               0           0  0.589135  0.176636    0.549096     0.251562
53398       25
Logistic Regression IPC + GDELT + temporal (no geo)         155    1
119              35           0  0.618396  0.174637    0.517751     0.245088
53398       25
Logistic Regression       IPC + GDELT base (no geo)         120    1
119               0           0  0.623494  0.171926    0.492473     0.234150
53398       25
           CatBoost             GDELT only (no geo)         154    0
119              35           0  0.752064  0.158280    0.205516     0.142902
53398       25
Logistic Regression             GDELT only (no geo)         154    0
119              35           0  0.736126  0.141109    0.212265     0.140459
53398       25
      Random Forest             GDELT only (no geo)         154    0
119              35           0  0.801112  0.195286    0.126119     0.098264
53398       25

======================================================================
RANDOM FOREST: AVERAGE FEATURE IMPORTANCES
======================================================================

── IPC only (averaged over 25 splits) ──
Feature                                                    Importance
----------------------------------------------------------------------
  IPC_last                                                     0.0942
  ADMIN0_yemen                                                 0.0554
  ADMIN0_nigeria                                               0.0395
  ADMIN0_kenya                                                 0.0268
  ADMIN0_sudan                                                 0.0238
  ADMIN0_somalia                                               0.0220
  ADMIN0_chad                                                  0.0195
  ADMIN0_south sudan                                           0.0184
  ADMIN1_jigawa                                                0.0176
  ADMIN0_zimbabwe                                              0.0161
  ADMIN0_democratic republic of the congo                      0.0154
  ADMIN0_uganda                                                0.0145
  ADMIN0_mozambique                                            0.0142
  ADMIN1_akwa ibom                                             0.0133
  ADMIN1_western                                               0.0133
  ADMIN1_oyo                                                   0.0127
  ADMIN1_osun                                                  0.0114
  ADMIN0_liberia                                               0.0112
  ADMIN1_hadramaut                                             0.0111
  ADMIN1_al hudaydah                                           0.0103
  ADMIN1_kano                                                  0.0100
  ADMIN0_burundi                                               0.0098
  ADMIN1_katsina                                               0.0084
  ADMIN1_imo                                                   0.0081
  ADMIN1_taizz                                                 0.0079
  ... (3307 more features omitted)

── IPC + GDELT base (averaged over 25 splits) ──
Feature                                                    Importance
----------------------------------------------------------------------
  IPC_last                                                     0.0792
```

ADMIN0_yemen	0.0395
ADMIN0_nigeria	0.0253
ADMIN0_kenya	0.0224
ADMIN0_sudan	0.0200
ADMIN0_somalia	0.0145
ADMIN0_south sudan	0.0139
ADMIN0_chad	0.0107
ADMIN0_zimbabwe	0.0104
ADMIN0_uganda	0.0103
ADMIN1_jigawa	0.0100
ADMIN0_democratic republic of the congo	0.0091
ADMIN0_mozambique	0.0090
evt_compound_score_sum	0.0076
ADMIN1_osun	0.0074
ADMIN1_western	0.0071
evt_sentiment.compound_mean	0.0071
evt_pos_score_mean	0.0069
evt_neg_score_mean	0.0069
evt_sentiment.compound_sum	0.0069
evt_compound_score_mean	0.0068
evt_sentiment.neg_mean	0.0068
ADMIN0_burundi	0.0068
ADMIN1_oyo	0.0068
evt_neg_score_sum	0.0066
... (3426 more features omitted)	

── IPC + GDELT + temporal (averaged over 25 splits) ──

Feature	Importance
IPC_last	0.0733
ADMIN0_yemen	0.0385
ADMIN0_nigeria	0.0268
ADMIN0_kenya	0.0196
ADMIN0_sudan	0.0160
ADMIN0_somalia	0.0132
ADMIN0_south sudan	0.0124
ADMIN0_chad	0.0103
ADMIN1_jigawa	0.0098
ADMIN0_uganda	0.0096
evt_compound_score_mean_rolling_3	0.0092
ADMIN0_zimbabwe	0.0089
evt_displaced_freq_mean_rolling_6	0.0084
evt_sentiment.compound_mean_rolling_6	0.0082
evt_compound_score_mean_rolling_6	0.0081
ADMIN0_mozambique	0.0080
evt_sentiment.compound_mean_rolling_3	0.0077
ADMIN0_democratic republic of the congo	0.0073
evt_fatalities_freq_mean_rolling_6	0.0071
ADMIN1_akwa ibom	0.0070
evt_displaced_freq_mean_rolling_3	0.0070
evt_compound_score_sum	0.0068
evt_sentiment.compound_sum	0.0067
evt_compound_score_mean	0.0064
evt_fatalities_freq_mean_rolling_3	0.0063
... (3461 more features omitted)	

── GDELT only (averaged over 25 splits) ──

Feature	Importance
ADMIN0_yemen	0.0401
ADMIN0_nigeria	0.0266
ADMIN0_kenya	0.0213
ADMIN0_somalia	0.0147
ADMIN0_sudan	0.0146
ADMIN0_south sudan	0.0124
ADMIN1_jigawa	0.0119
ADMIN0_chad	0.0115
ADMIN0_democratic republic of the congo	0.0097
evt_sentiment.compound_mean_rolling_6	0.0093
ADMIN0_uganda	0.0093
evt_displaced_freq_mean_rolling_6	0.0092
evt_compound_score_mean_rolling_6	0.0090
evt_compound_score_mean_rolling_3	0.0090
ADMIN0_zimbabwe	0.0089
evt_sentiment.compound_mean_rolling_3	0.0089
evt_fatalities_freq_mean_rolling_6	0.0078
ADMIN0_mozambique	0.0070
evt_fatalities_freq_mean_rolling_3	0.0070
evt_displaced_freq_mean_rolling_3	0.0070
evt_sentiment.compound_sum	0.0069
ADMIN1_akwa ibom	0.0069
evt_neg_score_mean	0.0069
evt_sentiment.compound_mean	0.0068
ADMIN1_osun	0.0068
... (3460 more features omitted)	

── IPC only (no geo) (averaged over 25 splits) ──

Feature	Importance
IPC_last	1.0000

── IPC + GDELT base (no geo) (averaged over 25 splits) ──

Feature	Importance
IPC_last	0.3017
evt_compound_score_sum	0.0201
evt_sentiment.compound_sum	0.0197
evt_compound_score_mean	0.0195
evt_sentiment.neg_mean	0.0192
evt_neg_score_mean	0.0192
evt_sentiment.compound_mean	0.0190
evt_neu_score_mean	0.0189
evt_sentiment.pos_mean	0.0188
evt_sentiment.neu_mean	0.0187
evt_pos_score_mean	0.0186
evt_neg_score_sum	0.0176
evt_sentiment.neg_sum	0.0174
evt_sentiment.pos_sum	0.0164
evt_sentiment.neu_sum	0.0157
evt_neu_score_sum	0.0153
evt_pos_score_sum	0.0149
evt_fatalities_count_mean	0.0141

```
  evt_fatalities_freq_mean                                    0.0135
  evt_fatalities_count_sum                                    0.0126
  evt_displaced_freq_mean                                     0.0120
  evt_detained_count_mean                                     0.0120
  evt_fatalities_freq_sum                                     0.0107
  evt_detained_count_sum                                      0.0105
  evt_displaced_count_mean                                    0.0102
  ... (95 more features omitted)

── IPC + GDELT + temporal (no geo) (averaged over 25 splits) ──
Feature                                                   Importance
------------------------------------------------------------------------
  IPC_last                                                    0.2069
  evt_sentiment.compound_mean_rolling_6                       0.0235
  evt_compound_score_mean_rolling_6                           0.0227
  evt_sentiment.compound_mean_rolling_3                       0.0211
  evt_compound_score_mean_rolling_3                           0.0211
  evt_fatalities_freq_mean_rolling_6                          0.0204
  evt_displaced_freq_mean_rolling_6                           0.0195
  evt_fatalities_freq_mean_rolling_3                          0.0162
  evt_displaced_freq_mean_rolling_3                           0.0157
  evt_agriculture_freq_mean_rolling_6                         0.0154
  evt_sentiment.compound_mean_lag1                            0.0152
  evt_compound_score_mean_lag1                                0.0151
  evt_compound_score_mean_lag2                                0.0149
  evt_sentiment.compound_mean_lag2                            0.0145
  evt_sentiment.pos_mean                                      0.0131
  evt_compound_score_sum                                      0.0128
  evt_sentiment.compound_sum                                  0.0126
  evt_sentiment.neu_mean                                      0.0126
  evt_sentiment.neg_mean                                      0.0125
  evt_sentiment.compound_mean                                 0.0123
  evt_food_insecurity_freq_mean_rolling_6                     0.0121
  evt_pos_score_mean                                          0.0120
  evt_neg_score_mean                                          0.0119
  evt_neu_score_mean                                          0.0118
  evt_compound_score_mean                                     0.0118
  ... (130 more features omitted)

── GDELT only (no geo) (averaged over 25 splits) ──
Feature                                                   Importance
------------------------------------------------------------------------
  evt_compound_score_mean_rolling_6                           0.0281
  evt_sentiment.compound_mean_rolling_6                       0.0279
  evt_compound_score_mean_rolling_3                           0.0260
  evt_sentiment.compound_mean_rolling_3                       0.0255
  evt_fatalities_freq_mean_rolling_6                          0.0248
  evt_displaced_freq_mean_rolling_6                           0.0243
  evt_compound_score_mean_lag1                                0.0210
  evt_fatalities_freq_mean_rolling_3                          0.0201
  evt_sentiment.compound_mean_lag1                            0.0193
  evt_compound_score_mean_lag2                                0.0190
  evt_displaced_freq_mean_rolling_3                           0.0186
  evt_agriculture_freq_mean_rolling_6                         0.0186
  evt_sentiment.compound_mean_lag2                            0.0184
  evt_sentiment.compound_sum                                  0.0168
```

```
evt_compound_score_sum                                  0.0163
evt_sentiment.neu_mean                                  0.0163
evt_pos_score_mean                                      0.0162
evt_neu_score_mean                                      0.0161
evt_compound_score_mean                                 0.0157
evt_sentiment.compound_mean                             0.0155
evt_sentiment.pos_mean                                  0.0152
evt_sentiment.neg_mean                                  0.0150
evt_neg_score_mean                                      0.0147
evt_sentiment.neg_sum                                   0.0145
evt_food_insecurity_freq_mean_rolling_6                 0.0145
... (129 more features omitted)
```

Experimental Findings

1. Level Prediction (10_predictions_next_ipc.ipynb)

Task: Predict the exact IPC CS_score (1-5) at each assessment period.

Key finding -- GDELT as IPC substitute: Logistic Regression with GDELT + geographic features (no IPC history) achieves F1 = 0.797, outperforming IPC_last + geographic features (F1 = 0.782). GDELT can serve as a viable alternative to IPC history where IPC assessments are unavailable, delayed, or have gaps.

Model	IPC only	IPC + GDELT + temporal	GDELT only
Logistic Regression	0.782	0.781	0.797
CatBoost	0.697	0.702	0.690
Random Forest	0.704	0.704	0.633

Observations:

- Geographic dummies (location fixed effects) are essential context -- they encode structural IPC levels
- Without geographic features, GDELT alone collapses (~0.32 F1) -- news signals encode what is happening, not absolute IPC levels
- LR handles the high-dimensional feature space (3,400+ features) best due to implicit regularization
- Tree models (RF, CatBoost) struggle with 3,331 sparse geographic dummies

2. Transition Prediction (this notebook)

Task: Binary -- did CS_score worsen vs the previous assessment? (~17% positive rate)

Key finding -- GDELT improves worsening detection: GDELT features consistently improve transition prediction across models. CatBoost with IPC + GDELT + temporal

achieves the best overall F1 for detecting worsenings.

Model	IPC only F1	+ GDELT + temporal F1	Improvement
CatBoost	0.649	0.659	+1.0
Logistic Regression	0.550	0.582	+3.2
Random Forest	0.492	0.492	flat

Recall (catching actual worsenings -- the policy-relevant metric):

- CatBoost achieves ~89% recall across all feature sets -- catches 9 out of 10 worsenings
- LR recall improves from 43% to 47% with GDELT features
- No-Change baseline gets 0% recall (never predicts worsening)

Observations:

- CatBoost handles the ~17% class imbalance much better than LR or RF
- GDELT largest relative contribution is for LR (+3.2 F1 points), where the model is interpretable
- Without geographic features, CatBoost + GDELT still works (F1 = 0.366) while LR collapses

Overall Narrative

1. **GDELT as IPC substitute**: When IPC history is unavailable, GDELT + location context can produce predictions as good as (or better than) IPC history alone. Relevant for new regions, coverage gaps, and between-assessment monitoring.
2. **GDELT for transition detection**: GDELT features capture escalation signals that improve detection of IPC worsening. The improvement is modest but consistent, and matters most for the policy-relevant metric (recall).
3. **Autocorrelation challenge**: For level prediction, food security autocorrelation makes it hard for any feature to outperform same as last time. Transition prediction strips this out and reveals GDELT contribution.
4. **Model selection**: CatBoost is best for transitions (handles imbalance), LR is best for levels (handles high dimensionality, interpretable).

Potential Further Experiments

1. **Longer forecast horizon**: Predict 2-3 assessments ahead -- GDELT early warning value should increase
2. **Crisis detection**: Binary classification (CS >= 3 vs < 3) with focus on recall
3. **SHAP / feature importance analysis**: Show which specific GDELT features drive predictions

4. **Subgroup analysis**: GDELT value for areas that transition vs stable areas

Previous notes (CS≥3) vs non-crisis (CS<3) rather than 5-class classification

2. **Feature Engineering**:
    - **Temporal Patterns**: Using lagged and rolling features (as we just added) rather than only current-period values
    - **Anomaly Detection**: Focusing on deviations from historical patterns
    - **Interaction Features**: Combining GDELT with other data sources (weather, prices, etc.)
3. **Data Coverage**:
    - **Spatial Aggregation**: Some studies aggregate to ADMIN0 or ADMIN1 level where coverage is better
    - **Temporal Aggregation**: Using quarterly rather than monthly data
    - **Filtering**: Focusing on regions/periods with good GDELT coverage
4. **Evaluation Metrics**:
    - **Recall for Rare Events**: Focusing on detecting crises (high recall for CS≥3) rather than overall accuracy
    - **Early Detection**: Measuring how early they detect transitions, not just accuracy
5. **Baseline Comparison**:
    - **Weaker Baselines**: Some studies don't include `previous_CS` in baseline, making improvement easier to show
    - **Different Baselines**: Comparing against simpler models or using different evaluation windows

Our Current Approach:

- ✅ Now includes temporal GDELT features (lags, rolling windows, anomalies, escalation)
- ✅ Uses same framework as previous work (allowing fair comparison)
- ⚠️ Still predicting exact CS_score level (highly autocorrelated)
- ⚠️ Using monthly data with CS_score available every 4 months

Potential Improvements:

1. **Predict Transitions**: Predict if CS_score will worsen (CS_t+1 > CS_t) rather than exact value
2. **Forecasting Horizon**: Predict 2-4 periods ahead instead of next period

3. **Crisis Detection**: Binary classification (crisis vs non-crisis) with focus on recall
4. **Feature Selection**: Use only temporal GDELT features, filter out low-importance current-period features

Feature Sets Used

| Feature Set | IPC | GDELT Base | GDELT Temporal | Geographic | Total | |---|---|---|---|---|---| | IPC only | 1 | 0 | 0 | 3,331 | 3,332 | | IPC + GDELT base | 1 | 119 | 0 | 3,331 | 3,451 | | IPC + GDELT + temporal | 1 | 119 | 35 | 3,331 | 3,486 | | GDELT only | 0 | 119 | 35 | 3,331 | 3,485 | | (no geo) variants | same as above but without the 3,331 geographic dummies |

- **IPC**: `IPC_last` (forward-filled last observed IPC score)
- **GDELT base** (119): aggregated event features (`evt_` prefix) -- counts, sentiment, NER, topic frequencies
- **GDELT temporal** (35): rolling windows, lags, and escalation indicators from GDELT features
- **Geographic** (3,331): one-hot encoded ADMIN0/ADMIN1/ADMIN2 location dummies